

Actively Passive: The Rise of Market Volatility

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Abstract

We show that U.S. stock market volatility has roughly doubled since the 1950s. The increase in volatility is strongest for daily returns and weaker for longer horizons, which points against changes in the volatility of fundamentals and toward a larger transitory component in returns. We trace the increase in volatility to two related shifts in market structure: investors increasingly trade the market index, through futures, index funds, and ETFs, and can trade over more hours of the day. We exploit changes in NYSE trading hours, and in the hours and access of index futures trading, as natural experiments. In each case, expanding the opportunity to trade brings more trading and more volatility, so volatility accrues in trading time as well as calendar time. Together, the estimates from these experiments account for roughly 70% of the rise in volatility. A model in which arbitrageurs absorb investors' demand shocks, increasingly expressed through the index and over more trading hours, accounts for these facts, including the shift in returns from continuation to reversal.

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1. Introduction

We document that stock market volatility has roughly doubled over the last 70 years. Figure 1 shows this rise in the rolling 5-year standard deviation of daily US market returns. Stock market volatility, constructed from daily returns, is a central measure in macroeconomics and finance (e.g., Bloom, 2009; Martin, 2017; Berger, Dew-Becker, and Giglio, 2020).¹ The goal of this paper is to document and understand this fact.

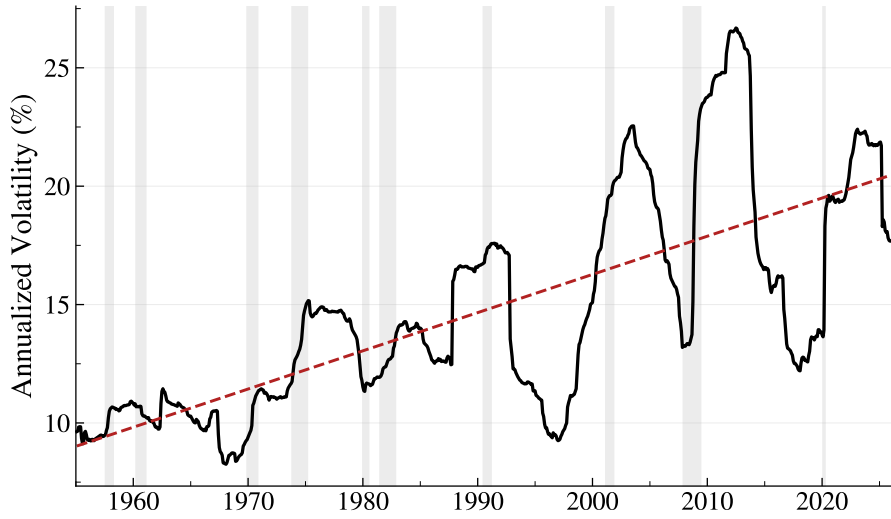


Figure 1: Stock Market Volatility. 5-year rolling standard deviation of daily CRSP value-weighted market return.

The structure of the increase already rules out some explanations. Daily volatility has risen 127%. As we cumulate returns to lower frequencies the increase shrinks, to 77% for weekly returns and 38% for monthly returns. Because a monthly (log) return is the sum of the daily returns inside it, this pattern means that the autocorrelation of returns has fallen substantially. If cash flow volatility were driving the increase, volatility would rise similarly across horizons, so the steep frequency gradient points to something about how markets operate day to day and against changes in

¹Bloom (2009) uses realized stock market volatility to identify uncertainty shocks that drive business cycles. Martin (2017) argues for a tight link between the conditional volatility of the stock market and the conditional market risk premium. Berger, Dew-Becker, and Giglio (2020) show it is specifically realized volatility — not forward-looking implied volatility — that predicts contractions.

the volatility of fundamentals. Further, we show that the rise in stock market volatility has been a rise relative to the realized volatility of fundamentals (profits, GDP, industrial production).

We argue that a large part of this increase is not about fundamentals but about increased trading activity that affects the short-horizon dynamics of returns. In the 1950s, an investor who wanted exposure to the market had to effectively buy stocks one at a time, and only during the exchange's trading day. Today hundreds of billions of dollars change hands around the clock in products that track the market index, including index futures, ETFs, and funds. This paper provides stylized facts and causal evidence that shifts towards (1) increased trading of the market as a whole and (2) increased ability to trade more often (i.e., overnight futures) can help account for a meaningful share of the increase in stock market volatility.

To measure the shift toward index trading directly, we first construct a market volume beta: the slope from a daily cross-sectional regression of each stock's share of total volume on its weight in the market portfolio. If all trading were index trading, this slope would equal one. It has doubled from 0.4 to 0.8 over our sample, tracking the rise in volatility closely. Thus, investors are trading relatively more along the market dimension than they used to. This change occurs slowly over time, consistent with the gradual move towards increasing trading of a market index. This started early in the 1960s when institutional investors became more dominant, concentrating trade in the Nifty Fifty. Wells Fargo started an institutional index fund in 1971, Vanguard a retail index fund in 1976, and S&P500 futures began trading in 1982. Overnight index trading became possible when S&P500 futures were listed on CME's electronic GLOBEX system in 1993, took hold with the E-mini introduction in 1997, and has continued increasing relative to intraday trading in S&P500 futures until the end of the sample. The trend in volatility shown in Figure 1 appears to start in the 1970s, which is consistent with the introduction and gradual adoption of these index trading instruments.

This correlation does not establish causation — volatility could drive trading rather than the reverse, or a third factor could drive both. We provide causal evidence from

natural experiments. The first builds on French and Roll (1986), who study what happens when the New York Stock Exchange closed every Wednesday for six months in 1968. If price movements reflect information, weekly returns should be unaffected — the same news arrives whether or not the exchange is open on Wednesday. Instead, weekly volatility falls by roughly one-fifth, implying that volatility tracks trading time more than calendar time. We extend this logic to nine changes in NYSE trading hours between 1952 and 1985, including the elimination of Saturday trading, extensions and reductions in daily hours, and the Wednesday closures. Using trading hours as an instrument for volume, we estimate that a 10% increase in trading volume raises market volatility by about 12%.

Much of the rise in daily volatility since the early 1990s has come from the overnight portion of market returns, the period in which overnight index trading was introduced and grew. Our second set of experiments uses this window to isolate index trading. Individual stocks could not trade overnight, but the index could once CME opened an overnight electronic session for S&P 500 futures in September 1993, and the E-mini made overnight index trading far easier when it arrived in September 1997. Comparing overnight-to-intraday price movements for the market versus individual stocks around each change, we find overnight market variance rises at both, by about 3% per hour of overnight index trading in 1993 and by about 50% at the E-mini, and futures volume rises with it, modestly in 1993 and sharply in 1997. The market became more volatile precisely when and where index trading expanded.

The third design turns index trading on and off within a single day. On NYSE holidays no individual stock can trade, but S&P 500 futures trade through the morning and then halt until the evening, so one day supplies a window in which only the index can be traded and a window in which nothing can, back to back. In 2014 CME moved the halt from 10:30 to noon, so the same clock window is dead on holidays before 2014 and index-only after. If prices moved only because news arrives, an hour in which the index can be traded would carry the same variance as an hour in which nothing trades. Instead it carries more than ten times as much. Finally, in 2003 CME moved its nightly one-hour shutdown from 10pm to 4:30pm, so on a single date one

window went from shut to open and another from open to shut. Variance rose in the hour that opened and fell in the half hour that shut.

We end by sizing up the experiments. We apply the estimates from the trading-hours and futures experiments to the actual history of trading hours, across all of the NYSE schedule changes and the changes in index futures hours and access, and ask how much of the rise in volatility they would have produced on their own. The answer is roughly 70%.

In all of these experiments, expanding the ability to trade results in both increased trading activity (volume) and in increased volatility. In contrast, in canonical asset pricing models, volatility is based on calendar time and not trade time. That is, if the stock market was only open one hour per week, the volatility of weekly compounded returns should be the same as in the case when the market is open 24 hours a day. In the former case, traders would collect all relevant information through the week and then move prices in large amounts during the short time when the market was open, rather than continuously as information arrived. But price moves over a calendar week would be the same. Our findings instead imply that expanding the ability to trade leads to more trading and larger price movements: volatility and volume track trade time rather than only depending on calendar time. Because of this, much of the rise in volatility we document in the later part of our sample (from the early 1990s onwards) comes from a rise in volatility of overnight returns as investors are now easily able to trade market indices nearly around the clock.

To interpret these patterns, we develop a simple model of index trading, along the lines of Campbell et al. (1993). In a market without index trading, investors hold concentrated portfolios (following Merton (1987)), while risk-averse arbitrageurs that trade in all stocks absorb investors' idiosyncratic demand shocks and lean against the systematic component of investors' demand shocks. When index trading is introduced, investors instead trade the market portfolio, which allows a more aggressive allocation to stocks, due to its higher Sharpe ratio relative to undiversified portfolios. Further, we let demand shocks scale with trade-time rather than calendar-time: more trading opportunities within a calendar day generate a larger total flow of demand

shocks, motivated by the idea that trading and observing transaction prices generate new signals, attention, or sentiment shocks, consistent with the spirit of French and Roll (1986) and with our empirical findings. Arbitrageurs are thus required to absorb larger systematic demand shocks than before, which they only partially accommodate. This shift in trading structure raises market volatility, reduces return autocorrelation, and increases the share of variance explained by market-wide factors. The predictions of the model are borne out in the data, which show both a decline in return autocorrelations and a rise in the systematic component of stock returns over time. Further, the model matches increased volume in market index trades and the large increase in market volatility.

Related literature and implications. Our findings bear on the rise of passive investing. Index investing is passive in the cross-section, but trading in index products is anything but. As David Booth puts it: *“There’s been an incredible movement to indexing, and yet there’s been an incredible increase in trading volume. [...] Index funds are the ideal market timing vehicle. [...] Instead of individual stock selection, it’s kind of like a big gambling casino.”*² The passive literature mostly studies cross-sectional efficiency (Sammon, 2025; Haddad et al., 2025; Brightman and Harvey, 2025) and the growth of passive ownership (French, 2008; Chinco and Sammon, 2024); we study what trade in index products does to aggregate volatility.

Our findings imply that volatility reflects how trading is organized as well as economic uncertainty. This matters for measuring macroeconomic uncertainty from daily stock returns (Bloom, 2009; Martin, 2017; Berger et al., 2020), for portfolio choice as systematic risk rises (Moreira and Muir, 2017; Lochstoer and Muir, 2022), and for how market macrostructure shapes asset prices (Haddad and Muir, 2025; Harvey et al., 2025). It also raises a policy question: proposals to extend trading hours, such as NYSE Arca’s 22-hour trading day planned for December 2026, could raise volatility further.

Three literatures inform our analysis. The first is excess volatility: Shiller (1981)

²Odd Lots Podcast, Eugene Fama and David Booth on the Birth of Modern Finance, minute 27.

showed that stock prices move too much to be justified by subsequent dividends, and Schwert (1989) established the benchmark relationship between market volatility and macroeconomic variables that we seek to explain. Our model builds on the demand-shock framework of Campbell et al. (1993), the noise traders of De Long et al. (1990), the limits of arbitrage in Shleifer and Vishny (1997), and the disagreement models surveyed by Hong and Stein (2007); we add a new source of excess volatility, index-level trading that forces arbitrageurs to absorb systematic rather than idiosyncratic demand shocks. The second is index inclusion and passive ownership: demand curves for stocks slope down (Shleifer, 1986), stocks added to the index co-move more with it (Barberis et al., 2005), ETF ownership raises stock-level volatility (Ben-David et al., 2018), and indexing is linked to negative index serial dependence (Baltussen et al., 2019); see also Da and Shive (2018); Greenwood and Thesmar (2011); Israeli et al. (2017); Basak and Pavlova (2013); Sullivan and Xiong (2012). These designs exploit cross-sectional variation in exposure to index demand; we ask whether the aggregate growth of index trading has raised the level of market volatility, using discrete changes in trading time and overnight index-trading access. The third is trading and volatility: French and Roll (1986) showed that trading itself induces volatility, volume and volatility move together (Karpoff, 1987) in a way usually attributed to information arrival (Admati and Pfleiderer, 1988), and Bessembinder and Seguin (1992) find mixed effects of futures activity on stock volatility in a sample with no overnight futures trading. We add a decomposition into overnight and intraday variance that compares systematic and idiosyncratic variance within the overnight window, where the index trades via futures but individual stocks cannot (see also Lockwood and Linn, 1990; Barclay and Hendershott, 2003; Hasbrouck, 2003; Lou et al., 2019).

A natural objection is that landmark launches (index futures in 1982, the Vanguard index fund in 1976, the SPY ETF in 1993) produce no immediate jump in market volatility. This is consistent with our story: these products had negligible trading volume at launch and took years or decades to gain adoption, so in instrumental variables language a launch has a weak first stage. Our experiments have strong ones: closing a trading day removes activity that was already happening at full volume,

extending trading hours gives immediate access to people who already know how to trade, and the E-mini made an existing contract cheaper and electronic. Event studies at the 1993 opening of overnight index trading and the 1997 E-mini anchor the slower buildup, and the within-day design needs no first stage at all: it compares hours in which the index can be traded with hours in which nothing can.

The rest of the paper is organized as follows. Section 2 documents the rise in market volatility. Section 3 presents the causal evidence, from changes in NYSE trading hours, from the opening of overnight index futures trading and the E-mini, and from turning index trading on and off within a day, and adds up how much of the rise the experiments explain. Section 4 develops the model. Section 5 concludes.

2. Market Volatility: Broad Facts

We take the CRSP value-weighted market return in excess of the risk-free rate from Ken French at the daily and weekly frequency. Volatility is the square root of the annualized sum of squared returns over a rolling 5-year window, expressed in percent per year.³

Table 1 regresses this series on a linear time trend over 1955 to 2025:

$$\sigma_t = a + b\tilde{t} + \varepsilon_t \quad (1)$$

where $\tilde{t} = (t - 1)/(T - 1)$ runs from 0 at the start of the sample to 1 at the end and T is the number of observations. The constant a is the level of volatility at the start of the sample, b is the percentage point increase over the sample, $a + b$ is the level at the end, and b/a is the percentage increase. The linear trend is a way to gauge the size of the rise. We do not mean that volatility will keep rising at this rate, or that the rise from 1955 to 2025 was linear.

For daily data the trend coefficient is 11.4% on a constant of about 9%: volatility has more than doubled over the sample. Because volatility is highly persistent,

³Computing volatility at shorter horizons, e.g., monthly, and then averaging these values over a 5-year period produces qualitatively similar results.

conventional standard errors would overstate the significance of a trend. We report Newey-West standard errors with 260 lags (5 years of weekly observations) and compute p -values using the fixed- b critical values of Lazarus et al. (2018).⁴⁵ Section 2.3 shows the result is robust to outliers, the sample start date, stale prices and other microstructure issues, and alternative ways of measuring volatility.

Table 1: Volatility Time Trends by Return Frequency

	Daily (1)	Weekly (2)	4-Week (3)
Constant	9.00 (0.68)	10.75 (0.91)	12.77 (0.98)
Trend	11.44 (2.03)	8.24 (1.94)	4.85 (1.99)
p -value	<0.001	<0.001	0.024
N	3,705	3,704	3,704
R^2	0.517	0.370	0.168
Vol increase	127%	77%	38%

Notes: Each column regresses 5-year rolling annualized volatility (in %) on a constant and a linear time trend normalized to zero at the start and one at the end of the sample. The dependent variable is the square root of the annualized 5-year realized variance. Column (1) uses daily excess returns from the Fama–French market factor, with weekly RV computed as the sum of squared demeaned daily returns within each calendar week, and 5-year RV as the sum of 260 weekly RVs. Column (2) uses weekly Fama–French excess returns. Column (3) uses 4-week excess returns, constructed by geometric compounding of weekly gross market and risk-free returns. All regressions use weekly overlapping observations from January 1955 to December 2025. Standard errors (in parentheses) use the Newey–West estimator with Bartlett kernel and $S = 260$ lags. p -values use fixed- b critical values with $\kappa = \lfloor 3/(2b) \rfloor$ degrees of freedom, where $b = S/T$. The last row reports the percentage increase in trend volatility from the beginning to the end of the sample.

Column 2 uses weekly returns: the trend is about 8 percentage points, a 77% rise over the sample. Column 3 uses compounded overlapping 4-week returns: about 5

⁴See also Vogelsang (1998) for the general fixed- b framework. The fixed- b approach accounts for the well-known size distortions of standard HAR tests when the bandwidth is a non-negligible fraction of the sample size, as it is here.

⁵Inference is also robust to a moving-block residual bootstrap: with 10-year blocks (seven effective blocks across the sample) the 95% bootstrap confidence interval for the daily-column trend is $[+7.9, +15.2]$; with 20-year blocks (only three effective blocks) it is $[+8.6, +14.6]$.

percentage points, a 38% rise. Volatility rises at every horizon, but the rise is concentrated in short-horizon returns. Because a compounded (log) return over a longer horizon is the sum of the daily log returns inside it, a trend that differs across horizons must reflect a change in the autocorrelation of returns. If autocorrelation at all horizons were zero, the expected squared weekly return would equal the expected sum of squared daily returns within the week.

The pattern across horizons points to changes in financial markets and trading itself. Time-varying volatility of fundamentals cannot produce a rise that differs by return horizon. That requires a rise in “discount rate” volatility, the volatility of a mean-reverting component of returns, and because the pattern sits at horizons of days and weeks the natural candidates are based on trading.⁶ We return to fundamentals shortly.

2.1 Horizon Effects and Return Autocorrelations

Figure 2 plots the trend in volatility by return horizon, from 1 day out to 1 quarter (13 weeks). The point estimate is positive at every horizon and significant out to around 6 weeks. The gradual decay is consistent with short-run reversals in returns.

Figure 3 plots the ratio of 5-year return variance from daily returns to 5-year return variance from monthly returns, alongside daily volatility from Figure 1. The ratio measures autocorrelation: when daily-based variance exceeds monthly-based variance the market price mean-reverts, because short-term price pressure that reverts quickly shows up in daily variance but much less in monthly variance. The two series track each other closely, and the variance ratio moves from around 0.6 to over 2, crossing one in the late 1990s: returns went from continuation to reversal. The variance ratio breaks at March 2000, the same date as the level of volatility (Appendix A.4), and the same rise appears in 35 of 35 international markets (Appendix A.7).

Section A.8 plots the daily and weekly return autocorrelations directly and extends the volume-conditioned regression of Campbell et al. (1993) (CGW) to the modern

⁶Typical discount rate volatility in traditional macro finance models (say, with time-varying risk-aversion) focuses on long-run mean-reversion at the horizon of years rather than weeks.

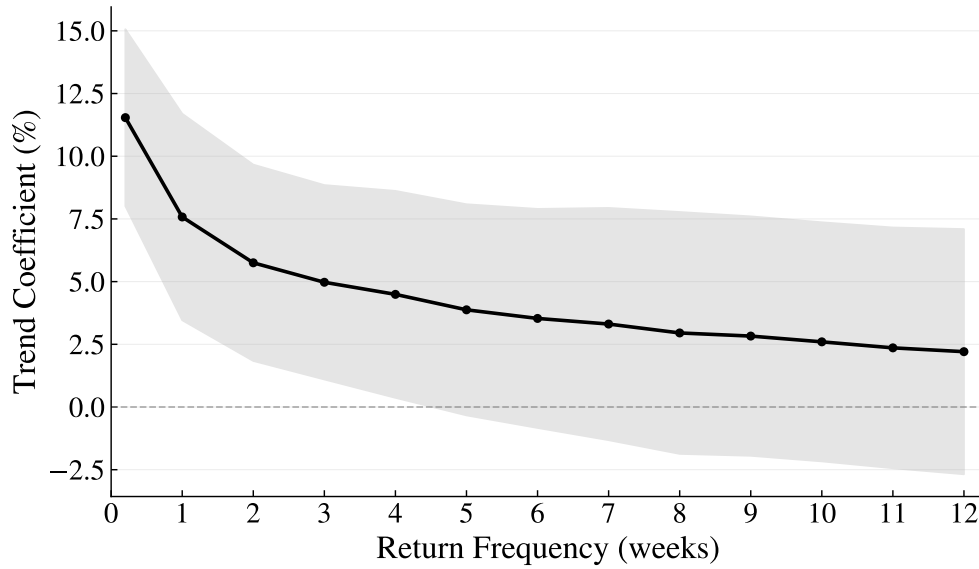


Figure 2: Time Trend in Realized Volatility by Return Frequency

Notes: Trend coefficient from regressing 5-year rolling annualized volatility on a linear time trend, by return horizon: daily (plotted at 0.2 weeks) and 1–12 week overlapping returns, constructed as in Table 1. The shaded band is a 95% confidence interval from Newey–West standard errors with $S = 624$ lags and fixed- b critical values. Sample: 1955–2025.

sample. The daily autocorrelation falls from roughly $+0.30$ in the 1970s to -0.15 by the 2020s, crossing zero around 2003; the weekly series crosses zero around 1990 and settles near -0.10 , and both trends are strongly significant under fixed- b inference.

2.2 Volatility, Volume, and Fundamentals

Figure 4 plots market volatility against total stock market volume as a share of market capitalization over 5-year windows from 1955 to 2025. Since the 1960s the two move together, in the low-frequency trend and in the cycle. A natural reading is that more trading produces more volatility, especially at short horizons, which fits the horizon pattern above. But other factors could move both series, and high volatility could itself raise the demand for trading. Section 3.1 addresses causality with changes in trading hours as an instrument for volume and with the introduction of the E-mini futures.

Figure 5 puts stock market volatility against the volatility of fundamentals from

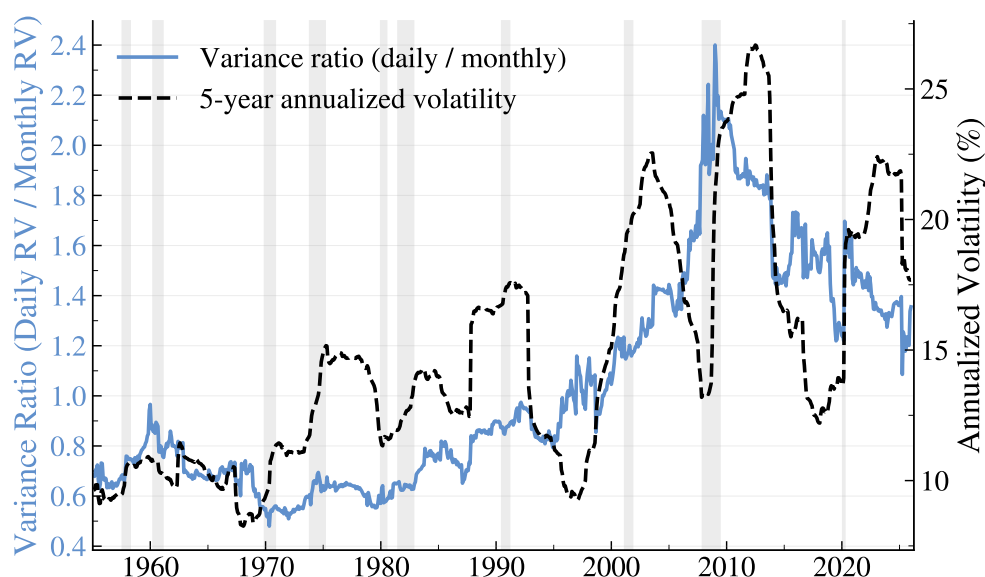


Figure 3: Variance Ratio and Volatility

Notes: Left axis (solid): variance ratio defined as the ratio of 5-year cumulative daily realized variance to 5-year cumulative monthly realized variance. Daily realized variance within each month is the sum of squared demeaned daily returns; monthly realized variance is the squared demeaned monthly return. Both are demeaned using full-sample means. A ratio above 1 indicates negative return autocorrelation (mean reversion); below 1 indicates positive autocorrelation. Right axis (dashed): 5-year rolling annualized volatility as in Figure 1. Shaded regions indicate NBER recessions. Sample: 1955–2025.

1926 on, with the variance ratio beside it. The black line is stock market volatility divided by the volatility of industrial production growth, both over ten-year rolling windows, so a rise means the market has become more volatile relative to the real economy. The blue line is the variance ratio of Figure 3 on the same windows. Both are log deviations from their 1950–1974 means.

The Great Depression is the natural test. Stock volatility averaged 24% through the Depression and the war, the highest in the sample, but the volatility of the macroeconomy was also extremely high. Scaled by the volatility of fundamentals, the Depression looks like the calm 1950s and 1960s. The variance ratio makes the same point on a separate axis: it was 0.74 in the Depression and 0.74 in 1950–1974. Both series trend up over the last fifty years. Since 2000 stock volatility has averaged more than eight times industrial production volatility, against about two in the 1930s and the 1950s, and the variance ratio has averaged 1.5. Real GDP and corporate profits give the same picture

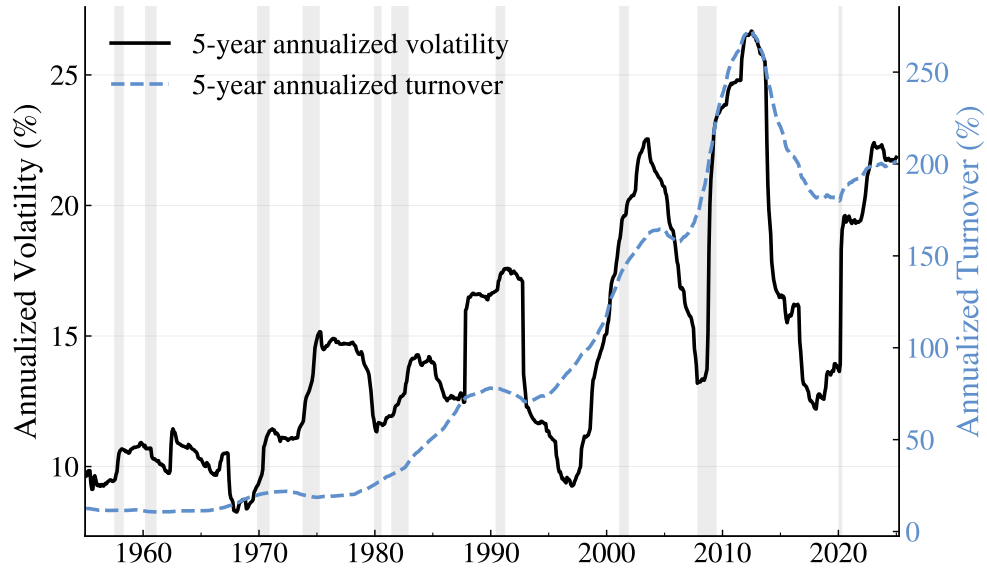


Figure 4: Volatility and Turnover

Notes: Left axis (solid): 5-year rolling annualized volatility of the Fama–French market excess return, computed from daily returns as in Figure 1. Right axis (dashed): 5-year rolling annualized turnover. Daily turnover is total dollar volume divided by total market capitalization across all CRSP common stocks; it is averaged within each month, then over a 60-month rolling window, and multiplied by 252 to annualize. Shaded regions indicate NBER recessions. Sample: 1955–2025.

as industrial production (Figure 17 in Appendix A.2).

This echoes a long literature starting with Schwert (1989) that struggles to tie stock market volatility to fundamentals. Controlling for industrial-production volatility sharpens the point: on the full 1931–2025 sample a linear trend in stock volatility is statistically zero, because Depression-era volatility offsets the modern rise, but adding 5-year rolling industrial-production volatility as a control flips the trend to +11.0 percentage points (s.e. 4.3), while on the post-1960 subsample the trend is +10.8 (s.e. 2.2) without the control and +10.9 (s.e. 2.1) with it.⁷ Together with the horizon pattern, this makes fundamentals a poor explanation for the rise in volatility and trading activity a promising one.

⁷Newey-West HAC with 60 monthly lags; the appendix also reports a structural-break version with a 1960 trend break.

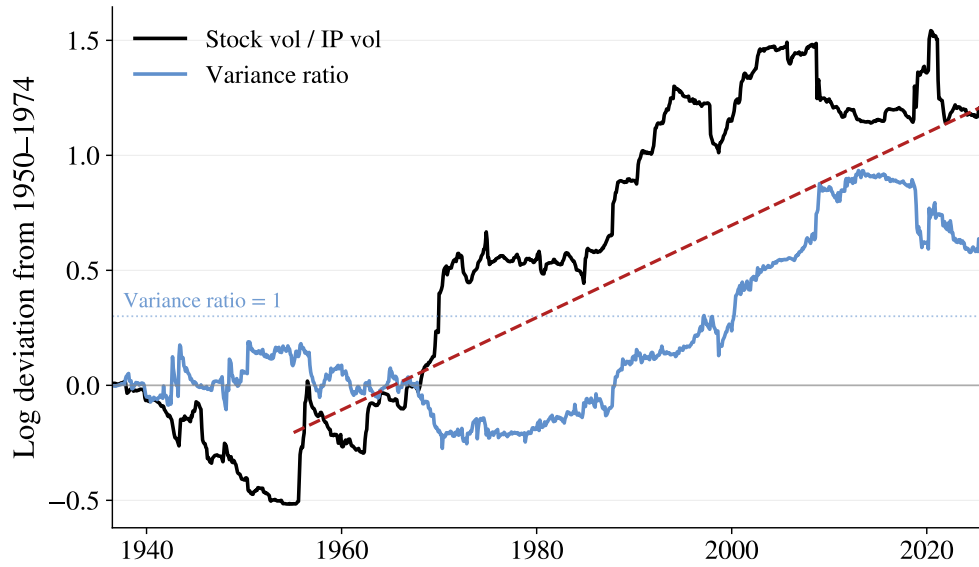


Figure 5: Stock Market Volatility Relative to Fundamental Volatility, and the Variance Ratio, 1926–2025

Notes: Solid black: annualized volatility of the daily Fama–French market excess return divided by the annualized volatility of monthly industrial-production growth (FRED INDPRO), both over 10-year rolling windows. Solid blue: the variance ratio of Figure 3, the ratio of cumulative daily to cumulative monthly realized variance, over the same 10-year windows. Both series are plotted as log deviations from their 1950–1974 means; the dotted line marks a variance ratio of one. The dashed line is a linear trend from 1955 fitted to the average of the two series. Data begin in 1926, so the first window ends in mid-1936. Industrial-production growth for March–July 2020 is excluded: the shutdown and reopening of the economy produce a swing in measured growth that would otherwise sit in every window for ten years, but it was a temporary pause rather than a change in the volatility of fundamentals.

2.3 Robustness of the rise in volatility

Table 2 shows that the rise in volatility survives outliers, a later start date, and stale prices.

2.3.1 Outliers

Volatility is right skewed, with extreme events such as the 1987 crash. Column (1) of Table 2 therefore replaces the rolling variance with the interquartile range of weekly realized variances within each 5-year window, rescaled to annualized volatility; the trend is almost identical at 11.3%, so events like the 1987 crash or Covid are not driving the rise.

2.3.2 Trend robustness

Institutions were increasingly holding and trading market portfolios from the 1960s, the first retail index fund (Vanguard) opened in 1976, and index futures and options followed in the early 1980s, so the start date could matter. Column (2) starts the regression in 1980 with the trend variable unchanged, so the coefficient is still scaled as a 70-year change, and finds a slightly stronger rise at a little over 12%.

We do not read the trend literally; as in Campbell et al. (2001), it is a convenient summary of a slow-moving rise. Estimating structural breaks in the mean of volatility instead gives two breaks, around 1975 and 2000, with the mean rising substantially at each and the overall increase consistent with the trend (Appendix A.4). The trend is stable across start dates, rolling-window lengths, and 30-year rolling subsamples (Appendix A.5).

2.3.3 Concerns about stale prices

Volatility can look low when many stocks do not trade on a given day. Column (3) weights stocks by dollar volume instead of market capitalization, so stocks that do not trade get zero weight, and column (4) uses only the 100 largest stocks, the most liquid and least exposed to stale prices; both give a nearly identical trend. Campbell et al. (1993) reach the same conclusion on nonsynchronous trading by re-running their volume and autocorrelation results on the Dow Jones Industrial Average and on individual large stocks. The rise also holds under equal weighting, across size segments, and for the S&P 500 directly, so it is not an artifact of value-weighting or mega-cap concentration (Appendix A.6).

2.3.4 Systematic vs. idiosyncratic volatility

The rise in market volatility is systematic, not firm-specific. We follow Campbell et al. (2001) (henceforth CLMX) exactly, decomposing average-stock variance into market, industry-relative-to-market, and firm-specific components on the full CRSP universe with their 49 industries, starting in 1962 to match them and extending through 2024;

Table 2: Volatility Trend Robustness

	IQR (1)	Post-1980 (2)	Volume-Weighted (3)	Top 100 (4)
Constant	8.51 (0.67)	8.45 (2.25)	12.47 (0.94)	10.28 (0.84)
Trend	11.28 (1.90)	12.15 (3.75)	12.48 (3.22)	10.04 (2.13)
p -value	<0.001	0.006	<0.001	<0.001
N	3,705	2,401	3,653	3,653
R^2	0.542	0.260	0.360	0.447
Vol increase	133%	93%	100%	98%

Notes: Each column regresses 5-year rolling annualized volatility (in %) on a constant and a linear time trend. The trend is normalized to zero at the start (January 1955) and one at the end of the full sample for all columns, ensuring comparability of slope coefficients. Column (1) uses the interquartile range (IQR) of weekly realized variances based on daily returns within each 5-year window, scaled by the $\chi^2(5)$ IQR to recover annualized volatility. Column (2) restricts the sample to January 1980 onward using the baseline daily-return RV measure from Table 1. Column (3) replaces the market-capitalization-weighted market portfolio with a dollar-volume-weighted portfolio constructed from individual CRSP stocks, using lagged dollar volume as weights. Column (4) uses a value-weighted portfolio of the 100 largest stocks by lagged market capitalization. Standard errors (in parentheses) use the Newey–West estimator with Bartlett kernel and $S = 260$ lags. p -values use fixed- b critical values with $\kappa = \lfloor 3/(2b) \rfloor$ degrees of freedom. The last row reports percentage increase in trend volatility from the beginning to the end of the sample.

our implementation replicates their headline numbers within tolerance on their 1962–1997 sample.⁸ The market share of average-stock variance nearly doubles, from 0.18 in 1962–1971 to 0.31 in 2015–2024, and the total systematic share (market plus industry) rises from 0.31 to 0.45. Both trends are significant: 2.50 percentage points per decade for the market share ($t = 3.95$) and 3.25 for the systematic share ($t = 4.01$).

The variance-ratio regime change in Figure 3 above also sits in the systematic component. The market-component variance ratio more than doubles from 0.72 in 1961–1995 to 1.51 in 1996–2024, the industry component shifts the same way but less, and the firm-specific variance ratio shows no regime change, only a steady decline consistent with falling microstructure noise (narrower spreads, decimalization). Our

⁸See Appendix A.1 for the validation table, figures, and additional results.

1962–1997 numbers reproduce the CLMX result on rising firm-specific volatility, and the reversal after 2001 that Campbell et al. (2023) document appears in our extended sample too. What persists across 1962–2024 is the rise in the market component; firm-specific volatility has fluctuated around its long-run level.

2.3.5 Implied volatility

The VIX, the most widely cited measure of equity volatility, has not trended up since 1990, and Dew-Becker and Giglio (2026) make this unsurprising: they document a large decline in the variance risk premium over the same period and attribute it to declining intermediary frictions in the options market. In our data realized volatility rises by roughly 1.8 percentage points per decade since 1990, the variance risk premium falls at an almost exactly offsetting rate, and implied volatility is flat. The implied-volatility sample is about 30 years rather than 70, so a trend test on it alone has little power, but the joint pattern across the three series is consistent with the rise we document (Appendix A.3).

3. Volume and Volatility: Stylized Facts and Causal Evidence

We propose that the market is more volatile because investors trade the overall market a lot more than they used to. That is, with the rise of S&P500 index funds, S&P500 futures, or similar products, it has become increasingly easy and prevalent over the past 70 years to trade in and out of the overall market rather than in individual stocks.

This concentration of trading volume in the overall stock market means that investors facilitating trade – e.g., arbitrageurs and liquidity providers – will have to absorb market risk, rather than idiosyncratic risk of individual stocks. Their compensation is given by the degree of short-term reversal, with higher reversal leading to more volatility of high frequency returns.

To explore the prevalence of “trading the market” we propose a simple measure

that assesses how much volume for individual stocks can be explained by volume of trade in the overall stock market. More specifically, suppose all volume activity in a given day came from flows in and out of the overall stock market – for example flows in and out of market index funds. Then volume on that day for all stocks would be proportional to their weights in the market index – that is, their market capitalization.

This motivates the following cross-sectional regression. For a given time period t , we run

$$\frac{Volume_{i,t}}{Total\ Volume_t} = a_t + b_t \times MktWeight_{i,t-1} + \varepsilon_{i,t} \quad i = 1, \dots, N \quad (2)$$

where $Volume_{i,t}$ represents the dollar volume in stock i on day t , $Total\ Volume$ represents the total dollar volume across all stocks, and $MktWeight_{i,t-1}$ represents stock i 's weight in the market portfolio. If all trading on a given day was trading by market index funds, then we would have $b_t = 1$ because volume in every stock would be proportional to that stock's market capitalization.

We run the regression in (2) daily, collect estimates b_t , and then compute rolling averages of the time-series of b_t .

Figure 6 plots the rolling 5 year average of volume coming from trading the market along with market volatility. Market volume betas double over this period from around 0.4 in 1960 to around 0.8 today, coinciding with the rise in market volatility. Recall that a market volume beta of 1 is consistent with all trades being trades in and out of the market portfolio, which suggests that today a huge fraction of trading volume in the stock market may be driven by trading the market index.

3.1 Causal Evidence: Volume and Volatility

The link between market volatility and market trading is suggestive, but it is not causal. Our claim is that the ease of trading index products drove trading in the market itself, and that this trading raised volatility. We test it with three sets of quasi-natural experiments. The first uses changes in the trading hours of the NYSE as an instrument for trading volume. The second uses the S&P 500 futures market. Two

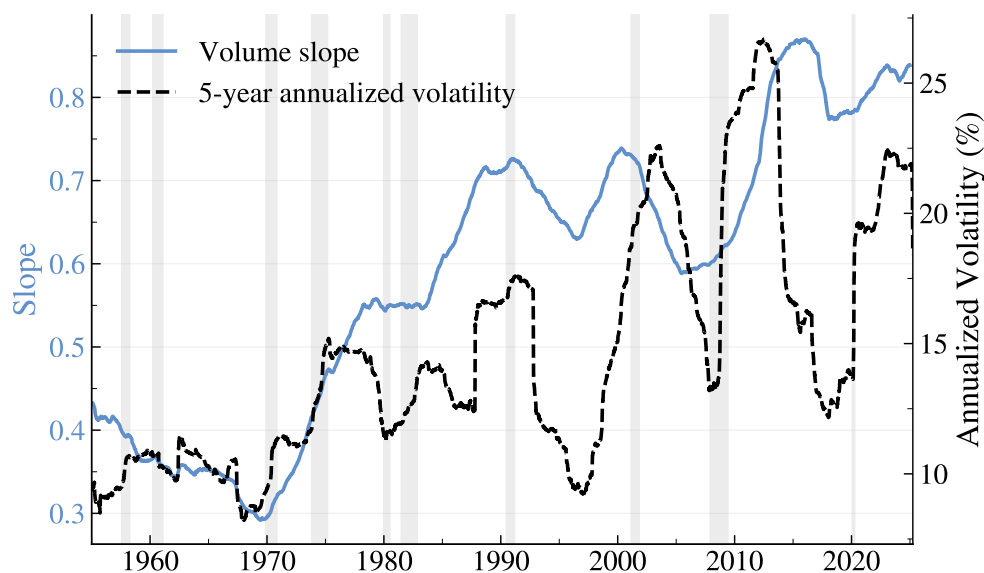


Figure 6: Market Trading vs Market Volatility

Notes: Left axis (solid): 5-year rolling average of the daily cross-sectional OLS slope from regressing volume shares on value weights within the top 500 stocks by market capitalization. A higher slope indicates that trading volume is more concentrated in stocks with larger market-cap weights, consistent with index-linked trading. Right axis (dashed): 5-year rolling annualized volatility as in Figure 1. Shaded regions indicate NBER recessions. Sample: 1955–2025.

changes altered when and how easily the index could be traded: an overnight electronic session opened in 1993 and the E-mini arrived in 1997, while individual stocks could not trade overnight either way. The third turns index trading on and off within a single day: on NYSE holidays futures trade through the morning and then halt, and CME’s schedule changes moved the boundaries between open and shut. We close by chaining the estimates together to ask how much of the rise in volatility they can account for.

An important consideration in choosing these experiments is the strength of the first stage. The most visible events in the history of index trading (Vanguard’s index fund in 1976, S&P 500 futures in 1982, the SPY ETF in 1993) did not produce immediate surges in volume. They started small and took decades to catch on, so a short-window event study around their launch dates has little power. Our experiments instead exploit changes to existing trading behavior: closing a trading day that was already active, extending hours for traders who already know how to trade, or

making an existing futures contract cheaper and electronic via the E-mini. In each case the first stage is strong because the change is immediate. The within-day design goes one step further: it needs no first stage at all, because the comparison is between hours in which the index can be traded and hours in which nothing can.

3.1.1 Changes in Trading Hours as an Instrument for Volume

We exploit changes in NYSE trading hours and trading days as instruments for trading activity. The basic idea goes back to work by French and Roll (1986). They exploited a paperwork backlog at the NYSE in 1968 which led the NYSE to close on Wednesdays from June-December 1968. The amount of information content relevant for stock prices presumably did not change because of this closure. During the closure period, the return on Thursdays is now a 2-day cumulative return – that is, the return Thursday reflects the price movements from the close on Tuesday to the close on Thursday.

Strikingly, however, French and Roll (1986) find that the *variance* of the Thursday return did not change due to the closure, when in fact it should double.⁹ As a result, total weekly variance (the sum of squared returns in a week) *falls* during the closure period by around 20% (one-fifth, given one in five trading days is removed). This decrease in variance is relative to the period of January through May of 1968 (the first half of the year) before the Wednesday closure was in place. Further, the variance on other days (i.e., Tuesday) did not change during the closure period, so that it does not appear there were any general changes in the level of volatility.

We take these results and push them further. Specifically, we exploit 9 changes in trading hours on the NYSE, 1952–1985. These include: the 1952 cut from 6 to 5 trading days per week (trading used to occur on Saturdays), the 1968 Wednesday closure, and the subsequent extensions of the trading day from 4 hours of trading per day in 1969 to eventually 6.5 hours today.

We present two versions of this analysis. The cleanest test is at the daily level

⁹If the returns on any two days are roughly uncorrelated, variance of a 2-day return will be twice the variance of a 1-day return. Letting r_t be the log daily return, we have $var(r_t + r_{t+1}) = 2var(r)(1 + \rho)$ where ρ is the autocorrelation of returns.

(Panel A of Table 3), where we can decompose total elapsed time into *trading hours* and *calendar hours*. This decomposition lets us ask the question: does trading time matter for volatility above and beyond calendar time? The weekly aggregate (Panel B) confirms the same elasticity persists when we integrate over the full calendar week, where trading hours and calendar hours are mechanically conflated and the test is correspondingly less sharp.

Panel A: Daily IV. Each NYSE schedule change generates a specific prediction about which days are affected. The 1968 Wednesday closure means Thursdays now span two calendar days instead of one, while Tuesdays are unaffected. The elimination of Saturday trading in 1952 means Mondays now span the full weekend rather than returning from a half-day Saturday session. We exploit this within-week heterogeneity by working at the daily level, separating two distinct components of the time between market closes: *trading hours* (hours the market is open) and *calendar hours* (total hours elapsed from the previous close to the current close). These vary independently across event types. When the NYSE eliminates Saturday trading, Monday's calendar hours jump while trading hours are unchanged. When the NYSE extends closing time, trading hours increase but calendar hours remain at 24 for weekdays.

Under the null hypothesis that information arrival drives variance, daily squared returns should scale with calendar time regardless of trading hours – a two-day return should have twice the variance of a one-day return. Under the alternative that trading generates variance, more trading hours should increase variance beyond what calendar time predicts. We estimate:

$$\ln |r_t| = fe + b \times \ln \text{turnover}_t + \beta_c \times \ln \text{calendar hours}_t + \varepsilon_t \quad (3)$$

$$\ln \text{turnover}_t = fe + \gamma \times \ln \text{trading hours}_t + \gamma_c \times \ln \text{calendar hours}_t + \nu_t \quad (4)$$

where $|r_t|$ is the absolute daily market return, turnover is daily dollar volume divided by market capitalization, and log trading hours instruments for log turnover. We use log absolute returns rather than log squared returns so that coefficients are elasticities

of volatility, directly comparable to Panel B. We include fixed effects fe , for each event, day-of-the-week, and individual holidays (a separate dummy for each post-holiday day, which prevents holidays from contaminating the calendar hours coefficient). We use ± 22 trading day windows (roughly one month) around each schedule change, yielding 9 events and 403 daily observations. Newey-West standard errors with 10 lags account for serial correlation.

Panel A of Table 3 presents the results. The first stage (column 3) is very strong: a 10% increase in trading hours increases turnover by about 13%, with an F -statistic of 57 on the excluded instrument. The IV estimate in column 2 is 1.2, significant at the 5% level, meaning a 10% increase in turnover raises daily volatility by about 12%. The OLS estimate in column 1 is similar at 1.3, suggesting minimal attenuation bias at the daily frequency. Column 4 presents the reduced form: log trading hours enters with a coefficient of 1.5, significant at the 5% level, while the coefficient on log calendar hours is 0.7 and statistically insignificant. Under the information-only null, the trading hours coefficient should be zero and the calendar hours coefficient should be 0.5 (since the dependent variable is log volatility, not log variance). We test this joint null directly and reject it: $F(2, 369) = 3.61$, $p = 0.028$. Trading hours raise volatility above and beyond what calendar time, and so information arrival alone, would generate.

Panel B: Weekly aggregate. We now aggregate to the week, with total trading hours per week as the instrument. The first stage runs log stock market volume (scaled by market capitalization) on log trading hours. The second stage estimates the relationship between log volatility and log volume. Volatility is computed as the square root of the sum of squared daily returns within the week, annualized.

$$\ln \sigma_t = a_m + a_e + b \times \ln VM_t + \varepsilon_t \quad (5)$$

$$\ln VM_t = a_m + a_e + \gamma \times \ln \text{trading hours}_t + \nu_t. \quad (6)$$

Table 3: IV: Instrumenting volume with trading hours

Notes. Panel A uses daily data. The dependent variable is log absolute daily market return, so that coefficients are elasticities of volatility (not variance). Turnover is daily dollar volume divided by market capitalization. Trading hours and calendar hours (hours from previous close to current close) are measured for each trading day. The sample uses ± 22 trading day windows around 9 NYSE schedule changes, 1952–1985. Panel B uses weekly data. The dependent variable is log realized volatility (annualized) of the weekly value-weighted market return. Volume is weekly dollar volume divided by market capitalization, as in Panel A. Trading hours is total trading hours per week. The sample includes 6-month windows around each of the 9 changes in NYSE trading hours. Standard errors in parentheses: Newey-West with 10 lags in Panel A (with event, day-of-week, and individual holiday FEs); Newey-West with 2 lags in Panel B (with event and month-of-year FEs). ***, **, * denote significance at the 1%, 5%, and 10% level.

<i>Panel A: Daily</i>				
	(1)	(2)	(3)	(4)
	OLS	IV	First Stage	Reduced Form
Dep. variable:	$\ln r $	$\ln r $	$\ln \text{Turnover}$	$\ln r $
Log Turnover	1.30*** (0.26)	1.16** (0.47)		
Log Trading Hours			1.27*** (0.17)	1.47** (0.61)
Log Calendar Hours	−0.05 (0.57)	0.08 (0.64)	0.55*** (0.13)	0.72 (0.56)
Observations	403	403	403	403
R-squared	0.37		0.92	0.32
First-stage F		57.0		
Joint F (info-only null)				3.61**
FE: Event, DOW, Holiday	Y	Y	Y	Y

<i>Panel B: Weekly</i>				
	(5)	(6)	(7)	(8)
	OLS	IV	First Stage	Reduced Form
Dep. variable:	$\ln \sigma$	$\ln \sigma$	$\ln \text{VM}$	$\ln \sigma$
Log Volume	0.72*** (0.12)	1.42* (0.82)		
Log Trading Hours			1.02*** (0.34)	1.45** (0.67)
Observations	326	326	326	326
R-squared	0.54	0.49	0.90	0.49
First-stage F		9.0		
FE: Event & Month	Y	Y	Y	Y

We use trading hours *only* in 6-month windows before and after each change as the instrument for weekly volume (VM). Restricting to shorter windows is important because trading hours have trended up over time along with volatility and volume; further, the levels of volume and of volatility are different around each of these changes. We include event dummies, a_e (equal to 1 in the 6 months before and after a change), and month-of-year fixed effects, a_m , to control for seasonal patterns. Newey-West standard errors with 2 lags account for serial correlation; 2 weekly lags is the same autocorrelation horizon as the 10 daily lags used in Panel A.

Our identifying assumptions are as follows. First, increases in trading hours only affect volatility through increased trading activity. Second, the decision to increase trading hours does not depend on an expected increase in volume 6 months before / after the official change. Demand for trading surely affects trading hours in the long run. What we need is that a permanent change in hours was not timed to a predicted spike in volume inside the event window, which is most plausible when the window is narrow. We use 6 months as a compromise with the precision of realized volatility. Results are robust at 2-month windows (IV coefficient 2.18, significant at 5%) and at 12-month windows (1.69, significant at 1%).

Panel B of Table 3 presents the weekly results. Column 3 is the first stage. Trading hours increase volume nearly one to one – when the stock market is open longer, people trade more. The first stage shows that trading hours are a strong instrument for volume. Column 2 finds a positive causal effect of volume on volatility: a 10% increase in trading hours increases volume by 10% and leads to about a 14% increase in volatility. The point estimate is significant at the 10% level. Column 1 presents OLS estimates of volatility on volume and finds a similarly strong positive relationship. Column 4 regresses volatility directly on trading hours; since the first-stage coefficient was 1, the coefficient in column 4 is about the same as that in column 2 and significant at the 5% level. The weekly elasticity of 1.4 is close to the daily Panel A estimate of 1.2, confirming the result is consistent across frequencies.

Overall, more trading causes more volatility in the market index. The daily specification delivers a first-stage F of 57 and rejects the information-only null: trading

hours matter in their own right, not just calendar time, and the weekly specification gives the same elasticity at the calendar-week aggregate.

3.1.2 The E-mini Futures as a Natural Experiment

The E-mini S&P 500 futures contract launched on September 9, 1997, and it did not add trading hours: the big S&P 500 contract had traded overnight on CME's GLOBEX system since September 1993, and the E-mini's own overnight session added at most half an hour to that (Section 3.1.3 and Table 15 in the appendix give the record). What the E-mini changed was access. It was a fraction of the big contract's size, it traded electronically around the clock, and it ran alongside the pit during regular hours, so an investor with a screen could trade the index at retail size at any hour without going through a floor broker. Volume in S&P 500 futures rose sharply within weeks of the launch. Individual stocks could not trade overnight either before or after.

We compare the market's overnight-to-intraday variance ratio with the same ratio for individual stocks (idiosyncratic vol), before and after the E-mini, since more overnight trading should raise overnight variance for the market but not for individual stocks. The stock ratio controls for news that arrives at night and moves stocks in general; the intraday denominator controls for anything that moves overall volatility in this period.

We use close to open and open to close returns as measures of overnight and intraday returns, respectively.¹⁰ The market return is the value-weighted return of the top 500 stocks by prior-December market capitalization. For each individual stock i , we use the simple market-adjusted return $r_{i,t} - r_{m,t}$ as a proxy for the idiosyncratic component. This imposes a unit market beta on each stock, following Campbell et al. (2001). We compute the sum of squared returns within a rolling 22 day period (roughly 1 month). For individual stocks, we aggregate into a single measure of idiosyncratic volatility by value weighting the resulting overnight and intraday

¹⁰Thus, we also use weekend returns. Returns are constructed using CRSP's split- and dividend-adjusted daily total return combined with raw same-day open/close prices, which gives correct overnight and intraday returns through corporate actions.

idiosyncratic variances at the stock level. We then compute

$$\frac{\text{var}(r_{mkt}^{night})/\text{var}(r_{mkt}^{day})}{\text{var}(r_{stock,idio}^{night})/\text{var}(r_{stock,idio}^{day})},$$

the ratio of the two variance ratios.

Figure 7 plots the results. Panel A shows the increase in this variance ratio. The first vertical line represents the introduction of the E-mini in September 1997. Since our variance computations use 22 day rolling windows, the full effect is not felt until the second vertical line, which then only uses data since the E-mini introduction. The red dashed line estimates the means before and after using the 3-month window in the plot, with a linear transition between to reflect the transition window induced by rolling returns. It is clear from the plot that the relative variance of the S&P500 increased overnight after the introduction of the E-mini.

However, for our story to be correct, we also need that the E-mini generated significant trading in the S&P500 when it was introduced. Panel B adds the (log) total volume of S&P500 futures contracts and shows they indeed rise with the introduction of the E-mini. We use a 22 day rolling average of total volume in S&P500 futures contracts to match the window at which volatility is computed. Thus, we see the rise in relative volatility coinciding with a large increase in trading activity in the S&P500, consistent with our story.

Table 4 tests the increase formally. Because the dependent variables are 22-day rolling averages, a simple pre/post dummy gets the timing wrong: on September 10 only 1 of 22 observations in the rolling window is post-launch. We therefore use a ramp variable that equals zero before the E-mini launch, increases linearly from 0 to 1 over the next 22 trading days, and equals one thereafter. The dependent variable in column (1) is the log ratio of overnight-to-intraday variance for the value-weighted market return, controlling for the corresponding log ratio computed from idiosyncratic stock returns. The ramp coefficient is 0.43 ($t = 7.9$), meaning the market's overnight variance relative to intraday rose by about 50% ($e^{0.43} \approx 1.5$) after the E-mini launch, relative to the same ratio for stocks. Column (2) shows log futures volume

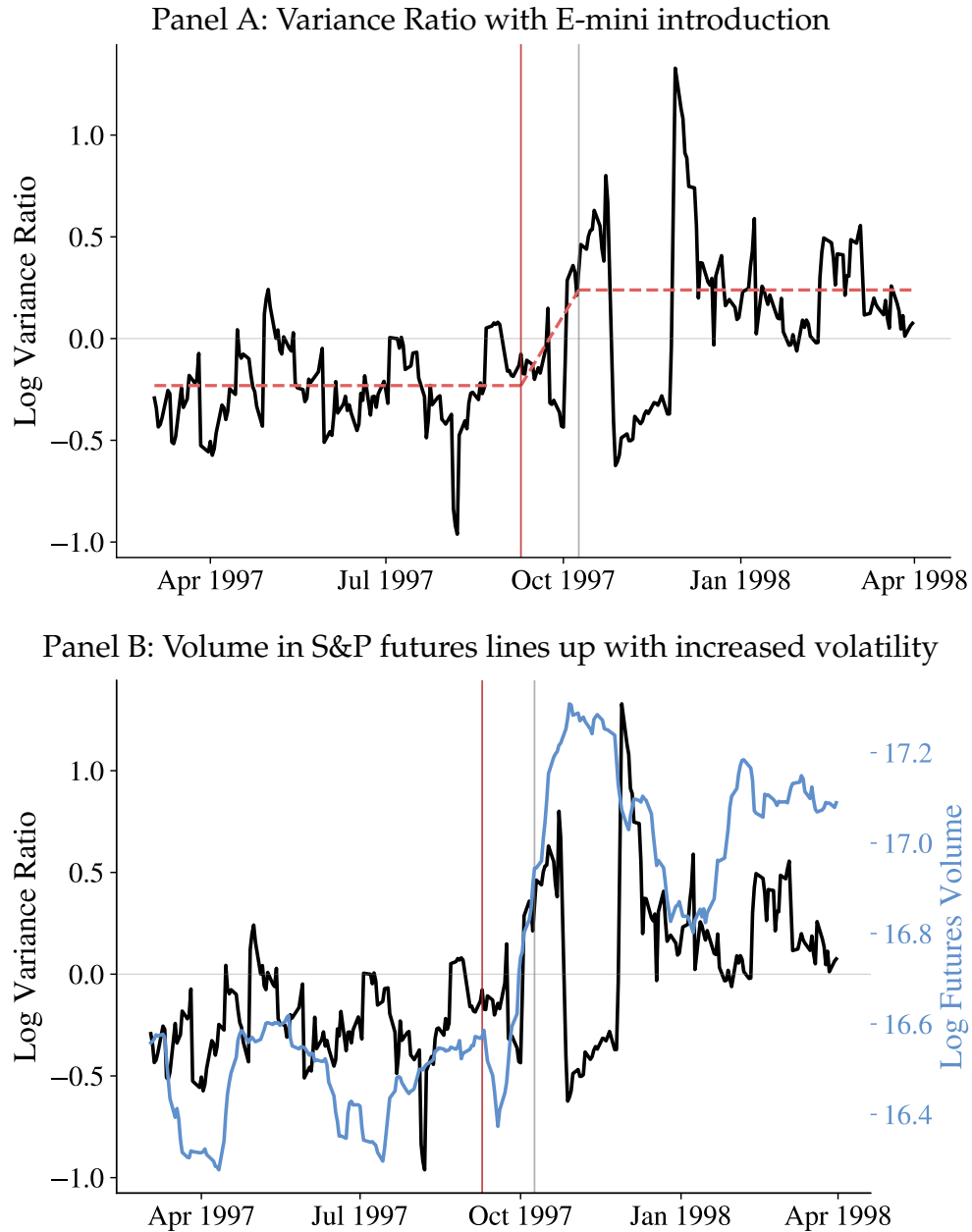


Figure 7: The Introduction of the E-mini Futures and the Effects on Volatility.

Panel A plots the log of the ratio of overnight-to-intraday variance for the market relative to individual stocks, using 22-day rolling windows around the E-mini introduction (Sept 9, 1997). The red dashed lines show pre- and post-event means with a ramp connecting them. Panel B adds the log of total S&P500 futures volume (22-day rolling average) on the right axis.

Table 4: The E-mini, Futures Trading Hours, and Overnight Market Variance

Notes. The dependent variable in column (1) is the log ratio of overnight-to-intraday variance for the value-weighted market return (top 500 stocks by prior-Dec market capitalization), controlling for the corresponding log ratio of idiosyncratic stock returns, value-weighted across stocks. The idiosyncratic return is defined in the text. Column (2) uses log total S&P 500 futures volume. Both dependent variables are 22-day rolling averages. The E-mini ramp variable equals 0 before September 9, 1997, increases linearly from 0 to 1 over the next 22 trading days, and equals 1 thereafter, matching the rolling window over which the effect builds. The sample window for columns (1)–(2) is March 1997 to March 1998. Column (3) pools four CME schedule changes in ± 6 -month windows with event fixed effects: the GLOBEX listing of the big contract (September 20, 1993, +16.25 overnight hours), the E-mini (September 9, 1997), the session expansion of September 29, 2003 (+0.75), and the maintenance window of November 19, 2012 (−0.25). The three hours changes enter as a linear dose in hours ($\text{Post} \times \Delta\text{Hours}$) and the E-mini as its own indicator; each event window keeps its own coefficient on the idiosyncratic control; Newey-West standard errors (44 lags) are computed within event blocks. The E-mini coefficient in column (3) differs from column (1) only because the window is ± 6 months (252 observations for that event) rather than March 1997 to March 1998. The per-event estimates are in Table 16. HAC standard errors (Newey-West, 44 lags) in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% level.

	(1) E-mini Log Market Var Ratio	(2) E-mini Log Volume	(3) Pooled, four CME changes Log Market Var Ratio
E-mini ramp (access)	0.434*** (0.055)	0.599*** (0.048)	0.430*** (0.058)
$\text{Post} \times \Delta\text{Hours}$ (per hour)			0.0301** (0.0120)
Log idio stock var ratio	0.858*** (0.169)		Y (event-specific)
Event FE			Y
Observations	273	273	1,006
R-squared	0.528	0.839	0.44

rose by 0.60 over the same window, an increase of roughly 80%.

3.1.3 Futures Trading Hours, Access, and Overnight Market Variance

The E-mini is one week in a thirty-year story. Figure 8 plots the volatility of the market return using 5-year rolling windows as in our main plots earlier, but splits returns into overnight (close to open) and intraday (open to close). A substantial amount of the increase in volatility over the past 30 years comes from overnight returns: overnight volatility rises from about 4% in the mid-1990s to about 14% in the most recent win-

dow, and the ratio of overnight to intraday volatility goes from under 60% around 2000 to over 90% today.

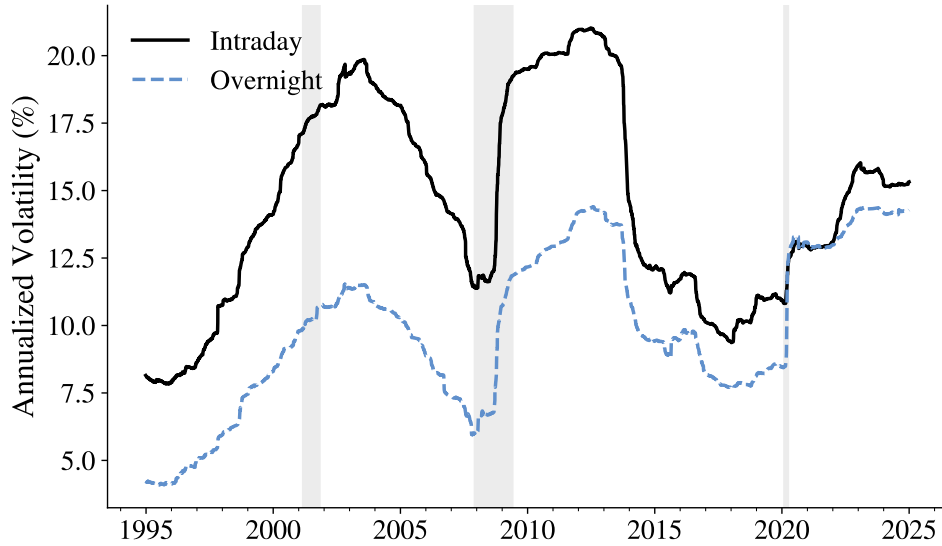


Figure 8: The Rise of Overnight Volatility on the Market.

5-year rolling standard deviation of the value-weighted market return, split into overnight (close-to-open) and intraday (open-to-close) components.

This rise coincided with the growth of overnight index trading: overnight activity in S&P 500 futures has grown steadily relative to daytime activity since the E-mini launch (Figure 35 in the appendix).

But is this rise in overnight volatility driven by the kind of trading our theory predicts? We now decompose overnight returns for individual stocks into a systematic component (driven by the market) and an idiosyncratic component (stock-specific). Our prediction is sharp: because index futures allow investors to trade the market overnight but not individual stocks, the systematic component of overnight variance should rise when overnight index trading expands, while the idiosyncratic component should not.

Setup. We construct four daily variance series from the 500 largest U.S. common stocks by market capitalization, re-selected at the start of each year from 1993 to 2024:

market overnight, market intraday, idiosyncratic overnight, and idiosyncratic intraday. The market return is the value-weighted return across the top 500 stocks, and idiosyncratic returns impose a unit market beta $(r_{i,t} - r_{m,t})$, exactly as in the E-mini event study; overnight and intraday returns use the same construction anchored on CRSP total returns. The object of interest is the daily triple-difference

$$TD_t = (\log V_{mkt,t}^{ovn} - \log V_{mkt,t}^{intra}) - (\log V_{idio,t}^{ovn} - \log V_{idio,t}^{intra}), \quad (7)$$

the log ratio of overnight-to-intraday market variance relative to the same ratio for idiosyncratic returns, the object plotted in Figure 7. Same-day intraday variance controls for anything that moves overall volatility on a given day; the idiosyncratic ratio controls for market-wide changes in overnight information flow that affect stocks in general. What remains loads specifically on the market's overnight window, the cell where index futures trade and individual stocks do not.

The record. Table 15 in the appendix gives the exchange's own record of trading hours, sourced line by line. Hours available for trading S&P 500 futures outside NYSE regular hours were essentially zero until September 20, 1993, when CME listed the big contract on GLOBEX and overnight hours went from a quarter of an hour (the pit's tail after the NYSE close) to 16.5 per weekday. The E-mini in September 1997 added at most half an hour. The September 2003 session expansion added three quarters of an hour and the November 2012 maintenance window removed a quarter. So the futures market gave the index two distinct things at two distinct dates: *hours*, in 1993, and *access*, in 1997.

Specification. We treat the two channels separately:

$$TD_t^{mkt} = a_e + \beta_H \text{Post}_t \times \Delta \text{Hours}_e + \beta_A \mathbf{1}[\text{E-mini}]_t + \gamma_e TD_t^{idio} + \varepsilon_t, \quad (8)$$

where the dependent variable is the 22-day rolling log ratio of overnight-to-intraday market variance, the control is the corresponding idiosyncratic ratio with its own co-

efficient in each event window, and Post_t is the ramp used in Table 4, rising from zero to one over the 22 trading days after each event. Each event gets a ± 6 -month window and an event fixed effect a_e . The three hours changes (1993, 2003, 2012) enter as a linear dose in hours, so β_H is the effect of one additional hour of overnight index trading. The E-mini gets its own indicator, so the hours channel is not asked to absorb an access event whose hours change is at most half an hour. Newey-West standard errors use 44 lags, since consecutive observations of a 22-day rolling series overlap by 21 days, and are computed within event blocks. The design mirrors Table 3: pool the events, absorb event fixed effects, identify from within-event variation, so that anything that evolves smoothly, such as growing demand for around-the-clock trading or the globalization of information flow, cannot produce a step at a CME schedule date.

Results. Column (3) of Table 4 pools the four CME schedule changes. The hours coefficient is $\beta_H = 0.030$ ($t = 2.5$): each hour of overnight index trading raises the market's overnight variance by about 3% relative to stocks, so the 16.25 hours that opened in 1993 raise it by $e^{0.49} \approx 1.6$, or 60%. The access coefficient is $\beta_A = 0.43$ ($t = 7.4$), the E-mini estimate of column (1) re-estimated inside the pooled design: access raised overnight market variance a further 50% with almost no change in hours. Both channels come with more futures trading, by 0.60 log points at the E-mini (column (2)) and by 0.17 at the GLOBEX listing (Table 16). The per-event estimates are in Table 16 in the appendix: the 1993 listing alone gives 0.48 ($t = 2.5$), while the 2003 and 2012 changes are a fraction of an hour each and imprecise.

The per-hour effect is small for a reason: in 1993 the overnight session existed but was thin, so the overnight variance ratio builds over the following years as volume arrives (Figure 8) rather than stepping at the hours change. Hours are necessary for overnight index trading, not sufficient. The pooled dose is therefore identified almost entirely by 1993. The appendix takes up the robustness of both large events (Table 17).

Over the full 1993–2024 sample, the market-overnight cell rises about 2.1% per year relative to the other three cells (SE 0.4%), from a stacked regression of the four log-variance cells on cell-specific trends. That is a description, not identification: over

three decades the growth of overnight index trading is collinear with everything else that changed in market structure. We read the event evidence as identifying the two channels locally and the trend as the shape of what they grew into.

Two measurement biases, the unit betas and the migration of earnings releases into after-hours trading, both push these estimates toward zero.

3.1.4 Turning Index Trading On and Off Within the Day

Where the experiments so far compare periods before and after a change in the ability to trade, the last design compares hours within a single day.

Start with the null. If prices move because news arrives, then variance accumulates on the clock whether or not a market is open. Closing the market defers the adjustment; the deferred moves show up as a jump when trading resumes. Under that view the squared jump across a closure should carry as much variance per hour as those same hours would carry if the market were open. Under our view it should carry much less, because part of the variance is generated by trading itself. French and Roll (1986) ran this test on the NYSE's 1968 Wednesday closures and found the jump small. We run it on the index futures market, where CME's schedule supplies closures inside a day, and where the thing that is switched on and off is the ability to trade the index and nothing else.

Two windows on one day. On a U.S. market holiday the NYSE is closed all day, so no individual stock can trade. S&P 500 futures open the evening before, trade through the morning, and halt at midday; from the halt until the evening reopen, nothing trades at all. So one holiday hands us two windows back to back. In the morning the index can be traded and its components cannot: the only claim anyone in the world can buy or sell is the market as a bundle. In the afternoon nothing can be traded. Same day, same news flow, same season, same state of the world. The only thing that changes is whether a market is open. Panel A of Figure 9 shows what that does to the price. On an ordinary weekday the afternoon is the busiest part of the session; on a holiday the afternoon collapses to a level far below the same day's

morning and below an ordinary weeknight.

Better still, the boundary moved: in 2014 CME moved the holiday halt from 10:30 to noon Chicago time. The clock window 10:30–12:00 on a holiday is therefore *dead* before 2014 and *index-only* after it, and fully open on ordinary days. Panel B of Figure 9 plots the hour-of-day profile of variance on holidays before and after the change, each scaled by its own overnight rate. The same clock window on the same kind of day sits at a small fraction of the overnight rate when it is shut and at or above the overnight rate when the index trades. This is the comparison the tables below turn into a number, and it kills the objection that mornings are simply busier than afternoons.

Specification. Let $V_{d,w}$ be the variance per hour of the futures price on day d in clock window w , measured in levels from five-minute bars. A shut window has no bars, so its variance is one squared increment, from the halt price to the price five minutes after the reopen, spread over the shut minutes plus five. We use five minutes as a conservative choice to avoid stale reopens. For every weekday from 2004 to 2019 we take the 10:30–12:00 window and subtract the same day’s overnight rate over 20:00–07:00 Central. In that overnight window only the index trades on every kind of day, and the news is Asian and European rather than U.S., so a quiet holiday is quiet in both windows and the level cancels within the day. We then estimate

$$V_{d,10:30-12:00} - V_{d,20:00-07:00} = a + \beta_I \mathbf{1}[\text{index tradeable}]_d + \beta_B \mathbf{1}[\text{ordinary business day}]_d + \varepsilon_d. \quad (9)$$

Dead is the baseline. The two indicators nest, so β_I is the increment from turning the index on, identified from holidays before and after 2014, and β_B is the further increment from an ordinary business day, on which the cash market is open and U.S. news arrives. Under the null $\beta_I = 0$: the jump carries the same variance per hour as live trading would. The sample is every scheduled holiday session with a same-day reopen (Monday through Thursday holidays; one-off closures such as Hurricane Sandy are out), which gives 58 dead afternoons and 33 index-only mornings, against 4,188

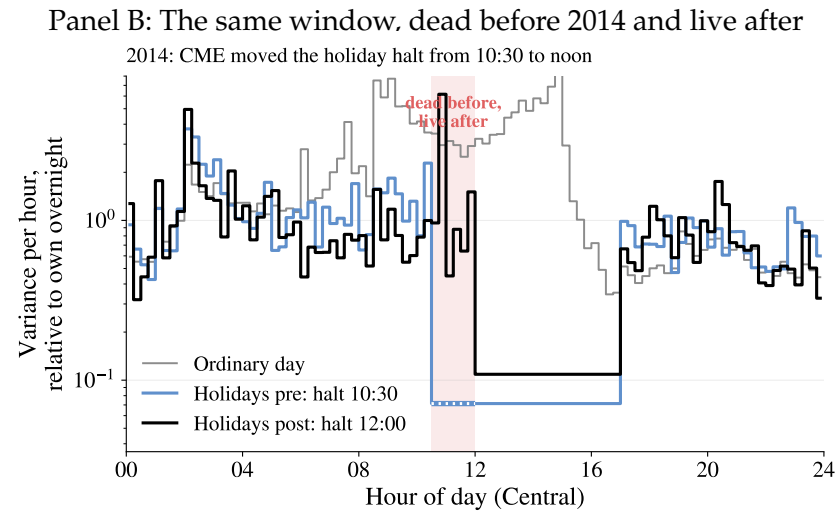
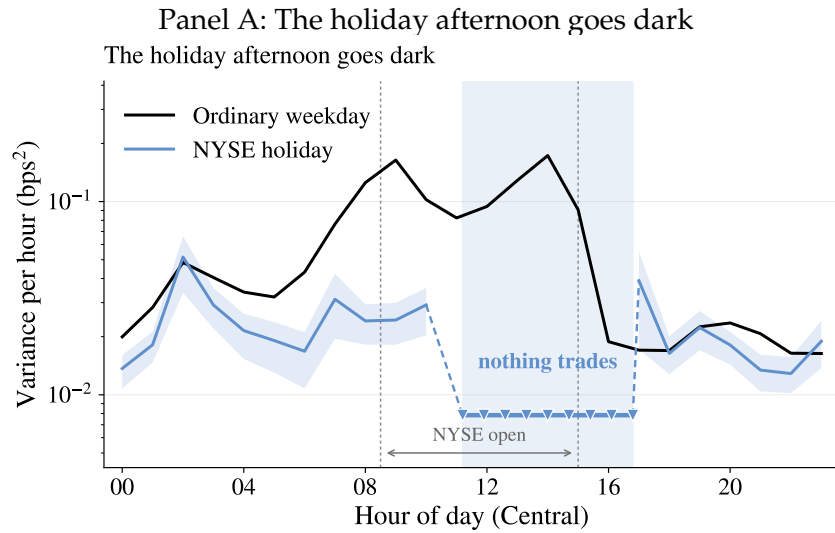


Figure 9: Index Trading Against a Shut Market.

Variance per hour of the E-mini S&P 500 futures price by hour of day (Central time), from five-minute bars. Panel A: hours-weighted average over ordinary weekdays and over NYSE holidays, July 2003 to March 2020, log scale, with NYSE regular hours marked. On holidays futures halt at 10:30 (before 2014) or noon (after) and reopen at 17:00; the shaded region marks the shut afternoon, whose variance is one jump from the halt to the reopen, drawn as its average rate; the band is a 95% interval. Panel B: mean profile in 15-minute blocks on Monday–Thursday holidays halted at 10:30 (2004–2013, 57 days) and at noon (2015–2019, 29 days), and on ordinary Tuesday–Thursday weekdays, each relative to its own 20:00–07:00 overnight rate; a shut window’s variance is one increment from the halt price to the price five minutes after the reopen. In the 10:30–12:00 window the dead shelf sits at 0.07 of the overnight rate and the index-only profile at 1.77. Panel B classifies holidays by calendar year, omits 2014, and keeps September 2, 2019; Table 5 keys the change to its date, May 26, 2014, includes the 2014 holidays, and excludes September 2, 2019, so the two samples differ slightly (58 and 33 days in the table). Without September 2, 2019 the index-only profile sits at 0.85.

ordinary days. We exclude Labor Day 2019, whose morning carried the September 1, 2019 tariff announcement and on its own accounts for half of the index-only cell's variance; with it in the estimate is larger, and the appendix reports it. Standard errors come from a bootstrap over days.

Results. Table 5 presents the results. Turning the index on multiplies variance per hour by 13 ($t = 7.2$), against a null of one: on the same holiday, an hour in which the index can be traded carries thirteen times the variance of an hour in which nothing trades. With day-of-week fixed effects the ratio is 12.6. Column (3) makes the comparison within each holiday, using a panel of 15-minute buckets with time-of-day and day fixed effects, so each holiday's index-only morning is set against its own dead afternoon and the day's level of news drops out; that gives $\times 11.3$ ($t = 3.3$). Column (4) is the 2003 schedule change described next. Across the columns the index margin runs from $\times 2.4$ to $\times 13$, and in each the null that the index adds nothing is rejected.

We do not report β_B as a clean effect, because an ordinary weekday differs from a holiday by a business day's worth of U.S. news as well as by the cash market being open, and this design cannot separate the two. The coefficient we are interested in is β_I , whose two cells are both holidays, so that the absence of U.S. news and of the cash market is common to both and only the futures market's state differs.

The 2003 schedule swap. On Monday, September 29, 2003, CME moved the E-mini's nightly one-hour shutdown from 22:00–23:00 to 16:30–17:00 Central. Two clock windows changed state on the same date, in opposite directions: the late-evening hour went from shut to open, and the half hour beginning at 16:30 went from open to shut.¹¹ That structure rules out the usual worry about hours changes, that the exchange expanded hours because demand was growing. Figure 10 plots the evening profile before and after: prices stop moving at 16:30 and start moving at 22:00. Column (4) of Table 5 compares the two switched windows with the same day's 18:00–

¹¹A third window also changed: the afternoon reopen moved from 15:45 to 15:30, so 15:30–15:45 went from shut to open. That window sits on the slot in which companies release earnings after the NYSE close, so it measures when earnings get priced rather than whether trading generates variance, and we exclude it.

Table 5: Index Trading Against a Shut Market

Notes: Each cell is one plus the estimated coefficient on the index-tradeable indicator divided by the dead cell's mean of the dependent variable, i.e. the ratio of variance per hour when only the index can be traded to variance per hour when nothing trades; a ratio of one means the index adds nothing. The t -statistic in parentheses is for the coefficient itself, against zero, so it tests that null. Variance per hour is from E-mini S&P 500 five-minute bars, in levels; a shut window's variance is one squared increment, from the halt price to the price five minutes after the reopen, spread over the shut minutes plus five. Columns (1)–(2): the 10:30–12:00 CT window on every weekday 2004–2019, which is dead on holidays before the 2014 halt change (58 afternoons), index-only on holidays after it (33 mornings), and fully open on ordinary days (4,188); the same day's overnight rate (20:00–07:00 CT) is subtracted; an ordinary-business-day indicator is included, so the reported coefficient compares holidays to holidays. Column (3): every 15-minute bucket of every scheduled holiday session, with time-of-day and day fixed effects, so the comparison is the index-only morning against the same day's dead afternoon. Column (4): the two windows the September 29, 2003 schedule change moved (16:30–17:00, open then shut; 22:00–23:00, shut then open) against the same day's 18:00–20:00, with day and window fixed effects, Monday–Thursday, July 2003 to March 2004. Holiday cells use scheduled sessions with a same-day reopen (Monday–Thursday; unscheduled closures excluded); September 2, 2019 is excluded from every column and Table 19 reports the estimate with it. t -statistics from a day bootstrap (2,000 draws) in columns (1), (2), and (4); clustered by day in column (3), where the bucket count overstates the effective sample. Appendix Table 18 reports the same columns under log, Poisson, and free-control specifications, together with the same bucket panel run on all weekdays, and Table 19 reports sample cuts.

	Matched window, day level one window per day, holidays		15-min bucket panel holidays	2003 swap day $\times$ window
	(1)	(2)	(3)	(4)
Index tradeable vs. nothing trading	$\times 13.1$ (7.2)	$\times 12.6$ (5.7)	$\times 11.3$ (3.3)	$\times 2.4$ (2.9)
Same-day overnight subtracted	Y	Y		
Day-of-week FE		Y		
Time-of-day FE			Y	
Day FE			Y	Y
Window FE				Y
Observations	4,279	4,279	8,736	372

20:00, with day and window fixed effects, Monday through Thursday, July 2003 to March 2004. Open versus shut is $\times 2.4$ ($t = 2.9$): the same clock window carries 2.4 times the variance per hour when the E-mini trades as when it is shut. The hour that opened rises by a factor of 4.5 ($t = 4.0$); the half hour that shut falls to two thirds of its former rate ($t = 0.9$).

The 2003 estimate is smaller than the holiday estimates because the windows are short and sit in the quietest hours of the day, so the shut side is bounded below by

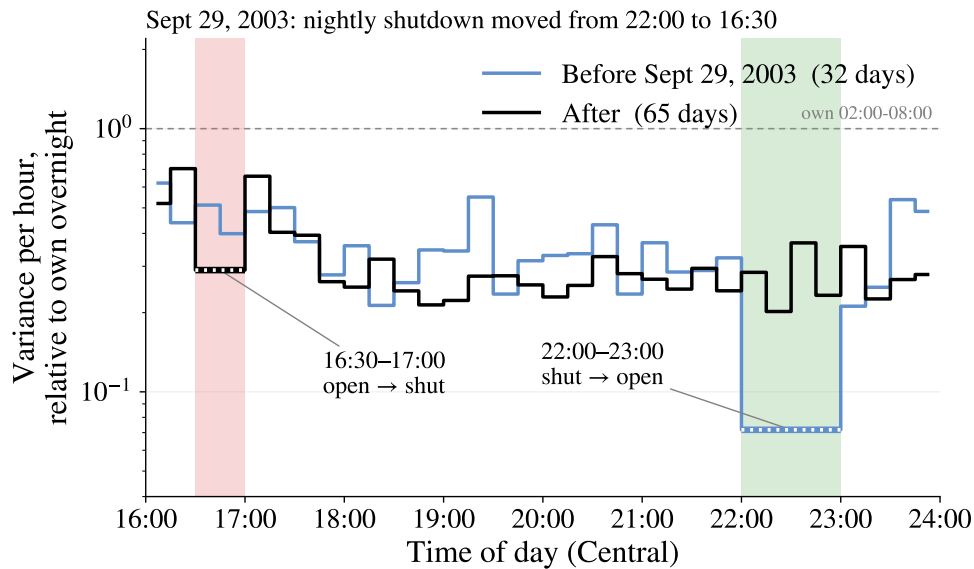


Figure 10: The 2003 Schedule Swap: CME Moved the Nightly Shutdown, and Variance Moved With It.

Variance per hour of the E-mini S&P 500 futures price by 15-minute block, 16:00–24:00 Central, log scale, each era divided by its own 02:00–08:00 rate, averaged over Tuesday–Thursday evenings before (July 1 to September 26, 2003, 32 days) and after (September 29, 2003 to March 26, 2004, 65 days) the September 29, 2003 schedule change, which moved the nightly one-hour shutdown from 22:00–23:00 to 16:30–17:00. A shut window’s variance is one increment from the halt price to the price five minutes after the reopen, drawn dotted. The August 14, 2003 blackout evening and the quarterly roll weeks are excluded.

price discreteness. What the 2003 swap establishes cleanly is *when* prices move: at the hour the exchange is open and not at the hour it is shut, in both directions on the same date.

Objections. The dead-zone comparison invites four objections, and each can be answered with the data in hand.

Since news is priced at the reopen, one concern is that dead afternoon variance could be undercounted. Our construction already charges the reopen’s first five minutes to the closure in full, and that is more than the catch-up warrants: charging only the reopen’s excess variance over an ordinary day’s reopen, all the way through 19:00, leaves the holiday morning at more than twenty times the dead afternoon’s rate.

10:30 Chicago is the European close; the pre-2014 afternoon is quiet because Europe went home. The 2014 move answers this: after 2014 the halt comes at noon, ninety minutes after Europe closes, so the two events separate on the same holiday. Holiday variance runs at 0.009 percent squared per hour with Europe open, *rises* to 0.021 after Europe closes at 10:30, and drops by an order of magnitude at the noon halt. Europe closing does not kill the variance; the halt does. Daylight saving puts a number on Europe: the U.S. and E.U. switch on different Sundays, so for a few weeks a year Europe closes at 11:30 Chicago while CME's clock stays put. Switching Europe's state is worth 0.48 log points of variance per hour against a pre-2014 holiday drop of 3.14, so Europe explains at most about 15% of it. Tokyo's close explains nothing, and foreign news reaching the tape while U.S. stocks are shut is in any case the mechanism, not an alternative to it: an index-only market prices information its components cannot trade on.

Holidays are quiet, so of course the afternoon is quiet. Then the morning would be quiet too, and it is not. In matched index-only hours, midnight to 8am Chicago on both holidays and ordinary days, holidays run at about 60% of a normal day's rate: quieter, not dead. And the within-day normaliser in (9) removes the level of holiday quiet entirely.

Variance measured from transaction prices includes bid-ask bounce, and bounce requires trades. Fong and Frino (2001) raise exactly this for the drop in index-futures volatility when the cash market closes. The bounce variance implied by the first-order autocovariance of five-minute returns is about 11% of the measured live rate, which cannot produce a ratio of 13.

What is known and what is new. The total open-versus-shut ratio is old. French and Roll (1986) report per-hour variance during trading hours at 13 times its value across mid-week holidays and 72 times across weekends; Stoll and Whaley (1990) report about 16 for daytime versus overnight on the NYSE. Our all-weekday appendix estimate is that number on the index (Table 18). Chang et al. (1995) show on S&P 500 futures that volatility falls after the NYSE close and name the behavior of futures prices while the underlying market is shut as an open question. The index-only cell

against a dead baseline is what is new. The closest magnitude comes from a cash market: Ito et al. (1998) find that yen variance over the Tokyo lunch hour doubled when lunchtime trading was allowed in 1994, with no change in public news. Our estimates for switching the index on are five to seven times that. The leading theory of periodic closure, Hong and Wang (2000), delivers a trading-to-nontrading ratio of 1.0 to 1.12 with constant information flow: a floor our estimates sit far above, not a rival that explains them.

3.1.5 Sizing Up the Experiments

Each experiment identifies a local effect: an elasticity to trading hours, a step at an access change, a within-day contrast. While none of them is a decomposition of the seventy-year rise, the first two can be chained. Figure 11 takes the estimated coefficients at face value, feeds them the actual history of trading hours and access, and asks how much of the rise in volatility they would have produced on their own.

The chain runs as follows. The NYSE experiments say that weekly volatility moves with weekly trading hours with an elasticity of about 1.45 (the reduced form in Table 3, Panel B). Weekly hours went from 27 to 32.5 across the nine schedule changes of 1952–1985, which takes a 10% baseline to about 13% by 1985. The futures experiments say that each overnight hour of index trading raises overnight market variance by 3% and that the E-mini raised it by a further 54%; the overnight cell was about a quarter of daily market variance in the mid-1990s. Applying the 1993 hours and the 1997 access step to that cell, and holding intraday variance fixed, multiplies daily market variance by about 1.4 and takes 13% to about 15.4%.

Actual five-year rolling volatility went from about 10% in the 1950s to about 18% in the most recent window, and the implied path covers 5.4 of those 7.7 percentage points: the experiments account for roughly 70% of the rise.¹² The identified channels are the right order of magnitude for the headline fact, and their timing lines up with the actual series.

¹²Against the fitted trend of Table 1, which ends near 20% because the five-year window ending in 2025 sits below the trend line, the share is about half; in variance rather than volatility units it is about 65%. The 10% starting point is the level of the actual series in the mid-1950s.

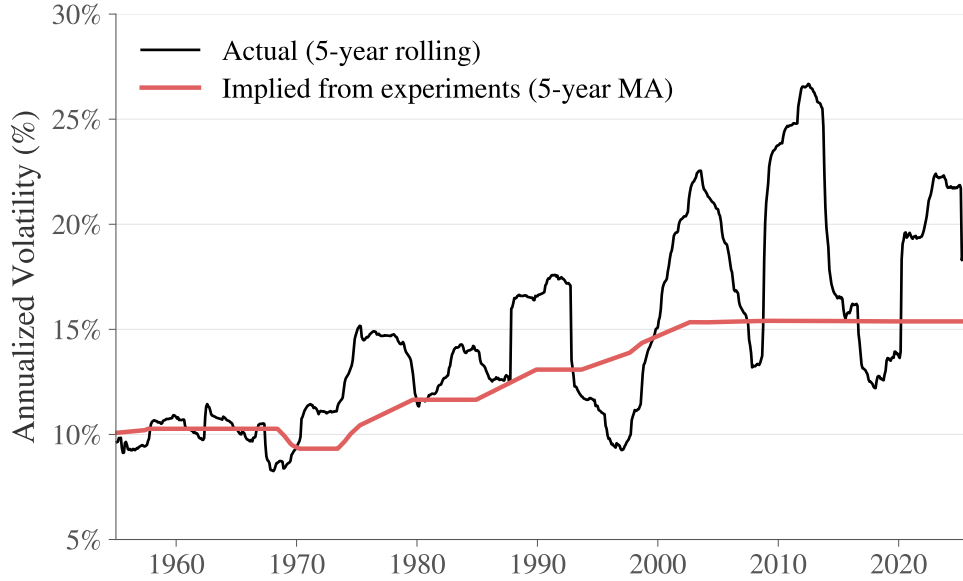


Figure 11: Volatility Implied by the Experiments, Against the Actual Series.

The black line is the 5-year rolling annualized volatility of the daily market return, as in Figure 1. The red line starts at 10% in 1950 and applies two sets of estimates to the actual history of trading hours and access. NYSE hours: the reduced-form elasticity of weekly volatility to weekly trading hours from Table 3, Panel B (1.45), applied to the nine schedule changes of 1952–1985, over which weekly hours went from 27 to 32.5. Index futures: the per-hour effect on overnight market variance from Table 4, column (3) (0.030 per hour), applied to the record of overnight futures hours in Table 15, plus the E-mini access effect (0.43) from September 1997, both applied to the overnight cell only, which is 25% of daily market variance measured over 1993–97 (after the 1993 change; the pre-1993 window is too short to use). Intraday variance is held fixed. The implied series is shown as a 5-year moving average to match the actual.

4. A model of index trading and market returns

The evidence above points to two related changes in how the market is traded. Investors increasingly trade market-wide positions rather than individual stocks, and index products increasingly allow those positions to be traded throughout the day. In this section, we present a simple model with both margins. Following Merton (1987), we consider an economy where investors initially trade only a subset of available financial assets. When an index-trading vehicle is introduced, investors can express the market-level component of their demand directly through the diversified market claim. As in Campbell, Grossman, and Wang (1993) (henceforth, CGW), risk-averse

arbitrageurs absorb the resulting order flow.

The second margin is trade time. Some investor demand evolves with trading opportunities rather than calendar time, so expanding the number of trading opportunities within a day can increase the amount of demand pressure that reaches prices. The model therefore links index trading and trade time to volatility, return dynamics, and trading volume.

4.1 Asset market without index trading

We first consider an economy without an index-trading vehicle. There are N risky assets with an $N \times 1$ vector of dividends $\mathbf{d}_t$,

$$\mathbf{d}_{t+1} = \mathbf{d}_t + \boldsymbol{\varepsilon}_{t+1}^F, \quad \boldsymbol{\varepsilon}_{t+1}^F \sim N(\mathbf{0}, \sigma_F^2 C_F), \quad (10)$$

where σ_F^2 is the variance of each stock's dividend innovation and C_F is the correlation matrix with constant pairwise correlation ρ_F .

There is a unit mass of behavioral investors and a competitive fully informed arbitrageur that can invest in all assets. All agents have CARA utility with risk aversion a over end-of-day wealth, and thus demands are myopic mean-variance. The risk-free asset has gross daily return $R_f > 1$ and is supplied elastically. We set supply $x_i = 1/N$ for each risky asset i .

The arbitrageur acts competitively, has rational expectations, $E_t^A(\cdot)$, and trades in all assets. Thus,

$$E_t^A[d_{i,t+1}] = d_{i,t} \quad (11)$$

for all i .

Behavioral investors are evenly distributed across N stock-specialist groups, with each investor trading only one stock in addition to the risk-free asset. These agents make two expectational errors: (1) they are slow to update their cash-flow expectations, and (2) their payoff expectations are subject to animal spirits whose variance is affected by the number of trading opportunities per day.

Let $E_t^B(\cdot)$ denote the average subjective expectation within the stock-specialist group relevant for the asset under consideration, and

$$\hat{d}_{i,t} \equiv E_t^B[d_{i,t+1}] \quad (12)$$

denote the average forecast among behavioral investors specialized in stock i . Write the dividend as the sum of its common and idiosyncratic components, $d_{i,t} = d_{m,t} + d_{i,t}^\perp$ with $d_{m,t} \equiv N^{-1} \sum_j d_{j,t}$. Stock investors observe $d_{i,t}^\perp$ as it arrives, but update their beliefs about the common component slowly. In particular, their forecast is

$$\hat{d}_{i,t} = d_{i,t}^\perp + \hat{d}_{m,t}^{\text{stk}}, \quad \hat{d}_{m,t}^{\text{stk}} = (1 - \alpha_{\text{stk}}) \hat{d}_{m,t-1}^{\text{stk}} + \alpha_{\text{stk}} d_{m,t}, \quad (13)$$

where $\hat{d}_{m,t}^{\text{stk}}$ is the forecast of the common component. When $\alpha_{\text{stk}} = 1$, the average behavioral forecast equals the rational forecast of next period's dividend. When $\alpha_{\text{stk}} < 1$, new permanent common dividend news is incorporated only gradually. The Appendix derives this structure as an outcome of costly monitoring, but here we will just take it as given.

The resulting difference between the average behavioral and rational forecasts of next period's dividend is

$$\delta_{i,t} \equiv E_t^B[d_{i,t+1}] - E_t^A[d_{i,t+1}] = \hat{d}_{i,t} - d_{i,t}. \quad (14)$$

Because idiosyncratic news is observed, $\delta_{i,t} = \hat{d}_{m,t}^{\text{stk}} - d_{m,t}$ is common across stocks: it is the forecast gap on the market component.

The second belief distortion is transitory, and the amount of new distortion generated within a day can depend on S . At the daily horizon, we write

$$z_{i,t+1} = \lambda z_{i,t} + u_{i,t+1}^z, \quad 0 < \lambda < R_f^{-1}, \quad (15)$$

where the innovations are serially independent, orthogonal to fundamental news, and

satisfy

$$\text{Var}(u_{i,t+1}^z | S) = \sigma_\eta^2 S^{1-\xi}, \quad \text{Corr}(u_{i,t+1}^z, u_{j,t+1}^z) = \rho_z, \quad i \neq j, \quad \xi \in [0, 1]. \quad (16)$$

The parameter λ governs the calendar-time persistence of existing demand pressure, while ξ controls how new demand shocks scale with trading opportunities. At $\xi = 0$, additional trading opportunities generate additional demand-pressure risk one-for-one; at $\xi = 1$, they only refine trading time and leave daily innovation variance unchanged. Thus lower ξ means a stronger trade-time shock-arrival channel. The Appendix constructs an intraday market with S explicit trading events that reproduces equations (15)–(16) when sampled at daily boundaries.

For notational simplicity, group i specializes in stock i . Let $P_{i,t}$ denote the end-of-day price of stock i . Its one-day excess payoff, which we henceforth refer to simply as its return, is

$$\Pi_{i,t+1}^e = P_{i,t+1} + d_{i,t+1} - R_f P_{i,t}. \quad (17)$$

The average subjective return expectation of investors specialized in stock i is

$$E_t^B(\Pi_{i,t+1}^e) = E_t^A(\Pi_{i,t+1}^e) + g_{i,t} + z_{i,t}, \quad (18)$$

where $E_t^A(\cdot)$ is the arbitrageur's rational expectation, $g_{i,t}$ is an expected-return distortion generated by stale fundamental information, and $z_{i,t}$ is a transitory non-fundamental demand distortion. The Appendix relates $g_{i,t}$ to the error in dividend expectations, $\delta_{i,t}$.

To characterize market prices, define the market-level component of stock investors' fundamental belief distortions, and market-level demand pressure as

$$g_{m,t}^{\text{stk}} \equiv \frac{1}{N} \sum_{i=1}^N g_{i,t}, \quad z_{m,t} \equiv \frac{1}{N} \sum_{i=1}^N z_{i,t}. \quad (19)$$

Because all stocks have equal supply and the covariance structure is symmetric, the equilibrium market price is a scalar function of these market-level states.

Proposition 1. *Let*

$$P_{m,t} \equiv \sum_{i=1}^N x_i P_{i,t} = \frac{1}{N} \sum_{i=1}^N P_{i,t}$$

denote the price of the market portfolio. The equilibrium market price in the economy without index trading is

$$P_{m,t} = p_S + \frac{d_{m,t}}{R_f - 1} + b_{m,S}^F g_{m,t}^{\text{stk}} + b_{m,S} z_{m,t}, \quad (20)$$

where

$$b_{m,S}^F = \frac{\psi_S}{1 + \psi_S} \frac{1}{R_f - (1 - \alpha_{\text{stk}})}, \quad b_{m,S} = \frac{\psi_S}{1 + \psi_S} \frac{1}{R_f - \lambda}, \quad (21)$$

and

$$\psi_S \equiv \frac{v_{m,S}}{v_S} < 1. \quad (22)$$

Here v_S is the common conditional one-day return variance of an individual stock and $v_{m,S}$ is the corresponding market return variance.

The term ψ_S captures the effect of investor underdiversification. Behavioral investors trade individual stocks rather than a diversified market claim. Their exposure to idiosyncratic risk makes their risky-asset demand less sensitive to fluctuations in subjective expected returns than it would be if they could trade a diversified portfolio without idiosyncratic risk. The market-level components of their belief distortions therefore reach the market portfolio with attenuated multipliers, with the degree of attenuation governed by ψ_S .

The return implications of the two belief distortions differ. Define

$$u_{m,t+1}^z \equiv \frac{1}{N} \sum_{i=1}^N u_{i,t+1}^z, \quad \varepsilon_{m,t+1}^F \equiv \frac{1}{N} \sum_{i=1}^N \varepsilon_{i,t+1}^F.$$

Substituting the equilibrium market price and the law of motion for demand pressure, the component of the market return generated by demand pressure is

$$b_{m,S} u_{m,t+1}^z - (R_f - \lambda) b_{m,S} z_{m,t}. \quad (23)$$

The first term reflects new demand shocks, which add variance. The second is the

reversal of past demand pressure. A high value of $z_{m,t}$ raises the current price, but because demand pressure mean reverts, it predicts a lower subsequent return.

Gradual incorporation of dividend news produces the opposite return dynamics. The Appendix shows that the component of the market return associated with the fundamental belief distortion is

$$-\frac{R_f}{R_f - 1}(1 - \alpha_{\text{stk}})b_{m,S}^F \varepsilon_{m,t+1}^F - [R_f - (1 - \alpha_{\text{stk}})] b_{m,S}^F g_{m,t}^{\text{stk}}. \quad (24)$$

A positive permanent dividend innovation therefore initially lowers the price relative to its full-information response and pushes $g_{m,t+1}^{\text{stk}}$ downward. Starting from zero belief distortion, $g_{m,t+1}^{\text{stk}}$ becomes negative; as this distortion subsequently closes, the second term in equation (24) generates positive expected returns. Thus gradual incorporation of permanent dividend news generates continuation, whereas transitory demand pressure generates reversal.

The empirics consider variance ratios, so the natural model object is the variance of cumulative returns over calendar horizons. For h calendar periods, let $V_h(S)$ denote the variance of the h -period market return and define

$$VR_{1,h}(S) = \frac{hV_1(S)}{V_h(S)}. \quad (25)$$

Demand pressure pushes $VR_{1,h}(S)$ above one, while gradual incorporation of permanent dividend news pushes it below one. For $\zeta < 1$, increasing S raises the amount of demand-pressure risk generated within the day while leaving the calendar-day distribution of fundamental news unchanged. Higher S therefore raises the relative importance of the mean-reverting demand-pressure component. As ζ increases, this trade-time shock-arrival channel is attenuated.

4.2 Asset market with index trading

We next introduce an index fund that allows investors to trade the diversified market portfolio directly. A fraction $\omega \in [0, 1]$ of behavioral investors, drawn symmetrically

from the stock-specialist groups, switches from individual stocks to the index fund; the remaining fraction $1 - \omega$ continues to trade stocks. We refer to ω as the *index-participation share*.¹³ The index fund is backed by the underlying stocks and therefore creates no new supply of risky cash flows. Normalizing one share of the fund to hold one unit of the market portfolio, its price and dividend are $P_{m,t}$ and $d_{m,t}$. Its one-day return is

$$\Pi_{m,t+1}^e = P_{m,t+1} + d_{m,t+1} - R_f P_{m,t}. \quad (26)$$

Write each stock's transitory belief distortion as

$$z_{i,t} = z_{m,t} + z_{i,t}^\perp, \quad (27)$$

where $z_{i,t}^\perp$ is the stock-specific component. Because the index fund holds the diversified market portfolio, the stock-specific components cancel, and its transitory belief distortion is $z_{m,t}$. Thus introducing the index fund allows investors to express the existing market component $z_{m,t}$ directly through a diversified claim rather than indirectly through individual stocks.

Let $E_t^{B,\text{idx}}(\cdot)$ denote the average subjective expectation among behavioral investors who trade the index fund. Then

$$E_t^{B,\text{idx}}(\Pi_{m,t+1}^e) = E_t^A(\Pi_{m,t+1}^e) + g_{m,t}^{\text{idx}} + z_{m,t}, \quad (28)$$

where $g_{m,t}^{\text{idx}}$ is the average fundamental expected-return distortion among index-fund investors.

As before, the arbitrageur observes realized dividends, and index investors observe the market dividend when they refresh their information. Since $d_{m,t}$ is the average of the individual-stock dividends, equation (10) implies

$$E_t^A[d_{m,t+1}] = d_{m,t}. \quad (29)$$

¹³We treat ω as exogenous; limited participation can reflect direct or nonpecuniary costs of switching.

Let

$$\hat{d}_{m,t}^{\text{idx}} \equiv E_t^{B,\text{idx}}[d_{m,t+1}] \quad (30)$$

denote the average forecast among behavioral investors who trade the index fund. Upon observing the current market dividend, this average forecast evolves according to

$$\hat{d}_{m,t}^{\text{idx}} = (1 - \alpha_{\text{idx}})\hat{d}_{m,t-1}^{\text{idx}} + \alpha_{\text{idx}}d_{m,t}. \quad (31)$$

The resulting difference between the average behavioral and rational forecasts is

$$\delta_{m,t}^{\text{idx}} \equiv E_t^{B,\text{idx}}[d_{m,t+1}] - E_t^A[d_{m,t+1}] = \hat{d}_{m,t}^{\text{idx}} - d_{m,t}. \quad (32)$$

As in the individual-stock case, the Appendix derives the mapping from $\delta_{m,t}^{\text{idx}}$ to the expected-return distortion $g_{m,t}^{\text{idx}}$.

We allow α_{idx} to exceed α_{stk} . Both groups process the same common dividend news; they differ only in the claim through which that information is acted on. The Appendix shows that this difference in updating rates can be rationalized by a monitoring problem: the same common information is more valuable to keep current when it can be acted on through a diversified claim, yielding $\alpha_{\text{idx}} > \alpha_{\text{stk}}$.

For compactness, define

$$\Delta_S(\omega) \equiv 1 + \omega + (1 - \omega)\psi_S. \quad (33)$$

The equilibrium return variances v_S and $v_{m,S}$, and hence ψ_S , can depend on ω ; as in the Appendix, we suppress this dependence in the notation.

Proposition 2. *The equilibrium market price with index trading is*

$$P_{m,t} = p_S + \frac{d_{m,t}}{R_f - 1} + b_{m,S}^{F,\text{stk}}(\omega)g_{m,t}^{\text{stk}} + b_{m,S}^{F,\text{idx}}(\omega)g_{m,t}^{\text{idx}} + b_{m,S}(\omega)z_{m,t}, \quad (34)$$

where

$$b_{m,S}^{F,\text{stk}}(\omega) = \frac{(1-\omega)\psi_S}{\Delta_S(\omega)} \frac{1}{R_f - (1-\alpha_{\text{stk}})}, \quad (35)$$

$$b_{m,S}^{F,\text{idx}}(\omega) = \frac{\omega}{\Delta_S(\omega)} \frac{1}{R_f - (1-\alpha_{\text{idx}})}, \quad (36)$$

$$b_{m,S}(\omega) = \frac{\omega + (1-\omega)\psi_S}{\Delta_S(\omega)} \frac{1}{R_f - \lambda}. \quad (37)$$

The no-index economy is nested as the special case $\omega = 0$.

The index mechanism is easiest to see in the loading on transitory demand pressure, $b_{m,S}(\omega)$. When behavioral investors trade individual stocks, their exposure to idiosyncratic risk attenuates the amount of market-level demand pressure that reaches prices. Trading the diversified index fund removes this idiosyncratic risk. As ω rises, a greater fraction of the same market-level demand signal is therefore expressed through the diversified claim, raising its impact on the market price. In the limiting case in which all behavioral investors trade the index fund,

$$b_{m,S}(1) = \frac{1}{2} \frac{1}{R_f - \lambda}, \quad \frac{b_{m,S}(1)}{b_{m,S}(0)} = \frac{1 + \psi_S(0)}{2\psi_S(0)} > 1, \quad (38)$$

where $\psi_S(0)$ denotes the no-index value of ψ_S . Thus index trading amplifies the market-price response to a given market-level demand distortion because investors can express that distortion through a diversified claim rather than through concentrated individual-stock positions.

In our benchmark, index trading also speeds the incorporation of fundamental information. With high ω , the market price places more weight on beliefs formed through the diversified claim and less on the more slowly updating stock-investor beliefs.

The two margins of market development therefore have distinct but complementary effects. A higher ω increases the sensitivity of market prices to transitory market-level belief distortions by allowing them to be expressed through a diversified claim. For $\xi < 1$, a higher S increases the amount of new transitory demand-pressure risk generated within a calendar day. Their effects reinforce one another: additional trad-

ing opportunities matter more when a larger fraction of behavioral demand can be expressed directly through the index fund.

4.3 Stock-level implications

The model also delivers stock-level implications. Its counterpart to the empirical market R^2 is the R^2 from regressing daily stock returns on the daily market return. Intermediated dollar volume is

$$V_{i,t}^{\text{model}} = P_{i,t} \left| n_{i,t}^A - n_{i,t-1}^A \right|. \quad (39)$$

By market clearing, $\Delta n_{i,t}^A = -\Delta n_{i,t}^{B,\text{net}}$. Higher ω routes more behavioral demand through the diversified claim and therefore raises the common component of stock-level order flow. Because the symmetric model has no cross-sectional size variation, we use the empirical volume beta to guide the historical direction of ω , not as a fitted model moment.

The model maps into three empirical predictions. First, higher S raises short-horizon volatility and pushes variance ratios toward reversal. This is the comparative static isolated by the trading-hours IVs, the E-mini overnight experiment, and the holiday tests. Second, because index investors trade a diversified claim, expanded index trading opportunities should raise systematic variance more than idiosyncratic variance. Third, the effect of an additional trading opportunity should be larger when more behavioral demand is expressed through the index. The Appendix adds a fourth implication: when index products were concentrated in the S&P 500, membership should have accelerated the incorporation of common information.

4.4 Benchmark parameterization and comparative statics

We next ask whether plausible changes in index participation and trading opportunities can reproduce the broad historical evolution of volatility and return dynamics. Primitive parameters are disciplined by external evidence; the paths for ω and S are

coarse historical mappings. The exercise is therefore about magnitudes and comparative statics rather than point identification.

Benchmark parameters. We set the number of assets to $N = 500$, so the index can be interpreted as a stylized S&P 500 index. We set the average correlation of fundamental innovations and the average correlation of demand-pressure innovations to $\rho_F = \rho_z = 0.25$, and the annual gross risk-free return to $R_f^{\text{ann}} = 1.02$. The annualized volatility of market-level fundamental news is 12.8%. With $\rho_F = 0.25$ and $N = 500$, this corresponds to annualized firm-level fundamental-news volatility of about 25%. The scale of transitory demand-pressure innovations, σ_η , is chosen to match early-sample daily market volatility and is held fixed thereafter. Thus the model's subsequent rise in volatility is generated by the historical changes in index participation and trading opportunities rather than by changes in shock magnitudes.

For fundamental information, we set $\alpha_{\text{stk}} = .30$, implying a roughly two-day half-life for common news processed through individual stocks, and $\alpha_{\text{idx}} = 1$, so common news is incorporated within the day through the index. The Appendix rationalizes these chosen rates from a monitoring problem. At the benchmark $\psi_s \simeq 0.21$, the same common information is about five times as valuable when acted on through the diversified claim; the Appendix also provides separate evidence that common information historically propagated slowly across stocks and that index membership later accelerated this process.

Demand-pressure persistence. The persistence of demand pressure governs the horizon of reversal. We set $\lambda = .90$, corresponding to a half-life of $\log(.5)/\log(.9) \approx 6.6$ trading days. This places the reversal at the weekly-to-monthly horizon observed in the modern data rather than at intraday or much longer horizons.

Trade-time scaling and market structure. The parameter ζ determines how strongly demand-pressure risk scales with trade time. Our trading-hours evidence points to a low value, and we set $\zeta = .1$.

Trading opportunities increased from roughly the regular cash-market day at the beginning of the sample to nearly around-the-clock access by the end. We normalize the low-trading-opportunity benchmark to $S = 1$ and use the historical path

$$S = (1, 1, 2, 4) \tag{40}$$

for 1950–70, 1971–90, 1991–2004, and 2005–24. Guided by the rise in market-directed trading documented in the data, we use

$$\omega = (.35, .70, .72, .90). \tag{41}$$

The volume-beta evidence disciplines the broad rise in ω , not its exact level in each era.

Quantitative implications. Figure 12 reports the comparative statics. Both index trading and expanded trade time raise the contribution of transitory demand pressure to daily market volatility. They also raise the variance ratio because demand pressure is mean reverting. The two mechanisms reinforce one another: additional trading opportunities have a larger effect when behavioral investors can express their market-level beliefs through the diversified market claim.

Figure 13 compares the model with the historical data, using the S&P 500 as the primary variance-ratio benchmark and the Dow as a nearby reference.

The parameterization captures both patterns that motivate the model. Daily market volatility rises from about 10% to about 20%, while return dynamics move from continuation to roughly unpredictable returns and then to reversal. With little change in trading hours between the first two eras, greater index participation moves the variance ratio toward one; the later expansion of trade time pushes volatility and reversal higher still.

Transitory demand pressure rises from 23% of daily market variance in the early economy to 61% in the modern economy and accounts for 82% of the modeled increase in variance. Permanent fundamental news remains important for the level of

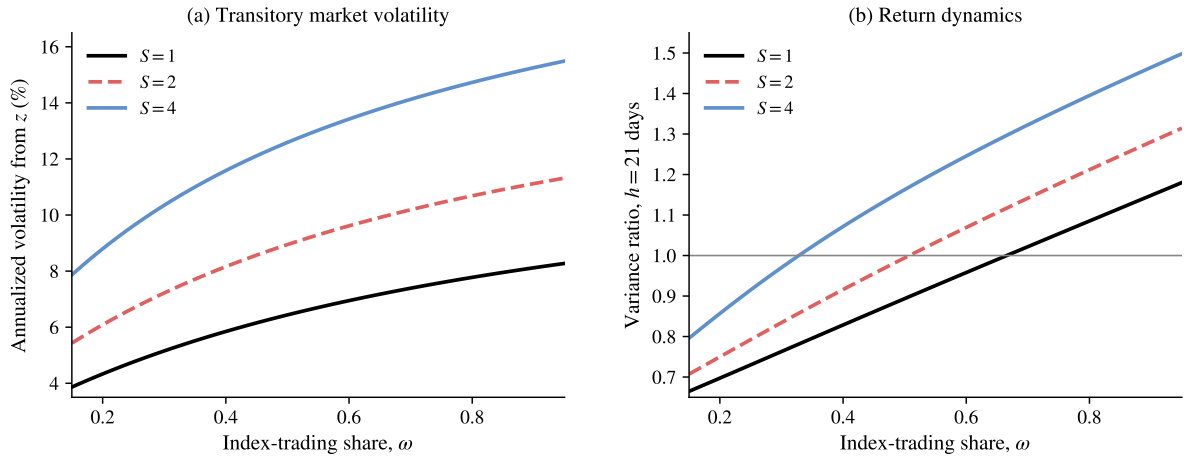


Figure 12: Market structure and transitory demand. Panel A plots the annualized contribution of the transitory z component to market volatility as the index-participation share ω varies, for $S = 1, 2, 4$ trading opportunities. Panel B plots the corresponding 21-trading-day variance ratio. All other parameters are held at their benchmark values.

market risk, but it does not drive the secular increase. Thus the two changes in market structure play distinct roles: index trading amplifies market-level demand pressure and speeds the incorporation of common information, while expanded trade time generates more transitory demand pressure. Together they reproduce the broad evolution of volatility and return dynamics documented in the data.

5. Conclusion

Stock market volatility has roughly doubled over the past 70 years. The increase is largest at short horizons and reflects a change in the autocorrelation of returns, from continuation to reversal. Over the same period the stock market transformed, with the proliferation of index products (index futures, index ETFs, and passive mutual fund products) and with substantially expanded trading hours for these products. We provide causal evidence that trading in these products has contributed substantially to the increase in volatility: wherever the opportunity to trade the index expands, trading and volatility rise with it, and together the experiments account for roughly

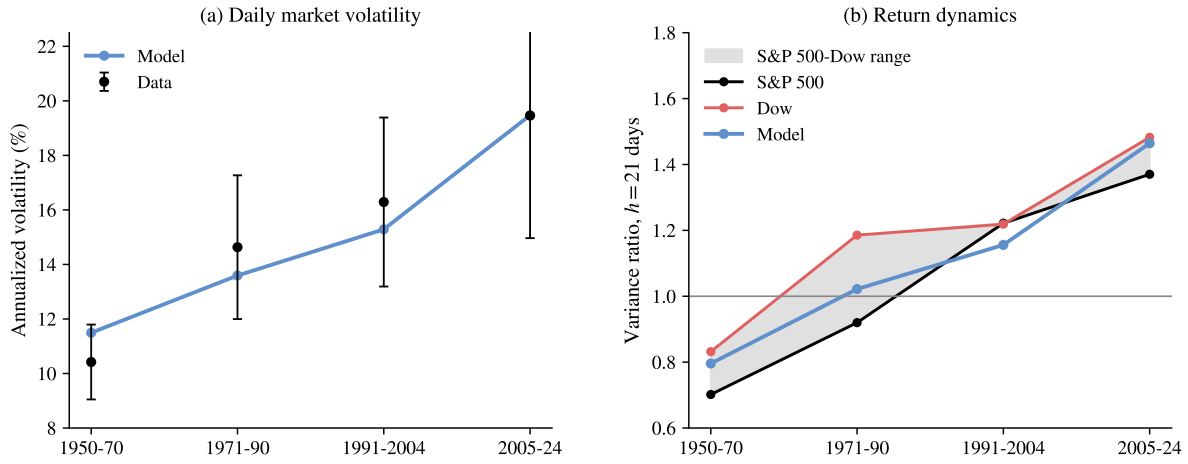


Figure 13: Historical volatility and return dynamics. Panel A compares broad-market daily volatility with the model; error bars show two standard errors from a year-block bootstrap. Panel B compares the model with the S&P 500 and Dow using a horizon-matched variance-ratio estimator. The shaded region spans the two index benchmarks; the horizontal line marks one. The S&P 500 first point is 1957–70; the final era is 2005–24.

70% of the rise. A model in which arbitrageurs absorb investors' demand shocks reproduces both the rise in volatility and the shift from continuation to reversal.

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A. Appendix

A.1 Systematic vs. Idiosyncratic Volatility

This appendix gives the construction behind the decomposition in Section 2.3.4 and reports the validation against Campbell et al. (2001) (CLMX), the variance-ratio cut, and the figures. We follow CLMX literally. Each stock return splits into a market component, an industry-relative-to-market component, and a firm-specific component through their unit-beta accounting identity: the industry residual is $R_{it} - R_{mt}$ and the firm residual is $R_{jit} - R_{i(j)t}$, with no betas estimated. We build industry returns by value-weighting CRSP firms within each of the 49 Fama–French industries; firm variances aggregate to the industry and market level by lagged market-cap weights; monthly variances are the within-month sum of squared daily residuals. The universe is all CRSP common stocks (share codes 10 and 11) from 1925 to 2024, roughly 82 million stock-days. We report the sample from 1962 to line up with CLMX and extend it through 2024.

Our construction reproduces their original numbers. Table 6 compares our implementation to their Table III over the 1962–1997 value-weighted daily sample they study. The firm-specific mean and trend – the moments CLMX build their paper around – match to within 3 and 8 percent, and our firm trend of 0.887 sits well inside their reported 90% confidence interval of $[0.55, 1.47]$.

Table 6: Validation against Campbell et al. (2001) Table III

	Our implementation	CLMX Table III	Δ
MKT mean $\times 100$	1.658	1.542	+7.5%
IND mean $\times 100$	0.957	1.032	−7.3%
FIRM mean $\times 100$	6.628	6.436	+3.0%
MKT trend $\times 10^5$	0.164	0.156	+5.4%
IND trend $\times 10^5$	0.094	0.062	+51.7%
FIRM trend $\times 10^5$	0.887	0.965	−8.1%

Notes: Mean and linear-trend coefficient of the market (MKT), industry-relative-to-market (IND), and firm-specific (FIRM) variance components, value-weighted, daily, over the 1962-07 to 1997-12 sample of Campbell et al. (2001). Means are average monthly variance $\times 100$; trends are the coefficient on a linear time index $\times 10^5$, matching their Table III units. CLMX report a 90% confidence interval of $[0.55, 1.47]$ for the FIRM trend.

The decomposition recovers both the original CLMX result and its later reversal. Firm-specific volatility rose from 16% in the postwar quiet to 26% by 2000–2010, the increase CLMX documented. It then fell back to 19% over 2011–2024 – the reversal Campbell et al. (2023) report, also studied by Chiah et al. (2023). The market component moved the other way: its share of average-stock variance reached a trough in the late 1990s, then rebounded.

Table 7 reports the share of average-stock variance carried by the market and by the systematic (market plus industry) component. The market share nearly doubles, from 0.18 in 1962–1971 to 0.31 in 2015–2024; the systematic share rises from 0.31 to 0.45. Both trends are strongly significant – +2.50 percentage points per decade for the market share ($t = 3.95$) and +3.25 for the systematic share ($t = 4.01$), under Newey–West HAC inference with 60 monthly lags.

Table 7: Share of average-stock variance in the systematic component

Period	Market share (= literal R^2)	Systematic share (MKT + IND) / Total
1962–1971	0.18	0.31
1988–1997	0.18	0.28
2000–2009	0.22	0.38
2015–2024	0.31	0.45
Trend (pp/decade)	+2.50 ($t = 3.95$)	+3.25 ($t = 4.01$)

Notes: Market share is value-weighted aggregate market variance divided by value-weighted aggregate total variance across all CRSP common stocks – the share-of-variance interpretation of average R^2 in Campbell et al. (2001). Systematic share adds the industry-relative-to-market component. Period rows are means of the 60-month rolling share; the trend row is the coefficient on a normalized linear time index, in percentage points per decade, with Newey–West HAC t -statistics (60 monthly lags).

The variance-ratio result in Section 2.1 sits in the same component. Decomposing the five-year variance ratio by CLMX component, the market-component ratio more than doubles, from 0.72 in 1961–1995 to 1.51 in 1996–2024, and the industry component shifts the same way by a smaller amount. The firm-specific ratio shows no regime change: it has always been above one – the mechanical signature of bid–ask bounce – and declines monotonically over the full sample as spreads narrow and microstructure noise dissipates. Equal-weighting the top 500 stocks gives the same market-component shift (0.62 $\rightarrow$ 1.42), so this is not a mega-cap artifact. Whatever changed the character of returns around 2000 changed the systematic component, not firm-specific behavior.

A.2 Macroeconomic Fundamentals Do Not Explain the Rise

The rise in stock volatility does not trace to more volatile fundamentals. This appendix shows the comparison with fundamentals for real GDP and corporate profits as well as industrial production, gives the industrial-production-volatility control behind Section 2.3, and updates the Schwert (1989) regression.

Figure 17 plots the ratio of annualized daily stock market volatility to annualized macro volatility over four non-overlapping 25-year periods from 1926 to 2025, for

60-month rolling std dev with linear trend: market, industry, firm-specific (VW)
 Literal CLMX construction, 1962–2024

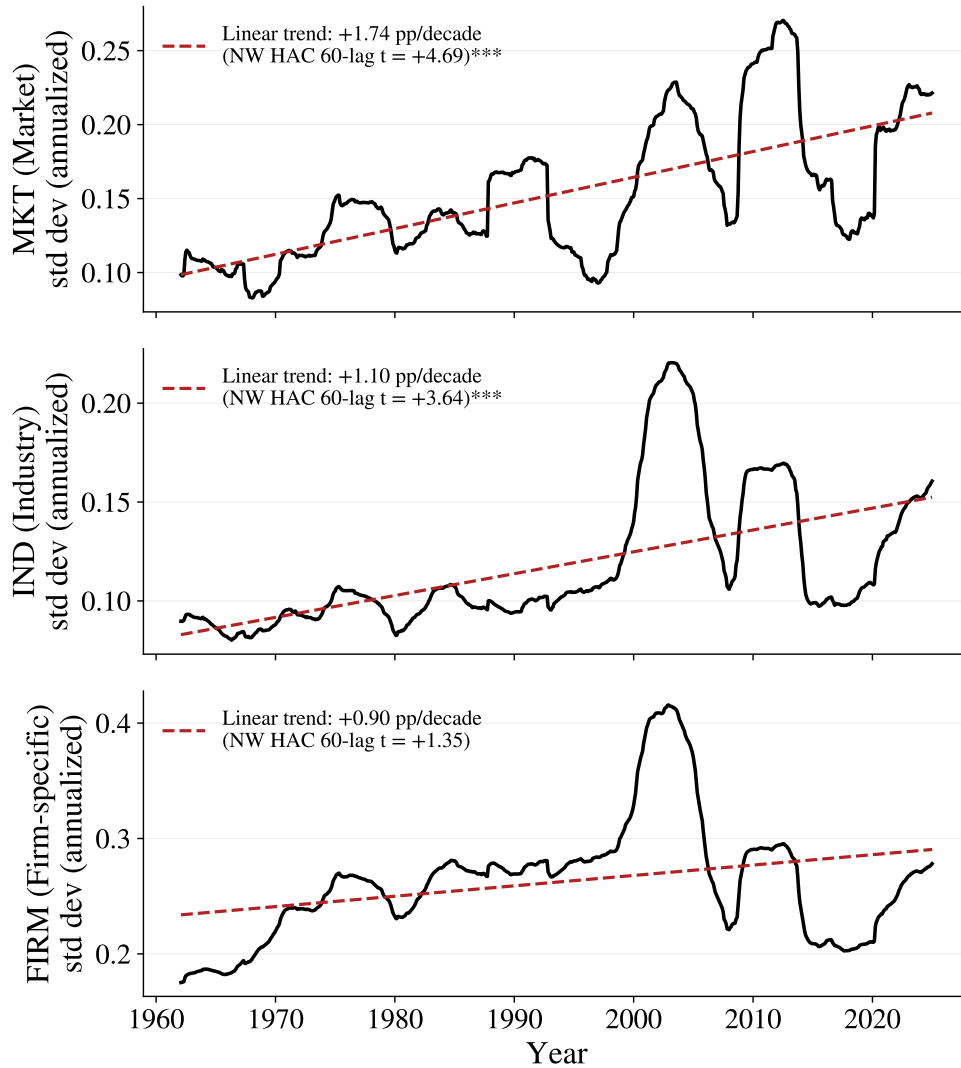


Figure 14: Market, industry, and firm-specific volatility, 1962–2024

Notes: Annualized standard deviation of the market (MKT), industry-relative-to-market (IND), and firm-specific (FIRM) components of average-stock variance, five-year rolling, value-weighted, 1962–2024, with fitted linear trends. The market and industry components trend up significantly; the firm-specific linear trend is positive but insignificant over the full sample because of the post-2001 reversal.

Share of systematic variance, VW, 60-month rolling
Literal CLMX construction, 1962–2024

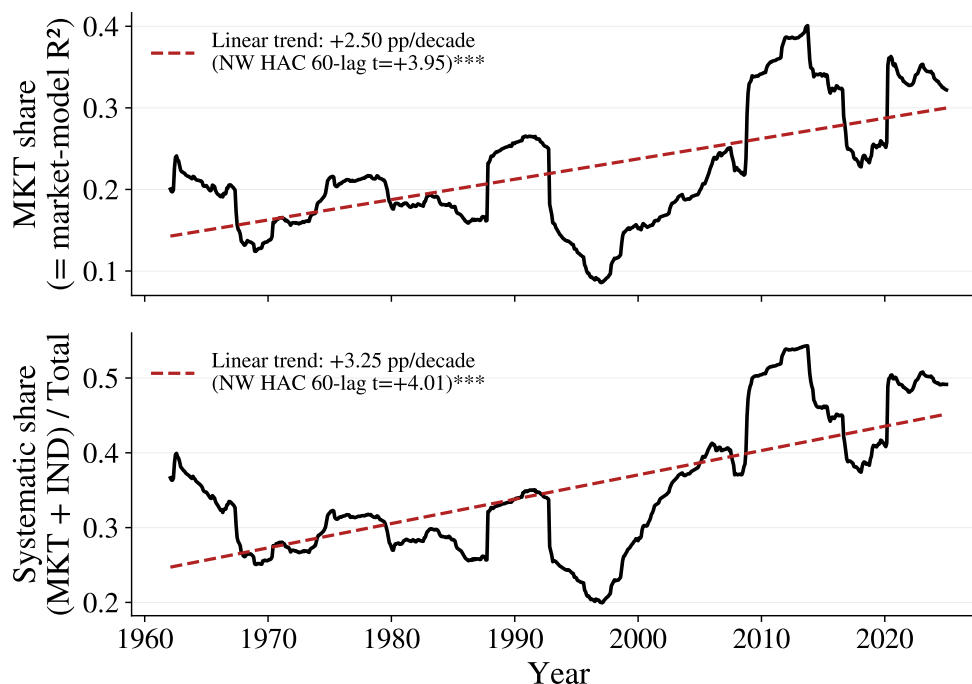


Figure 15: Systematic share of average-stock variance, 1962–2024

Notes: Market share (value-weighted aggregate market variance over total variance) and systematic share (market plus industry), 60-month rolling, 1962–2024. The late-1990s trough in the market share mirrors the firm-specific volatility peak in Campbell et al. (2001).

three macro series: industrial production growth, real per capita GDP growth, and corporate profit growth. Industrial production is monthly from 1920; the other two are quarterly after WWII, spliced with annual data back to 1929. Corporate profits are a levered cash flow, so they absorb changes in corporate leverage. For all three measures the ratio was lowest in the first period, rose slightly in 1951–1975, rose strongly in 1976–2000, and stayed high.

On the full 1931–2025 sample a simple linear trend in stock volatility is essentially zero – elevated Depression-era volatility offsets the modern rise. Adding five-year rolling industrial-production volatility as a control flips the trend to +11.0 percentage points on the normalized index, matching the paper’s 1955–2025 baseline (Table 8, Panel A). The reason is that IP volatility trended *down* over the sample, so partialling it out uncovers the upward trend in stock volatility it was masking. On the post-1960 subsample the trend is nearly identical with or without the control, +10.78 versus +10.91 (Panel B): industrial-production volatility absorbs essentially none of the modern rise. Allowing a slope break at 1960 (column 3b) places the change in slope at +28 points on the normalized trend, about +3 pp/decade ($p = 0.013$).

Variance ratio (daily-implied / monthly-implied variance)
60-month rolling, VW, 1931–2024

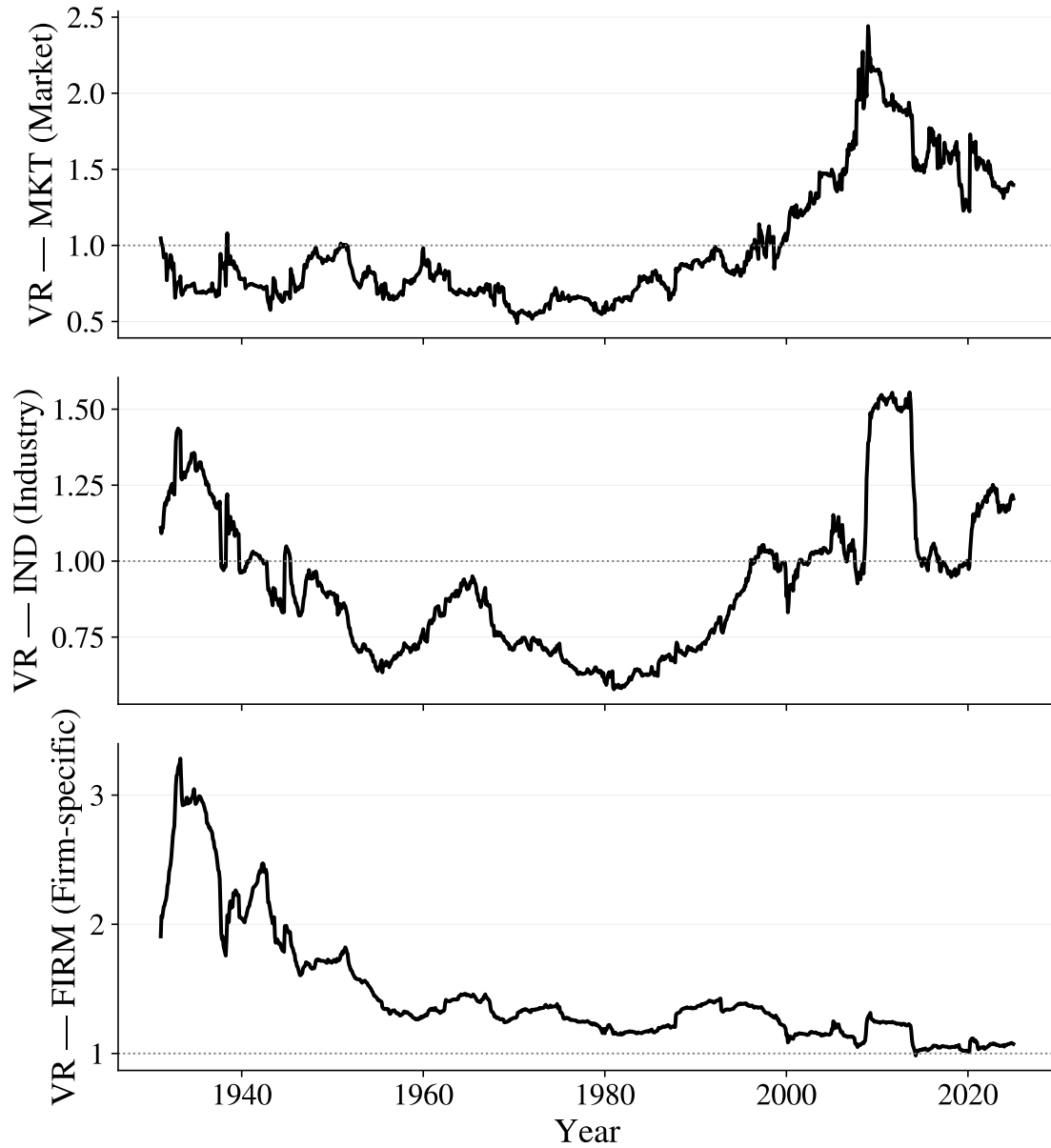


Figure 16: Variance ratio by CLMX component, 1931–2024

Notes: Five-year rolling variance ratio (daily-implied annualized variance over monthly-implied annualized variance) for the market, industry-relative-to-market, and firm-specific components. Note the differing vertical scales: the market and industry ratios shift upward through a regime change around 2000, while the firm-specific ratio declines monotonically with no regime change.

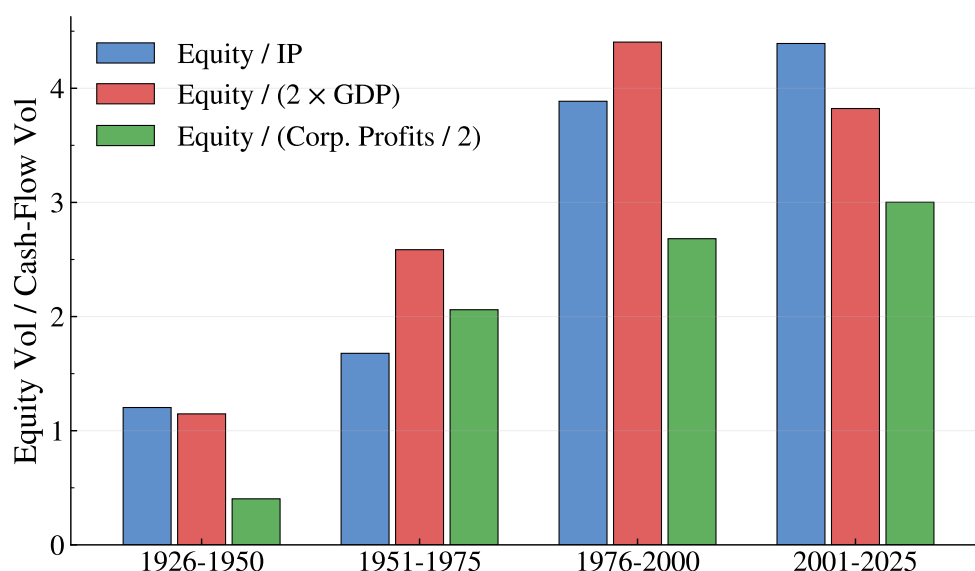


Figure 17: Equity Volatility Relative to Fundamental Volatility, by 25-Year Period

Notes: Ratio of annualized equity return volatility to annualized fundamental volatility, by 25-year sub-period. Equity volatility is the annualized standard deviation of daily Fama–French market excess returns within each sub-period. Fundamental volatility measures: industrial production (IP) uses 12-month overlapping growth rates; real GDP per capita uses quarterly data post-1947 and annual data pre-1947, spliced; corporate profits uses annual real growth rates. GDP volatility is scaled by 2 and corporate profits volatility is scaled by 1/2 for visual comparability. Data sources: Fama–French daily factors, FRED (INDPRO, real GDP per capita, real corporate profits).

The same conclusion holds in the classic Schwert (1989) regression of stock volatility on the volatilities of industrial-production growth, inflation, unemployment, and the short rate. Updating it to 1947–2024 gives an R^2 of 0.26 – close to the 0.38 we recover on his original window, and leaving roughly three-quarters of the variation in stock volatility unexplained by fundamentals (Table 9). The relationship has decoupled since the 1980s: macroeconomic volatility has fallen to historic lows while stock volatility has risen.

A.3 Implied Volatility and the Variance Risk Premium

A natural objection is that the VIX, the most-cited measure of market volatility, shows little trend over its post-1990 sample. The reason is that the variance risk premium has declined at almost exactly the rate realized volatility has risen. Splicing the VXO (1986–1989) to the VIX (1990 onward) and measuring realized volatility, implied volatility, and their difference as five-year rolling averages, realized volatility trends up at +1.84 pp/decade while the variance risk premium falls at –2.01 pp/decade, leaving implied volatility flat at –0.17 pp/decade, indistinguishable from zero (Table 10). The declining premium is the central finding of Dew-Becker and Giglio (2026), who attribute it to easing intermediary frictions in the options market. The flat VIX reflects

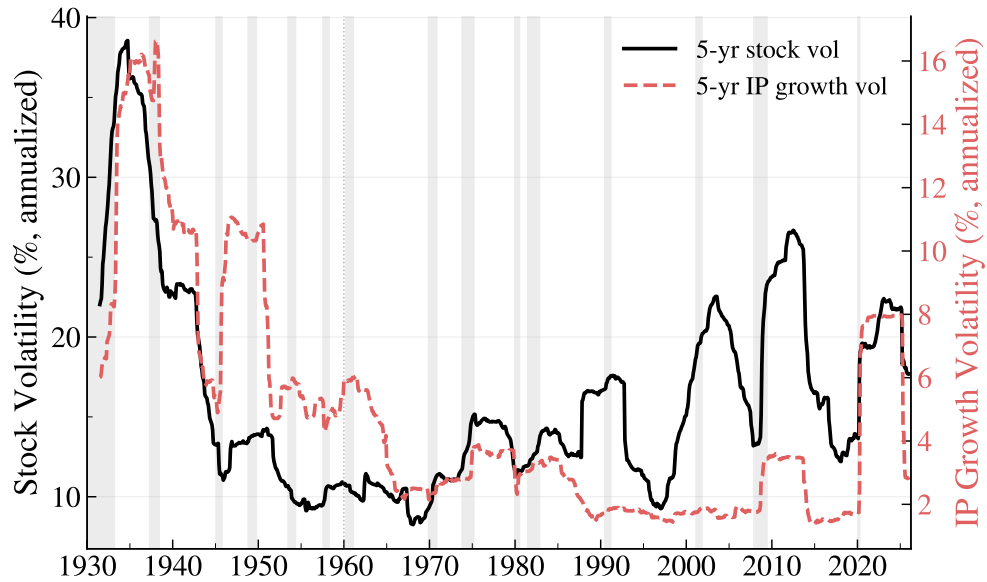


Figure 18: Stock volatility and industrial-production volatility, 1931–2025

Notes: Five-year rolling annualized stock volatility (left axis) and industrial-production-growth volatility (right axis). The two track each other through the Depression and early postwar period, then diverge after the mid-1980s: IP volatility falls to historic lows while stock volatility rises.

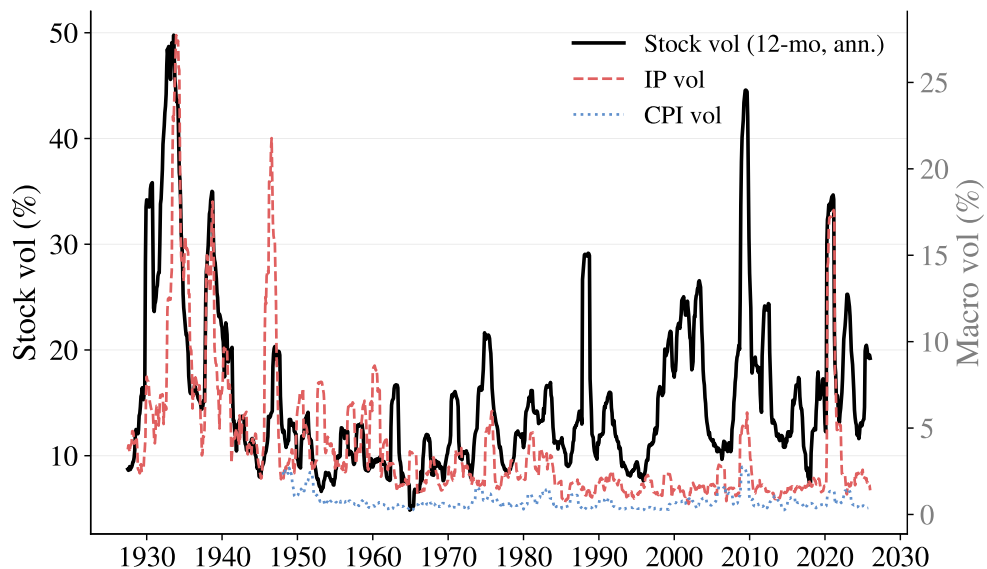


Figure 19: Stock volatility versus macroeconomic volatility, 1928–2025

Notes: Stock volatility (left axis) against industrial-production-growth and CPI-inflation volatility (right axis), all 12-month rolling annualized. Macroeconomic volatility was elevated in the Depression and the 1970s but has been near historic lows since the mid-1990s, even as stock volatility has risen.

Table 8: Stock volatility and industrial-production volatility

Panel A: Full sample, 1931–2025 (N = 1,135)

	(1) Trend only	(2) + σ_{IP}	(3b) Slope break
Constant	17.21*** (3.98)	3.21 (3.02)	6.56* (3.58)
Trend $\tilde{t}$	−1.97 (6.05)	11.01** (4.31)	−21.64 (15.32)
σ_{IP}		1.54*** (0.33)	1.57*** (0.26)
Post-1960 $\times \tilde{t}$			28.27** (11.37)
R^2	0.008	0.474	0.565

Panel B: Post-1960 subsample (N = 792)

	(4a) Trend only	(4b) + σ_{IP}
Constant	9.65*** (0.74)	8.16*** (1.17)
Trend (1960→end)	10.91*** (2.11)	10.78*** (2.20)
σ_{IP}		0.51* (0.27)
R^2	0.477	0.515

Notes: Regressions of five-year rolling annualized stock volatility (%) on a normalized linear trend $\tilde{t}$ and the five-year rolling annualized volatility of monthly industrial-production growth, σ_{IP} . Panel A is the full 1931–2025 sample; column (3b) adds a slope change interacted with a post-1960 indicator. Panel B restricts to post-1960 with the trend renormalized to zero in 1960. Newey–West HAC standard errors (60 monthly lags) in parentheses. *, **, *** denote $p < 0.10, 0.05, 0.01$.

a shrinking premium, not an absence of rising volatility.

A.4 Structural Breaks in the Level and Character of Volatility

A linear trend is a convenient summary, not a literal model. As an alternative, we estimate structural breaks in the mean of the daily five-year volatility series in Figure 1. Figure 22 allows two breaks and dates them to around 1975 and 2000, with the increase in mean volatility over the sample about the same as in the trend specification.

Extending the break analysis back to 1931 sharpens the picture. With four breaks the daily volatility series splits into a Depression peak (mean 33%), a late-Depression and wartime regime (23%), the postwar low (11%), a step up around 1973 (14%), and the modern era (19%), recovering breaks in 1937, 1943, 1973, and 2000 (Figure 23). The

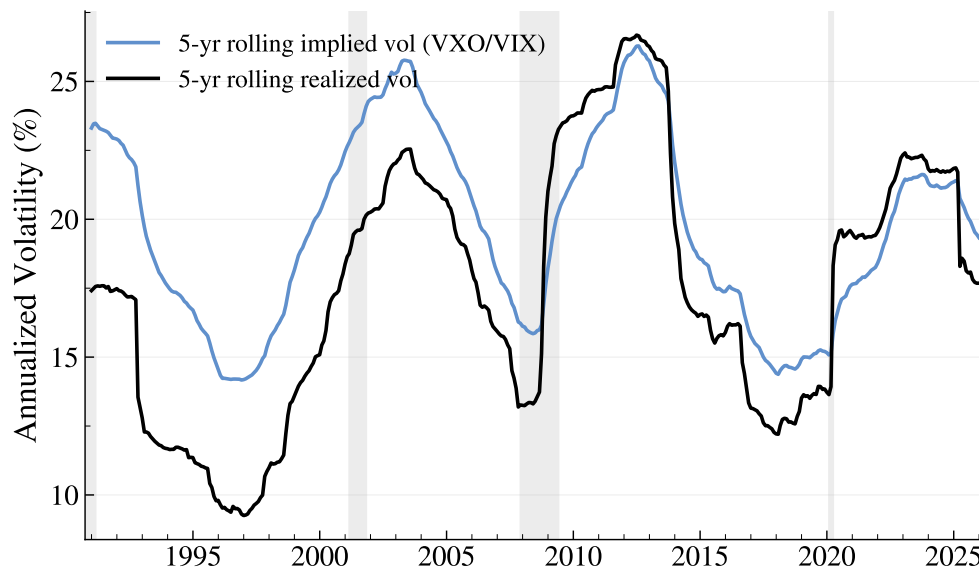


Figure 20: Implied and realized volatility, 1991–2025

Notes: Five-year rolling realized volatility and implied volatility (VXO spliced to VIX). Realized volatility trends up underneath a flat implied volatility; the gap between them – the variance risk premium – closes from roughly six points in the early 1990s to near zero by the late 2010s.

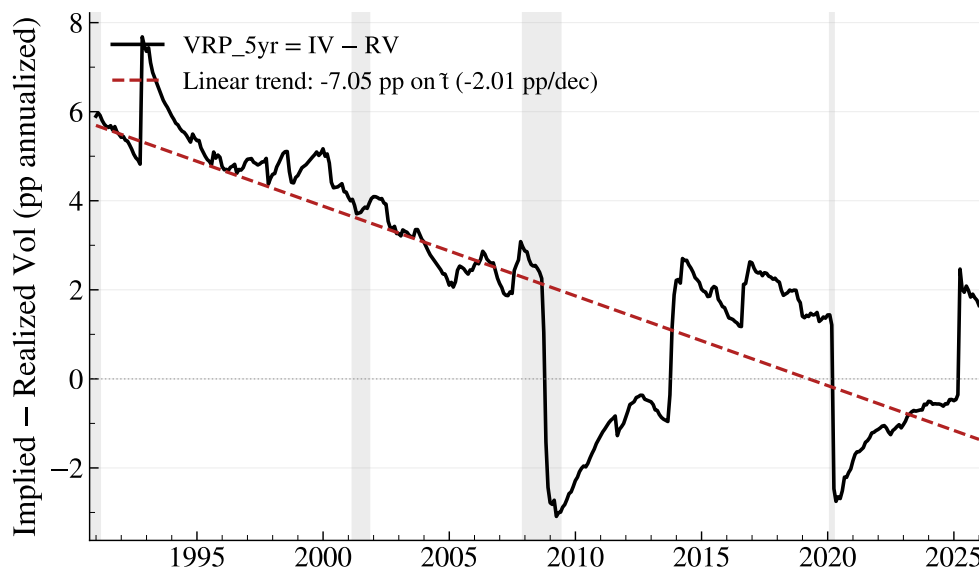


Figure 21: The variance risk premium, 1991–2025

Notes: Five-year rolling variance risk premium (implied minus realized volatility) with fitted linear trend. The premium falls from about six points in the early 1990s to near or below zero from 2018 onward, a decline of roughly two points per decade.

Table 9: Schwert (1989) regression, updated sample

Sample	N	σ_{IP}	σ_{CPI}	σ_{UNR}	σ_{RF}	R^2
Schwert original (1947–1987)	468	−0.29*	2.21***	0.00	0.13***	0.38
Updated full (1947–2024)	912	−0.93***	5.94**	0.03***	0.05	0.26
Pre-1985	432	−0.34**	2.39***	0.01	0.13***	0.46
Post-1985	480	1.58	6.63**	−0.01	0.11	0.39
Post-2000	300	1.30	6.08**	−0.01	0.26**	0.46

Notes: Joint regression of 12-month rolling annualized stock volatility on the 12-month rolling annualized volatilities of industrial-production growth, CPI inflation, unemployment-rate changes, and risk-free-rate changes, following Schwert (1989). Newey–West HAC (12 monthly lags). *, **, *** denote $p < 0.10, 0.05, 0.01$.

Table 10: Trends in realized volatility, implied volatility, and the variance risk premium

Series	Trend (pp on $\tilde{t}$)	Trend (pp/decade)	p -value	R^2
Realized vol (RV)	+6.44 (3.35)	+1.84	0.084	0.158
Implied vol (IV)	−0.61 (2.93)	−0.17	0.84	0.003
Variance risk premium (VRP)	−7.05 (0.86)	−2.01	9×10^{-6}	0.624

Notes: Trend regressions of five-year rolling realized volatility, implied volatility (VXO 1986–1989 spliced to VIX 1990+), and their difference (VRP = IV − RV) on a normalized linear trend, monthly, 1990–2025 ($N = 421$). Newey–West HAC standard errors (60 monthly lags) in parentheses; p -values use fixed- b critical values matching Table 1.

Depression had higher volatility than today, but it came with extraordinarily volatile fundamentals; the modern rise does not.

The level of volatility is not the only moment that breaks. The variance ratio – the ratio of daily-implied to monthly-implied variance, a measure of short-horizon mean reversion – sat near 0.74 from 1931 through the mid-1990s, identical in the high-volatility Depression and the quiet postwar period despite volatility levels differing by a factor of two and a half. It then doubles. A break test on the variance ratio places a single significant break at March 2000 (Figures 24 and 25), the same date as the modern break in the level of volatility. Two independent moments of returns shift at the same moment – harder to attribute to fundamentals than to a change in how the market trades.

A.5 Stability of the trend across samples and rolling windows

The trend coefficient in Table 1 is robust to the choice of sample window and rolling-window length. Figure 26 plots the trend coefficient on $\tilde{t}$ estimated separately for

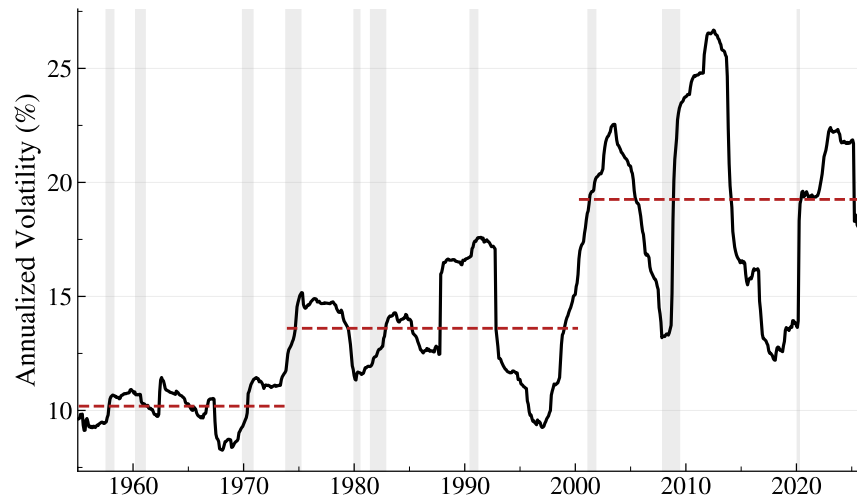


Figure 22: Structural Breaks in 5-Year Rolling Market Volatility

Notes: Annualized realized volatility of the Fama–French market excess return (same series as Figure 1). Dashed horizontal lines show estimated segment means from a Bai–Perron test for two structural breaks in the mean. The optimal break dates are October 1973 and March 2000, yielding three regimes with means of 10.2%, 13.6%, and 19.3%. Shaded regions indicate NBER recessions. Sample: 1955–2025.

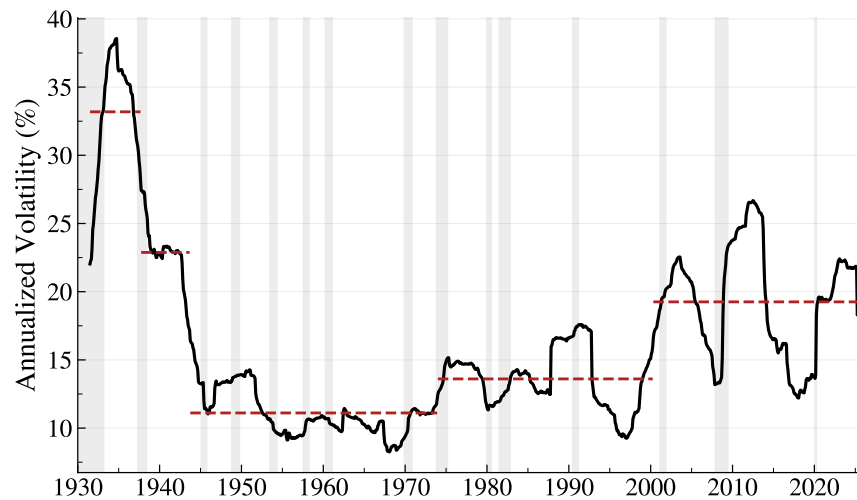


Figure 23: Structural Breaks in Volatility over the Long Sample, 1931–2025

Notes: Five-year rolling annualized volatility of the Fama–French market excess return, 1931–2025. Dashed horizontal lines show segment means from a Bai–Perron test with four breaks, dated August 1937, August 1943, November 1973, and March 2000, with regime means of 33.2%, 22.9%, 11.1%, 13.6%, and 19.3%. Shaded regions indicate NBER recessions.

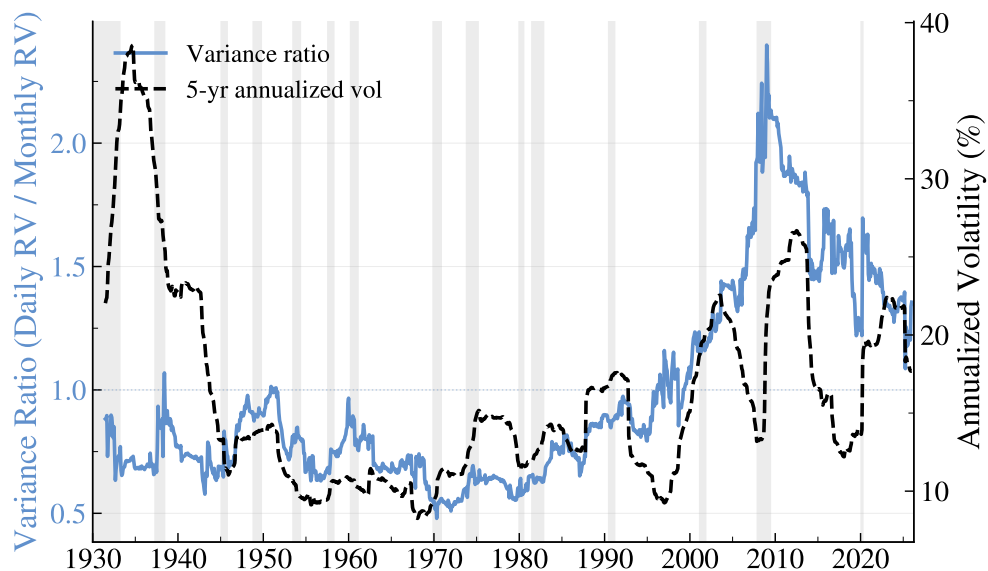


Figure 24: Variance ratio and volatility over the long sample, 1931–2025

Notes: Five-year rolling variance ratio (daily-implied over monthly-implied annualized variance, left axis) and annualized volatility (right axis), 1931–2025. The variance ratio is near 0.74 in both the Depression and the postwar period despite very different volatility levels, and rises only after the mid-1990s.

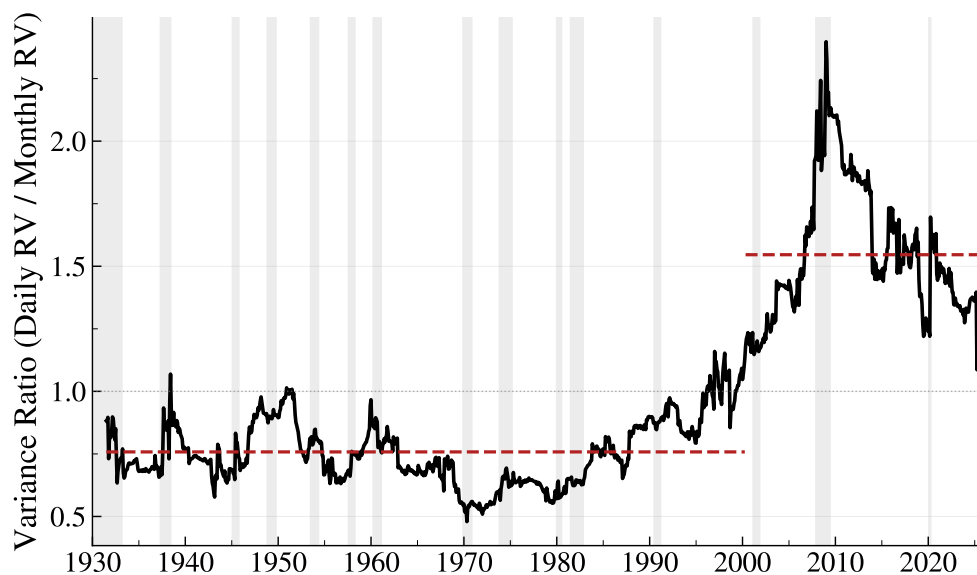


Figure 25: Structural break in the variance ratio, 1931–2025

Notes: Five-year rolling variance ratio with a single Bai–Perron break, dated March 2000. The segment means are 0.76 before and 1.55 after, doubling at the same date as the modern break in the level of volatility.

every start year from 1931 to 2010 (sample ends in 2025), with 90% fixed- b confidence intervals. The trend is positive at every post-WWII start year. For the daily column, the 90% confidence interval excludes zero for any start year from approximately 1945 through 1993; for the weekly column, through 1992. Start dates pre-1945 absorb Depression-era volatility, which pulls the trend toward zero or negative. Start dates after the mid-1990s cut off the rise itself, so the trend in a now-elevated regime is mechanically smaller and less precise.

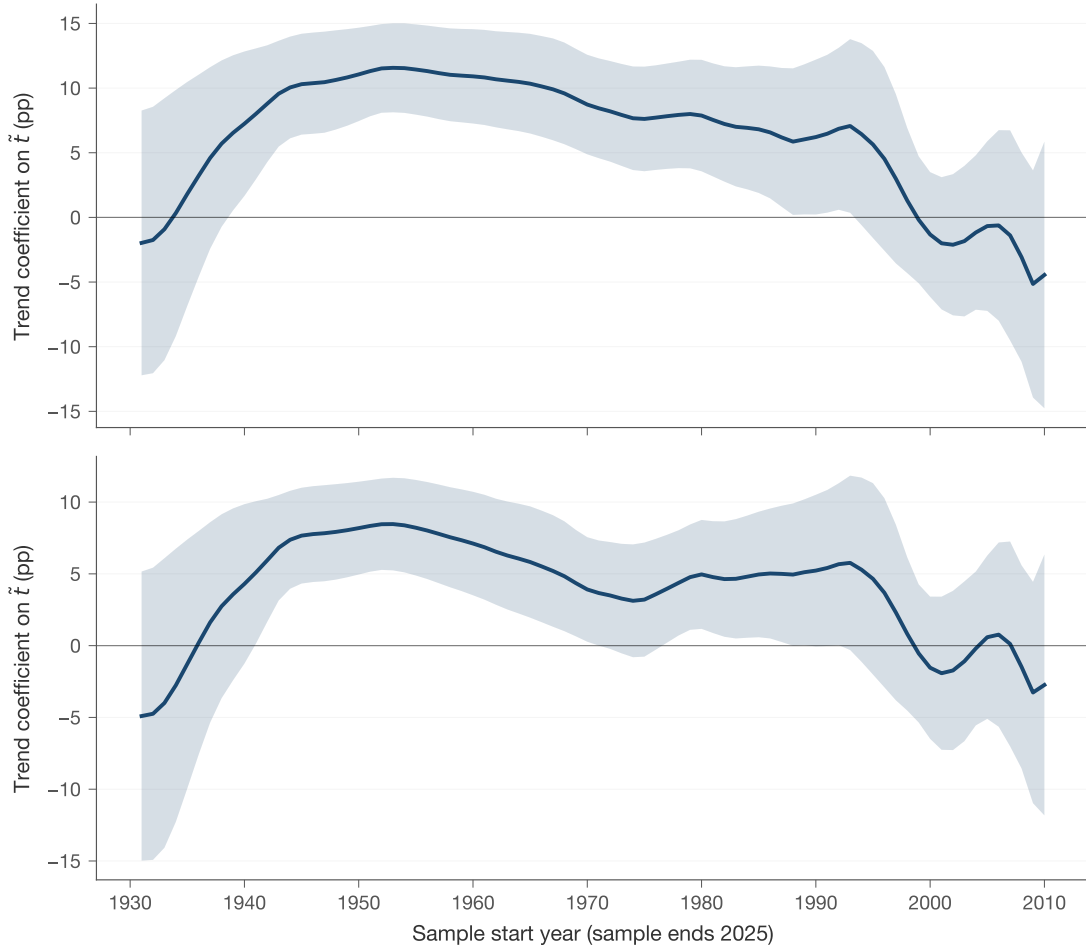


Figure 26: Trend coefficient by sample start year (sample ends 2025)

Notes: Each point is the trend coefficient b from the regression $\sigma_t = a + b\tilde{t} + \varepsilon_t$ estimated over $[\text{start year}, 2025]$, where σ_t is the 5-year rolling annualized volatility from daily (top panel) or weekly (bottom panel) Fama–French market excess returns, and $\tilde{t} \in [0, 1]$ over the sample window. The shaded band is the 90% confidence interval under Newey–West HAC with Bartlett kernel and $S = 260$ weekly lags, fixed- b critical values with $\kappa = \lfloor 3/(2b) \rfloor$ degrees of freedom.

Figure 27 plots the trend coefficient by rolling-window length, from 1 month (4 weeks) up to 10 years (520 weeks), on the 1955–2025 sample. The trend increases

monotonically with window length and asymptotes around 7 years; the paper’s 5-year window sits in the middle of this plateau. Even at 1-month windows the daily trend is +8.5 pp and the weekly trend is +5.6 pp, both with 90% confidence intervals excluding zero. The 5-year window choice is not a smoothing artifact.

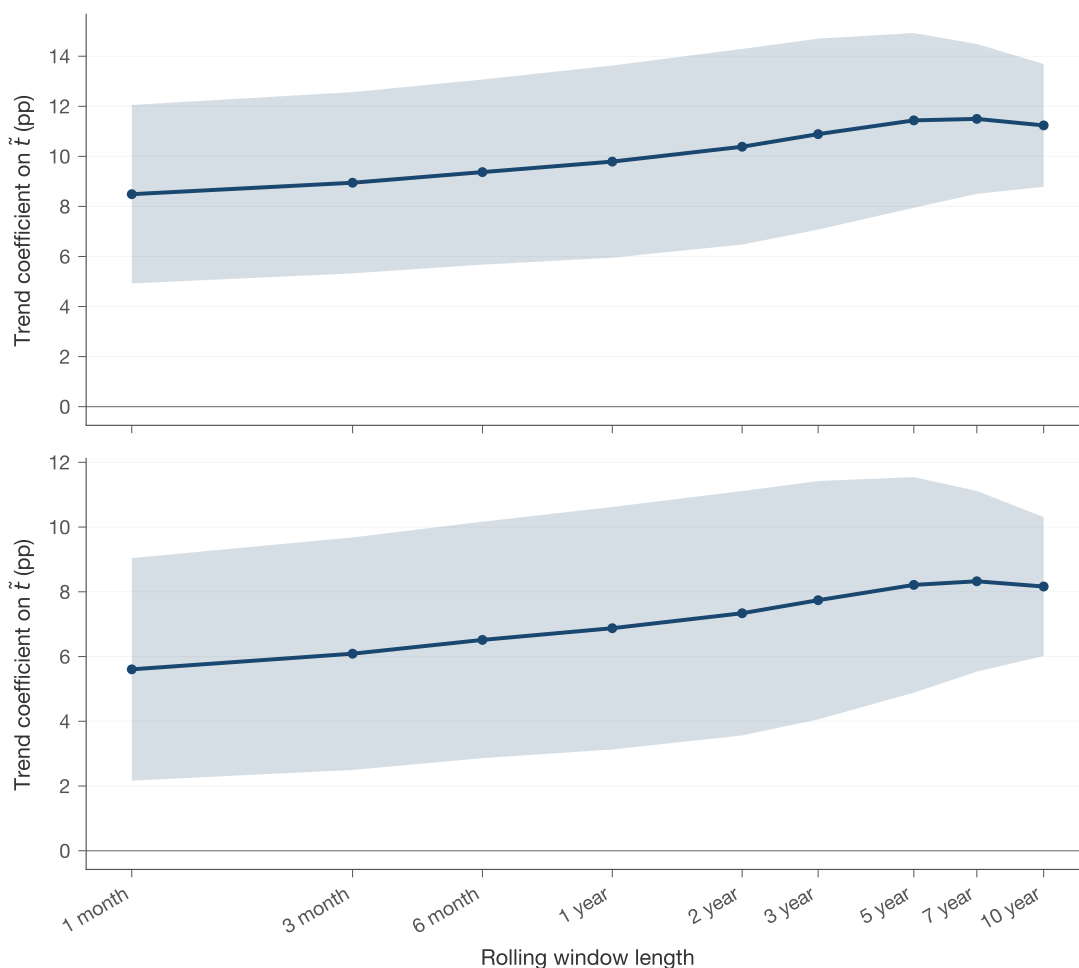


Figure 27: Trend coefficient by rolling-window length

Notes: Each point is the trend coefficient b from the regression $\sigma_t = a + b\tilde{t} + \varepsilon_t$ on the 1955–2025 sample, with σ_t constructed as the rolling annualized volatility of daily (top panel) or weekly (bottom panel) Fama–French market excess returns over the window indicated on the x-axis. The shaded band is the 90% confidence interval under Newey–West HAC with Bartlett kernel and $S = 260$ weekly lags, fixed- b critical values. X-axis is on a log scale.

A third cut holds the trend stable. Estimating the trend on rolling 30-year windows, every window ending after 1980 has a positive coefficient, and the windows ending in the 2010s are the steepest in the sample at about +2.8 pp/decade – above the +1.6 pp/decade full-sample headline (Figure 28). Windows ending in the 1960s are negative because they still contain the Depression-to-postwar decline. If anything,

the linear-trend summary understates the recent steepening.

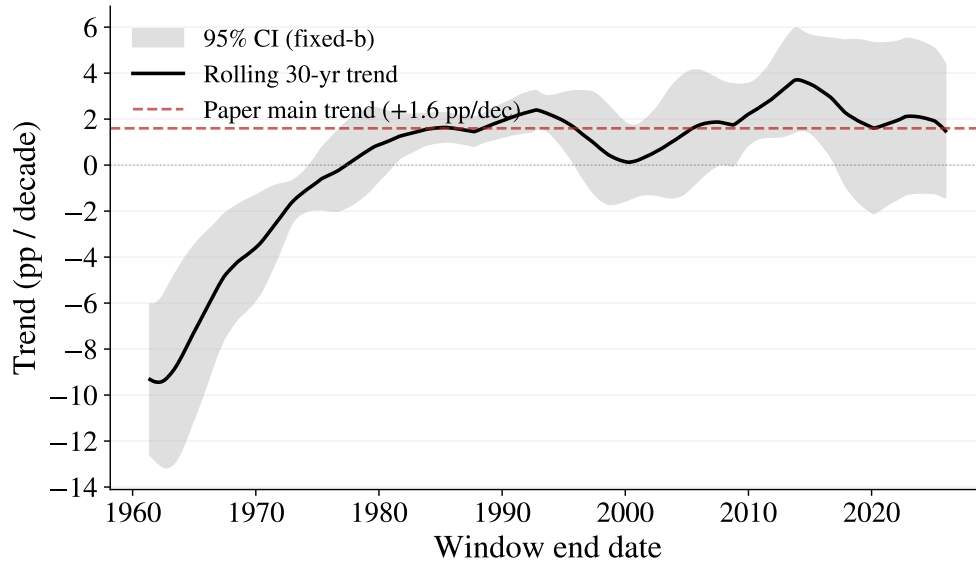


Figure 28: Trend coefficient on rolling 30-year windows

Notes: Trend coefficient (pp/decade) from estimating $\sigma_t = a + b\tilde{t} + \varepsilon_t$ on each rolling 30-year window, dated at window end, with 95% fixed- b confidence band. The dashed line marks the +1.6 pp/decade full-sample estimate from Table 1. Windows ending in the 1960s are negative because they still contain the Depression-to-postwar decline.

A.6 The Rise Is Not a Weighting or Concentration Artifact

The rise is broad-based across the size distribution, not a product of value-weighting or a handful of mega-caps. Table 11 reports trend coefficients for alternative market constructions. Equal-weighting the market gives a steeper trend than value-weighting (+2.16 versus +1.61 pp/decade), and decomposing by size, the top-decile trend (+1.44) sits below the full-market trend while the mid- and small-cap segments run steeper (+2.18 and +2.35). If a few large stocks drove the result, the largest stocks would show the steepest trend; they show the shallowest. Measuring volatility from the S&P 500 index directly – the index actually traded through futures and ETFs – gives +1.54 pp/decade against +1.71 for the value-weighted market over the matched 1967–2024 sample, with a correlation of 0.98.

A.7 The Variance-Ratio Rise Is Global

The variance-ratio rise is not specific to the United States. Building daily value-weighted market returns for 35 countries from Compustat Global firm-day data, the five-year variance ratio rises in all 35, with 29 significant at the 5% level under fixed- b inference and a mean trend of +0.189 per decade (Table 12). Every major developed

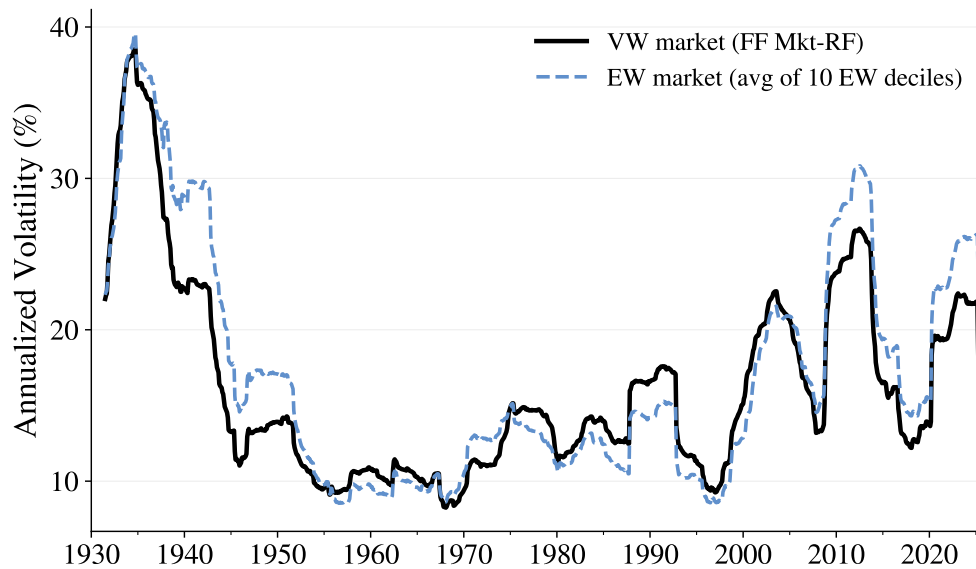


Figure 29: Equal-weighted and value-weighted market volatility, 1932–2025

Notes: Five-year rolling annualized volatility of the value-weighted and equal-weighted market, 1932–2025. The two track closely; the equal-weighted series runs slightly higher and trends slightly steeper in the modern era.

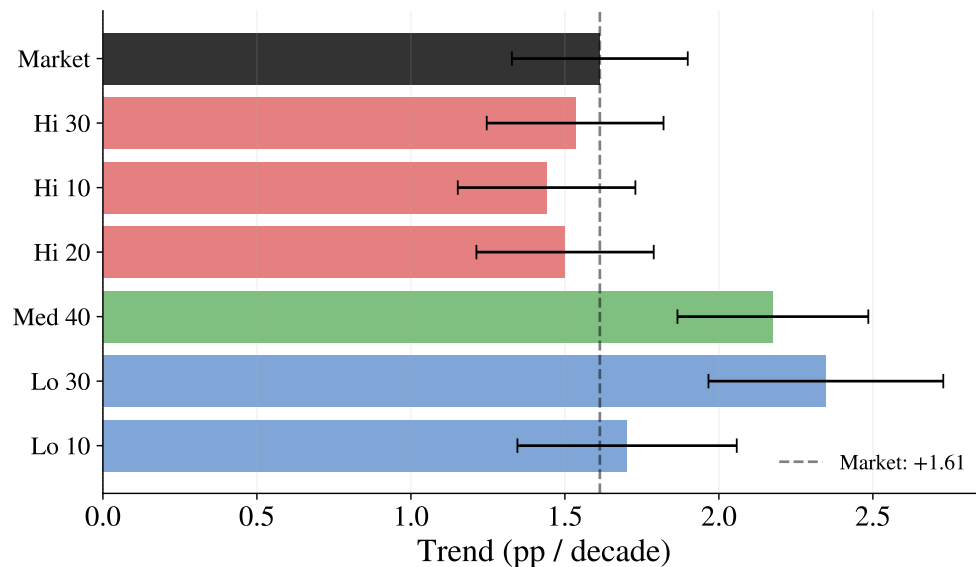


Figure 30: Volatility trend by size segment

Notes: Trend coefficient (pp/decade) of five-year rolling volatility by capitalization segment, 1955–2025, with 95% confidence intervals. The dashed line marks the full-market trend. The largest segments sit at or below the market trend; mid- and small-cap segments sit above it.

Table 11: Volatility trend by market construction and size segment

Construction	Trend (pp/decade)
Value-weighted market	+1.61***
Equal-weighted market	+2.16***
Top decile	+1.44***
Mid 40%	+2.18***
Bottom 30%	+2.35***

Notes: Trend coefficient (pp/decade) from regressing five-year rolling annualized volatility on a normalized linear trend, 1955–2025. The value-weighted market is the Fama–French market factor; the equal-weighted market averages Ken French equal-weighted size deciles; size segments are Ken French capitalization portfolios. Newey–West HAC (60 monthly lags). *** denotes $p < 0.01$ under fixed- b critical values.

market is significantly positive. Volatility *levels* are mixed across countries – emerging markets enter the sample at their 1990s crisis highs and decline through it – but the variance ratio, harder to pin on country-specific fundamentals, rises everywhere. A common, daily-frequency channel of the kind index trading produces fits this pattern; a fundamentals story does not.

Table 12: Variance-ratio trends across countries

	Variance-ratio trend
Countries with a positive trend	35 of 35
Significant at 5% (fixed- b)	29 of 35
Mean across countries (per decade)	+0.189
United States	+0.10***
United Kingdom	+0.22***
France	+0.29**
Germany	+0.15**
Japan	+0.29***

Notes: Five-year rolling variance ratio (daily-implied over monthly-implied annualized variance) for daily value-weighted market returns built from Compustat Global firm-day data, 35 countries, samples of 17–35 years (95 years for the US). Trends are the coefficient on a normalized linear time index, per decade, with Newey–West HAC (60 monthly lags) and fixed- b critical values. **, *** denote $p < 0.05, 0.01$.

A.8 Return Autocorrelations and Volume

The horizon decay in Table 1 is mechanically a statement about daily and weekly return autocorrelation. We show the underlying series and link them to volume, fol-

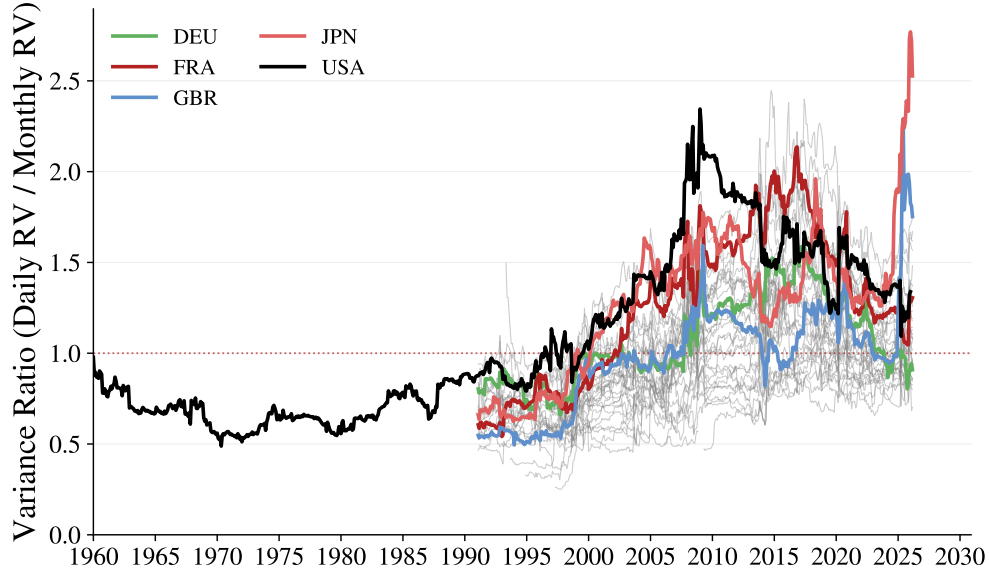


Figure 31: Variance ratio over time, 35 countries

Notes: Five-year rolling variance ratio for 35 countries built from Compustat Global firm-day data; the United States (longest sample) in black. By 2010 nearly every country has a variance ratio above one.

lowing Campbell et al. (1993) (henceforth CGW).

Figures 33a and 33b plot the rolling 5-year first-order autocorrelation of daily and weekly Fama–French market excess returns. Each window estimates $r_{t+1} = \alpha + \beta r_t + \varepsilon$ over the prior 60 calendar months; the shaded band shows ± 1.96 heteroskedasticity-consistent (HC1) standard errors. Daily autocorrelation peaks at roughly $+0.30$ in the mid-1970s, declines steadily, and crosses zero around 2003. The weekly series crosses zero earlier (around 1990) and settles near -0.10 . The linear time-trend coefficient on $\tilde{t} \in [0, 1]$ over 1955–2025 is -0.39 (NW HAC standard error 0.08) for the daily series and -0.30 (0.03) for the weekly, with fixed- b p -values below 10^{-3} in both cases.

CGW’s own decade-by-decade evidence (their Table VI) already showed that the daily index autocorrelation moved substantially across regimes: 0.015 in the 1930s, 0.11 in the 1940s, 0.13 in the 1950s, rising to 0.28 in 1962–1974 and declining to 0.17 in 1975–1987. The series here extends their evidence by nearly four decades; the trajectory continues, with the autocorrelation turning negative in the 2000s and reaching -0.16 in the 2020s. Replicating their original Sample B (1962–1974), we recover the same first AC of 0.28 that they reported.

Table 13 runs the CGW Table II regression

$$r_{t+1} = \alpha + \beta r_t + \gamma (V_t \cdot r_t) + \sum_{d=1}^4 \delta_d (D_d \cdot r_t) + \varepsilon_{t+1}, \quad (42)$$

where V_t is log aggregate market turnover (raw, or detrended by a backward 1-yr or 5-yr moving average as in CGW), and D_d are day-of-week dummies interacted with r_t . Standard errors are heteroskedasticity-consistent (HC1). The CGW prediction $\gamma < 0$

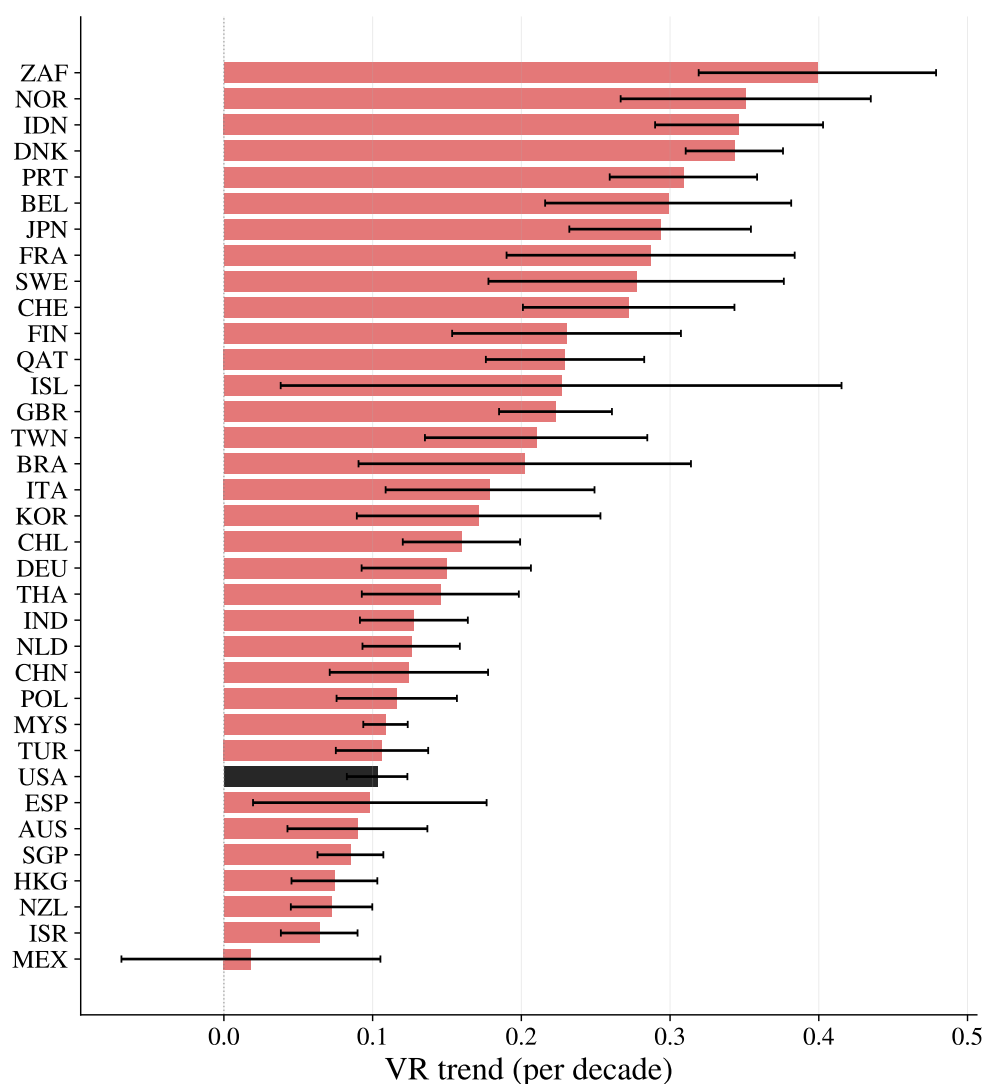
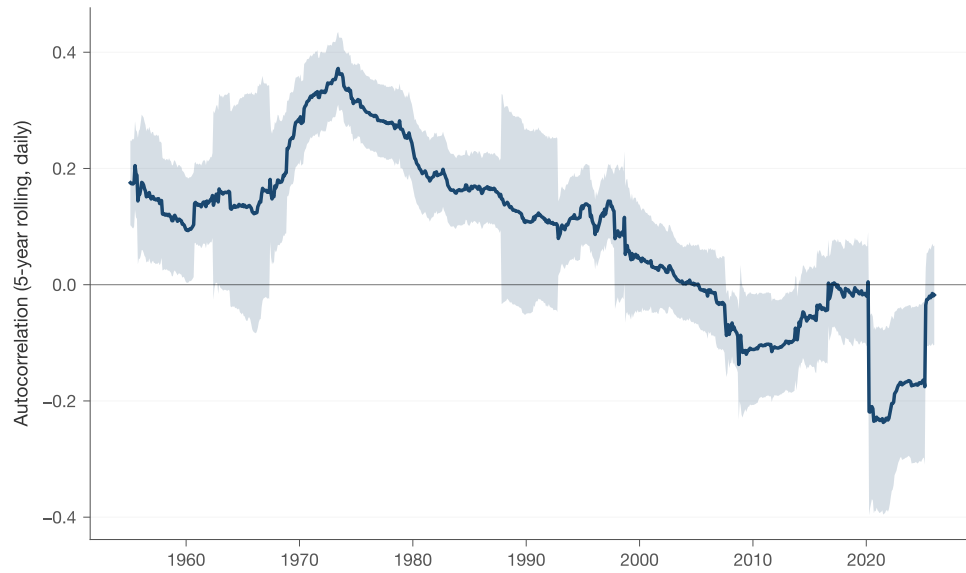


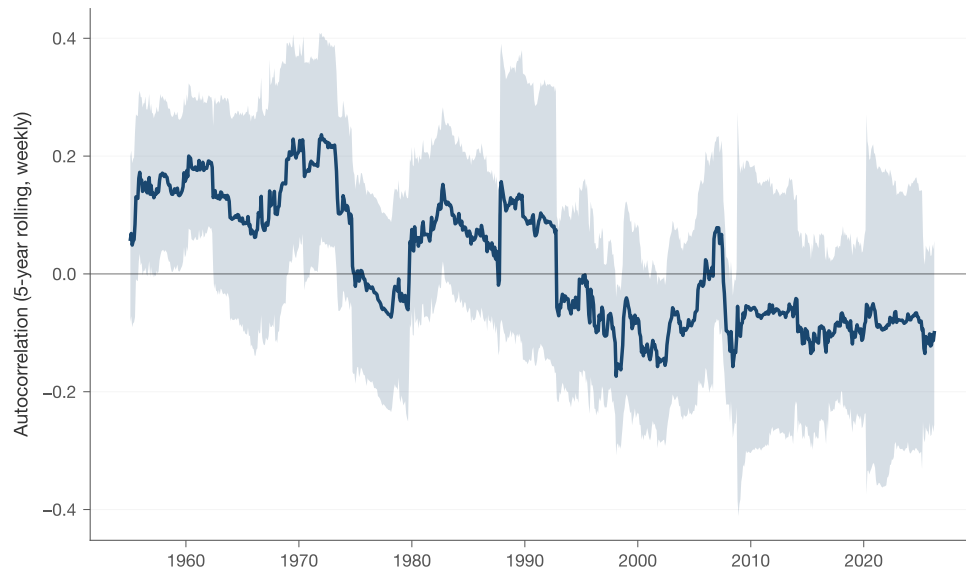
Figure 32: Variance-ratio trend coefficients by country

Notes: Per-country trend coefficient of the five-year rolling variance ratio (per decade), with Newey–West HAC standard-error bars. All 35 bars are positive; the United States sits in the middle of the distribution.

(high-volume returns more strongly reversed) holds on the full 1955–2025 sample under all three V_t specifications. On the CGW samples our estimates match theirs closely (their γ on Sample A was -0.328 with their detrended measure; ours is -0.349). On the post-CGW 1988–2025 sample γ remains strongly negative (-0.36 , $t = -3.2$ for the 1-yr detrended specification), so the volume-reversal mechanism continued operating after CGW's sample. The 1980–2000 sub-period attenuates toward zero – the period when the autocorrelation itself was near zero, leaving little to interact with volume – and the post-2000 sub-period returns to strongly negative estimates as the autocorrelation moves into negative territory.



(a) Daily returns



(b) Weekly returns

Figure 33: First-order return autocorrelation of the market, 5-year rolling

Notes: First-order autocorrelation of Fama–French market excess returns, estimated as the OLS slope β in $r_{t+1} = \alpha + \beta r_t + \varepsilon$ over each 5-year rolling window (60 calendar months). The shaded band is ± 1.96 HC1 standard errors estimated within each window. Panel (a) uses daily returns; panel (b) uses weekly returns. Sample: 1955–2025 (rolling window so first estimate uses data from 1950).

Table 13: Return autocorrelation and volume

Notes: Estimates of γ from Equation (42): $r_{t+1} = \alpha + \beta r_t + \gamma(V_t \cdot r_t) + \sum_d \delta_d(D_d \cdot r_t) + \varepsilon_{t+1}$, where V_t is log aggregate market turnover constructed from CRSP daily data following the construction underlying Figure 4. Detrended versions subtract a backward moving average from $\log V_t$. Standard errors in parentheses are heteroskedasticity-consistent (HC1). Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ (two-sided, normal asymptotics). Day-of-week interactions with r_t included.

Sample	Raw $\log V_t$	$\log V_t$ – 1-yr MA (CGW)	$\log V_t$ – 5-yr MA	N
Full (1955-2025)	–0.131*** (0.015)	–0.289*** (0.089)	–0.226*** (0.065)	17,619
CGW Sample A (1962-87)	–0.114*** (0.024)	–0.349*** (0.063)	–0.299*** (0.049)	6,344
CGW Sample B (1962-74)	–0.171** (0.076)	–0.452*** (0.112)	–0.294*** (0.079)	3,122
CGW Sample C (1975-87)	–0.104*** (0.035)	–0.226*** (0.077)	–0.276*** (0.076)	3,221
1988-2025 (post-CGW)	–0.197*** (0.049)	–0.361*** (0.112)	–0.257*** (0.082)	9,321
Pre-1980	–0.134*** (0.047)	–0.356*** (0.094)	–0.304*** (0.073)	6,274
1980-2000	–0.074 (0.065)	–0.043 (0.110)	–0.013 (0.100)	5,055
2000-2025	–0.229*** (0.083)	–0.354*** (0.123)	–0.235*** (0.088)	6,288

Figure 34 plots γ estimated on rolling 10-year windows. The interaction has been negative throughout, deepest in the 1970s and in the post-2008 / COVID period, and remains strongly negative at the end of the sample.

A.9 NYSE Trading Hours Changes

Table 3 exploits changes in NYSE trading schedules as instruments for trading volume. Table 14 lists all schedule changes used in the analysis. Before June 1952, the NYSE operated six days per week, with a shortened Saturday session (10:00 AM–12:00 PM). All subsequent changes affected weekday hours only, adjusting the closing time while keeping the open at 10:00 AM (until 1985, when it moved to 9:30 AM). The 1968 Wednesday closure temporarily reduced the trading week to four days.

Both specifications use all 9 events. The daily specification (Panel A) exploits the distinct within-week predictions of each schedule change; the weekly specification (Panel B) aggregates to calendar weeks.

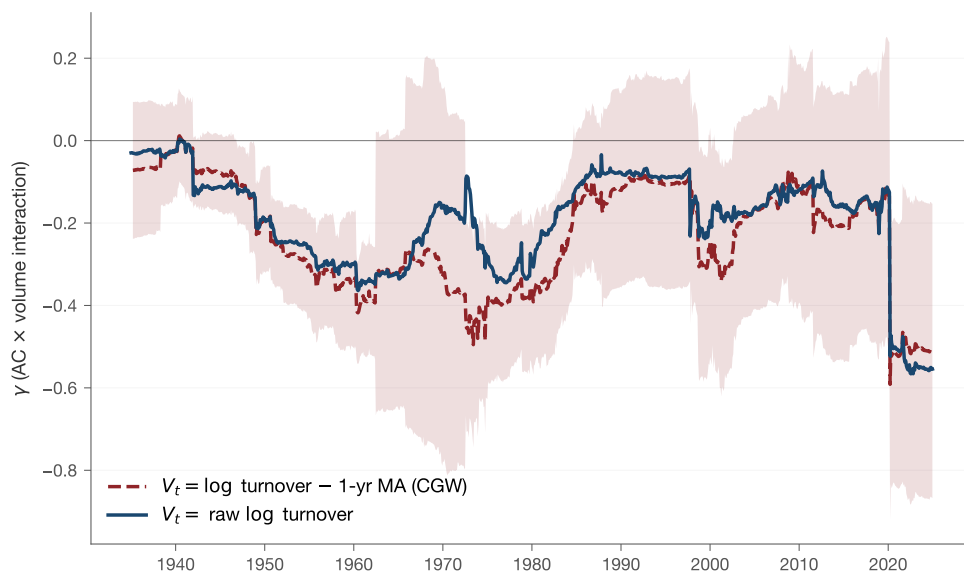


Figure 34: Rolling AC-volume interaction γ

Notes: Estimate of γ from Equation (42) on rolling 10-year windows, stepped monthly. Shaded band shows ± 1.96 HC1 standard errors for the detrended specification. Solid line uses raw log V_t ; dashed line uses CGW's 1-year backward MA detrending.

A.10 S&P 500 Futures Trading Hours

Several tests in this paper require knowing when S&P 500 index futures could be traded, particularly outside regular equity-market hours. Inferring the schedule from the coverage of a data vendor's 5-minute bar files dates the overnight session to September 1997 and holiday sessions to 2004; both are the vendor's coverage boundaries rather than exchange events: the big contract was listed on GLOBEX for overnight trading in September 1993, and CME's archived calendars show holiday sessions continuously from late 1998. Table 15 instead gives the exchange's own record, compiled from CME rule filings and notices, Federal Register publications, and contemporaneous accounts, with every statement labeled by its evidence status; the full sourced timeline accompanies the replication files. The four events in the table are the ones pooled in Table 4, column (3), and estimated one at a time in Table 16. Two later schedule changes are used in Section 3.1.4: the September 29, 2003 relocation of the nightly one-hour shutdown from 22:00–23:00 to 16:30–17:00 Central, and the 2014 move of the holiday-session halt from 10:30 to noon Central. Both are documented in the same record and both are visible in the bar data as a change in where prints stop.

A.11 Robustness of the 1993 and 1997 Events

Table 16 reports each of the four CME schedule changes on its own, and Table 17 re-estimates the two large events under a single change at a time, with both events run

Table 14: NYSE Trading Hours Schedule Changes

Notes: All times Eastern. Before June 1952, the NYSE was open six days per week (Monday–Saturday), with Saturday sessions from 10:00 AM to 12:00 PM. The “Daily Hours” column refers to weekday hours; Saturday hours were always 2.0 when applicable. “Weekly Hours” accounts for both the number of trading days and Saturday’s shorter session.

#	Effective Date	Description	Schedule	Daily Hours	Weekly Hours
	<i>Pre-change</i>	<i>6-day week, Mon–Sat</i>	<i>10:00–3:00 (Sat 10–12)</i>	<i>5.0</i>	<i>27.0</i>
1	Jun 1952	Saturday trading eliminated	10:00–3:00, Mon–Fri	5.0	25.0
2	Sep 1952	Closing time extended 30 min	10:00–3:30, Mon–Fri	5.5	27.5
	<i>Stable: Sep 1952 – Jun 1968</i>				
3	Jun 12, 1968	Wednesday closure (paperwork)	10:00–3:30, 4 days/wk	5.5	22.0
4	Jan 2, 1969	Wed reopens; hours shortened	10:00–2:00, Mon–Fri	4.0	20.0
5	Jul 3, 1969	Closing extended 30 min	10:00–2:30, Mon–Fri	4.5	22.5
6	Sep 26, 1969	Closing extended 30 min	10:00–3:00, Mon–Fri	5.0	25.0
7	May 1, 1970	Closing extended 30 min	10:00–3:30, Mon–Fri	5.5	27.5
8	Oct 1, 1974	Closing extended 30 min	10:00–4:00, Mon–Fri	6.0	30.0
9	Jan 2, 1985	Open moved 30 min earlier	9:30–4:00, Mon–Fri	6.5	32.5

through identical code.

Three things stand out. The Newey–West lag choice matters a great deal for the 1993 event, whose t -statistic runs from 7.1 with no correction to 2.2–3.0 with 22 to 132 lags; we report 44 lags, twice the rolling window, throughout. The 1993 estimate is stable across windows from ± 3 to ± 24 months, while the E-mini estimate is confined to ± 6 months and is zero at ± 12 . And the placebo grid is clean for 1993 but does not discriminate for the E-mini, which returns a significant coefficient of either sign at placebo dates one year on either side of the launch. Two probes not in the table cut the other way for 1993. On a quarterly grid of placebo dates over 1993–96, a no-event date in January 1996 returns 0.60 ($t = 4.3$), larger than the event itself, so the ramp specification is not clean in this era. And the 1993 estimate rests on the tail of large overnight moves: dropping the k largest overnight market moves inside the window takes the coefficient to 0.34 at $k = 3$, 0.25 at $k = 5$, and 0.21 ($t = 1.0$) at $k = 10$, while dropping the k most influential days by leverage makes it stronger and the E-mini estimate is unchanged by the same probe (0.38, $t = 4.8$ at $k = 10$). Whether tail concentration is thin identification or the mechanism itself is not something the ± 6 -month design can settle.

Table 15: S&P 500 Futures Trading Hours Outside NYSE Regular Hours

Notes: Hours available for trading S&P 500 index futures outside NYSE regular trading hours (9:30am–4:00pm ET), per weekday, taking the maximum availability across the pit tail, the big-contract (SP) GLOBEX session, and the E-mini (ES), net of documented halts and maintenance windows. Times are Central. Every row is sourced line-by-line in `sp_futures_trading_hours_timeline.md`, which labels each statement DOCUMENTED, CONFLICT, INFERENCE, or NOT ESTABLISHED. The 1997 E-mini row is an *access* event rather than an hours event: its change in outside-RTH hours is at most half an hour and is not established, while its documented innovation—simultaneous electronic and pit trading *during* regular hours, with a 30-contract limit on the electronic book—is invisible to a measure defined outside those hours.

Event	Date	Before (hours)	After (hours)	Δ (hours)
SP listed on GLOBEX <i>Month double-attested; day single-source</i>	Sep 20, 1993	0.25	16.50	+16.25
E-mini launch <i>Launch-day clock not established; $\Delta \in [0, 0.5]$</i>	Sep 9, 1997	16.75	17.25	+0.50
Globex session expansion <i>Documented</i>	Sep 29, 2003	16.00	16.75	+0.75
Maintenance window <i>Documented</i>	Nov 19, 2012	16.75	16.50	−0.25

A.12 Index Trading Against a Shut Market: Estimators and Sample Cuts

Table 18 reports every column of Table 5 under each of the estimators we considered, together with the same 15-minute bucket panel run on all weekdays rather than on holidays alone, and Table 19 re-estimates column (1) on alternative samples. The headline specification is ordinary least squares on variance per hour in levels, for two reasons. A shut window’s variance is one squared increment, so a log of it is the log of a single draw: its expectation sits about 1.27 log points below the log of its expectation, and a log-ratio regression that does not correct for this overstates the contrast by a factor of three or more. And a window whose price is unchanged across the closure has zero variance and no log at all; ten of the 58 dead holiday afternoons and 47 of the 124 shut windows in the 2003 sample are of this kind. Dropping them removes the quietest shut windows and biases the ratio down; flooring them biases it up by an amount set by the floor. Levels need neither a correction nor a zero rule. Poisson pseudo-maximum likelihood, which estimates the same multiplicative model without taking logs and keeps every zero, gives nearly the same answer as levels in every column.

The construction of a shut window’s variance is the one choice on which the numbers depend. The first print after a reopen is stale about half the time, and the post-halt price is discovered inside the first five minutes of trading; measuring the shut window to the first print therefore undercounts it, and charging it k live minutes in full caps the

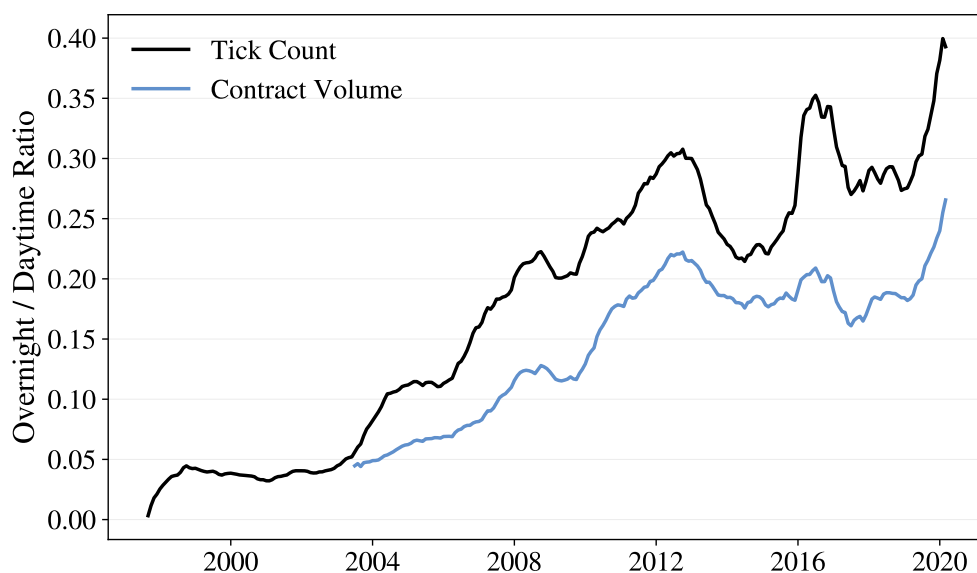


Figure 35: Overnight-to-Daytime Trading Activity in S&P 500 Futures

Notes: Ratio of overnight to daytime trading activity for S&P 500 E-mini futures, measured by tick count (black) and contract volume (blue). Overnight is defined as trading outside regular NYSE hours (9:30am–4:00pm ET). Both series are smoothed using rolling averages. Tick-count data begin in 1997; contract volume data begin in 2004. CME changed how trades are bundled into reported prints in 2011 and again in 2015, which affects the tick-count series but not volume; the volume series is the cleaner measure where the two are both available. Sample: 1997–2020.

open-to-shut ratio at $(\text{shut} + k)/k$. For the 2003 windows the ratio is $\times 12$ to the first print, $\times 2.4$ to five minutes, and $\times 1.2$ to fifteen minutes, the last close to its cap; for the six-hour holiday afternoon the same choice moves the ratio from $\times 56$ measured to the first print to $\times 13$ under the five-minute rule, because the reopen's first minutes are a small share of a long closure. Every table and figure in the paper uses the five-minute rule. The sample cuts in Table 19 show where the day-level number comes from: once Labor Day 2019 is out no single remaining holiday moves it much (dropping MLK 2008, the largest day in the dead cell, gives $\times 11.0$), it falls to $\times 3.8$ only if unscheduled closures are added back, and that fall is entirely December 5, 2018, a national day of mourning on which the Huawei arrest broke during the closure.

Table 16: Index Futures Trading Hours, Access, and Overnight Market Variance

Notes: The dependent variable is the 22-day rolling log ratio of overnight to intraday market variance, controlling for the corresponding idiosyncratic log variance ratio. The market is the value-weighted top 500 stocks by prior-December market capitalization; overnight and intraday returns use the CRSP-total-return-anchored construction, correct through splits and dividends; idiosyncratic returns impose a unit market beta following Campbell et al. (2001). The event ramp equals 0 before the event, rises linearly to 1 over the next 22 trading days, and equals 1 thereafter. The log futures volume row re-runs the same event-ramp regression with log S&P 500 futures volume (22-day rolling average) as the dependent variable. Windows are ± 6 months; Newey-West standard errors with 44 lags in parentheses. Column (5) pools all four events with event fixed effects, entering the three hours events through a linear $\text{Post} \times \Delta\text{Hours}$ dose and the E-mini through a separate access indicator, so that the hours channel is not asked to absorb an access event whose hours change is at most half an hour. Each event window keeps its own coefficient on the idiosyncratic control, so the access indicator in column (5) is the same object as the event ramp in column (2), re-estimated jointly, and reproduces it exactly. Because the dependent variable is a 22-day rolling average, consecutive observations share 21 of 22 days; the Newey-West correction is what handles that overlap, and Table 17 reports the lag sensitivity. In column (5) the Newey-West sum is computed within event blocks, so no autocovariance is taken across the multi-year gaps between events. Hours are measured in levels rather than logs: the two differ by whether the pre-1993 base is the 6.5-hour regular session or the 15-minute pit tail, and the level specification is invariant to that choice. Sample 1993–2024; the panel cannot begin earlier because CRSP opening prices are unavailable before mid-1992. ***, **, * denote significance at the 1%, 5%, and 10% level.

	(1) GLOBEX listing Sep 1993	(2) E-mini launch Sep 1997	(3) Globex expansion Sep 2003	(4) Maintenance Nov 2012	(5) Pooled
Channel	hours	<i>access</i>	hours	hours	
Δ Overnight hours	+16.25	+0.50	+0.75	−0.25	
Event ramp	0.478** (0.195)	0.430*** (0.058)	0.199 (0.154)	−0.118 (0.185)	
Log futures volume	0.169* (0.088)	0.597*** (0.054)	0.030 (0.058)	0.009 (0.097)	
$\text{Post} \times \Delta\text{Hours}$ (per hour)					0.0301** (0.0120)
E-mini access indicator (= col. 2)					0.430*** (0.058)
Log idio. var. ratio	0.887 (0.783)	0.819*** (0.222)	0.758** (0.355)	0.611* (0.320)	Y
Event FE					Y
Observations	253	252	252	249	1006
R-squared	0.15	0.46	0.14	0.14	0.44

Table 17: Robustness of the Two Large Events

Notes: Each row re-estimates the event-ramp specification of Table 16 under a single change to the baseline (top row). Both events are run through identical code. *t*-statistics in parentheses. The final row reports the range of the ramp coefficient across a leave-one-day-out jackknife in which each raw trading day in turn is removed and the rolling variances recomputed, so that no single day can be responsible for the estimate. The one-year-prior placebo is unavailable for the 1993 event because the panel begins in January 1993.

	GLOBEX listing Sep 1993	E-mini launch Sep 1997
<i>Baseline</i>	0.478** (2.46)	0.430*** (7.48)
Newey-West lags = 0	0.478*** (7.14)	0.430*** (10.55)
Newey-West lags = 22	0.478** (2.16)	0.430*** (4.81)
Newey-West lags = 44	0.478** (2.46)	0.430*** (7.48)
Newey-West lags = 88	0.478*** (2.93)	0.430*** (8.77)
Newey-West lags = 132	0.478*** (3.03)	0.430*** (10.03)
Window ± 3 months	0.823*** (4.13)	0.313*** (3.75)
Window ± 6 months	0.478** (2.46)	0.430*** (7.48)
Window ± 12 months	0.603*** (3.74)	-0.003 (-0.02)
Window ± 18 months	0.588*** (4.35)	-0.112 (-0.68)
Window ± 24 months	0.497*** (3.63)	-0.121 (-0.87)
Rolling window = 10 days	0.427** (2.10)	0.583*** (9.34)
Rolling window = 22 days	0.478** (2.46)	0.430*** (7.48)
Rolling window = 44 days	0.607*** (4.03)	0.245*** (2.76)
Rolling window = 66 days	0.547*** (2.64)	0.150** (2.05)
No idiosyncratic control	0.422** (1.97)	0.504*** (4.82)
Step indicator, not ramp	0.580*** (3.20)	0.386*** (5.35)
Placebo: event +1 year	-0.192 (-0.77)	0.410** (2.04)
Placebo: event +2 year	0.279 (1.49)	0.058 (0.32)
Leave-one-day-out range	[0.392, 0.625]	[0.344, 0.502]

Table 18: Index Trading Against a Shut Market: Estimators

Notes: Columns (1), (2), (4), and (7) are columns (1)–(4) of Table 5, same samples and construction; column (3) adds year-month fixed effects to column (2). Columns (5)–(6) run the 15-minute bucket panel of column (4) on all weekdays 2004–2019, with time-of-day fixed effects alone and with time-of-day and day fixed effects. There the dead time is mostly Friday evening and the weekend, set against the same clock hours on Monday through Thursday, so the comparison is the weekend comparison of French and Roll (1986) rerun on the index; it is reported here as a benchmark and not in the main table. Each cell is an implied ratio with the coefficient's t -statistic in parentheses. LEVELS, OLS is the specification in Table 5. Poisson PML estimates the same multiplicative model without taking logs and keeps every zero cell. The two LOG rows differ only in the zero rule (a window whose price is unchanged across the closure has no log): dropping biases the ratio down, a floor biases it up. The second block is the further increment from an ordinary business day, a benchmark rather than a clean effect (Section 3.1.4); it is not defined in the holidays-only panel or the 2003 design. Appendix A.12 discusses the estimators.

	Day level: 10:30–12:00 CT vs. same-day overnight			Day × 15-min buckets		2003 swap
	(1)	(2)	(3)	holidays	all weekdays	
<i>Index tradeable vs. nothing trading</i>						
LEVELS, OLS (Table 5)	×13.1 (7.2)	×12.6 (5.7)	×35.5 (8.3)	×11.3 (3.3)	×3.2 (12.7)	×4.0 (2.8)
LEVELS, OLS, overnight as a free control	×18.6 (5.9)	×18.1 (5.5)	×35.6 (8.3)	—	—	—
LEVELS, Poisson PML	×12.0 (5.9)	×12.4 (5.8)	×16.6 (6.3)	×13.2 (6.7)	×4.7 (10.8)	×2.4 (3.3)
LOG, zeros dropped	×4.9 (5.9)	×4.9 (5.8)	×5.7 (6.4)	×4.9 (7.7)	×7.9 (27.4)	×1.0 (−0.2)
LOG, zeros floored at tick ² /3	×8.0 (7.4)	×7.9 (7.4)	×9.5 (7.9)	×7.0 (8.7)	×8.8 (27.5)	×2.0 (5.0)
LOG, zeros dropped, overnight as a free control	×4.3 (5.6)	×4.3 (5.4)	×5.3 (6.5)	—	—	—
<i>Cash market open, on top of the index (benchmark)</i>						
LEVELS, OLS	×7.0 (7.6)	×6.5 (6.9)	×3.8 (4.2)	—	×4.5 (19.2)	×4.0 (17.2)
LEVELS, Poisson PML	×3.5 (3.8)	×3.3 (3.6)	×2.8 (3.4)	—	×7.2 (11.3)	×3.9 (9.8)
LOG, zeros dropped	×3.6 (8.2)	×3.3 (7.5)	×2.9 (6.7)	—	×4.8 (11.1)	×3.7 (20.9)
Same-day overnight subtracted	Y	Y	Y			
Day-of-week FE		Y	Y			
Year-month FE			Y			
Time-of-day FE				Y	Y	Y
Day FE				Y		Y
Window FE						
Cells	58/33/4,188 days	same	same	8,736	411,236	372
Zero-variance cells	10	same	same	—	15,800	49

Table 19: Index Trading Against a Shut Market: Sample Cuts

Notes: Column (1) of Table 5 (LEVELS, OLS, same-day overnight subtracted) re-estimated on alternative samples and normalisations. “No overnight adjustment” drops the subtraction of the same-day overnight rate. Unscheduled closures are the two Hurricane Sandy days (October 29–30, 2012), the December 5, 2018 national day of mourning, and MLK Day 2006, on which the session in the bar data does not match the published schedule. The 02:00–08:00 normaliser is the window used in an earlier version of this design; the 20:00–07:00 window used in the paper excludes the 17:00 reopen and the 07:00–08:00 U.S. release hour.

<i>Column (1), LEVELS OLS</i>	Ratio	<i>t</i>
As in Table 5 (58 dead afternoons, 33 index-only mornings)	×13.1	7.2
no overnight adjustment at all	×5.8	4.8
September 2, 2019 (Labor Day) back in	×19.4	6.7
unscheduled closures back in (Hurricane Sandy, December 5, 2018, MLK 2006)	×3.8	1.7
... all 62 dead days minus December 5, 2018 alone	×11.6	6.7
without MLK 2008 (world markets fell with the U.S. shut)	×11.0	5.5
dead afternoons 2010 onward only	×11.8	4.8
02:00–08:00 normaliser instead of 20:00–07:00	×17.7	6.0

B. Model details

This appendix derives Propositions 1 and 2, gives the intraday implementation of the daily model, rationalizes the updating rates through information acquisition, and reports the benchmark variance decomposition.

B.1 Equilibrium without index trading: Proposition 1

There are N stocks in supply $1/N$ each, a competitive fully informed arbitrageur, and a unit mass of behavioral investors distributed evenly across the N stock-specialist groups, so that group i has mass $1/N$. Dividends follow equation (10), beliefs follow equations (13)–(14) and (15)–(16), and the market-level states are those of equation (19).

B.1.1 From forecast gaps to expected-return distortions

Conjecture, and verify in Section B.1.3, that equilibrium prices are affine in the dividend levels with the full-information loading $1/(R_f - 1)$ on $d_{i,t}$. Under any forecast of next period’s dividend, the expected one-day return in equation (17) is $E[P_{i,t+1}] + E[d_{i,t+1}] - R_f P_{i,t}$. A unit increase in the forecast of $d_{i,t+1}$ raises the forecast of the dividend by one and, because the increase is permanent and priced at $1/(R_f - 1)$, raises the forecast of $P_{i,t+1}$ by $1/(R_f - 1)$. The behavioral and rational forecasts differ only in the dividend forecast, by $\delta_{i,t}$ of equation (14), so the fundamental component of the

subjective expected-return distortion in equation (18) is

$$g_{i,t} = \delta_{i,t} + \frac{\delta_{i,t}}{R_f - 1} = \frac{R_f}{R_f - 1} \delta_{i,t}. \quad (43)$$

Because stock investors observe the idiosyncratic component of their dividend, $\delta_{i,t} = \hat{d}_{m,t}^{\text{stk}} - d_{m,t}$ is the same for every stock, so $g_{i,t} = g_{m,t}^{\text{stk}}$ for all i : the vector $\mathbf{g}_t^{\text{stk}} \equiv (g_{1,t}, \dots, g_{N,t})'$ equals $g_{m,t}^{\text{stk}} \mathbf{1}$, and its stock-specific component is identically zero. Subtracting $d_{m,t+1} = d_{m,t} + \varepsilon_{m,t+1}^F$ from the updating rule in equation (13) gives $\delta_{i,t+1} = (1 - \alpha_{\text{stk}})\delta_{i,t} - (1 - \alpha_{\text{stk}})\varepsilon_{m,t+1}^F$, hence

$$g_{m,t+1}^{\text{stk}} = (1 - \alpha_{\text{stk}})g_{m,t}^{\text{stk}} - \frac{R_f}{R_f - 1}(1 - \alpha_{\text{stk}})\varepsilon_{m,t+1}^F, \quad E_t^A[g_{m,t+1}^{\text{stk}}] = (1 - \alpha_{\text{stk}})g_{m,t}^{\text{stk}}. \quad (44)$$

A positive permanent dividend innovation therefore pushes $g_{m,t+1}^{\text{stk}}$ down, and the gap closes geometrically at rate $1 - \alpha_{\text{stk}}$.

B.1.2 Demands and market clearing

Let $\Pi_{t+1}^e \equiv (\Pi_{1,t+1}^e, \dots, \Pi_{N,t+1}^e)'$ collect the stock returns, and define

$$\boldsymbol{\mu}_t \equiv E_t^A[\Pi_{t+1}^e], \quad \Sigma_S \equiv \text{Var}_t(\Pi_{t+1}^e), \quad \mathbf{z}_t \equiv (z_{1,t}, \dots, z_{N,t})', \quad q \equiv \frac{1}{N}\mathbf{1}. \quad (45)$$

The S subscript denotes equilibrium constants that depend on the number of trading periods per day, S . The vector q is both the weight vector of the market portfolio and the vector of supplies. Under symmetry every stock has the same conditional return variance v_S and every pair the same covariance c_S , so

$$\Sigma_S = (v_S - c_S)I_N + c_S\mathbf{1}\mathbf{1}', \quad v_{m,S} = q'\Sigma_S q = \frac{v_S + (N-1)c_S}{N} < v_S, \quad (46)$$

and v_S , $v_{m,S}$ are the variances that appear in Proposition 1. They are equilibrium objects derived in Section B.1.3.

The arbitrageur trades all stocks with the myopic mean–variance rule of the text, $\max_{\mathbf{n}} \{\mathbf{n}'\boldsymbol{\mu}_t - \frac{a}{2}\mathbf{n}'\Sigma_S\mathbf{n}\}$, so

$$\mathbf{n}_t^A = \frac{1}{a}\Sigma_S^{-1}\boldsymbol{\mu}_t. \quad (47)$$

An investor specialized in stock i has the subjective expected return of equation (18), variance v_S , and therefore holds $(\mu_{i,t} + g_{i,t} + z_{i,t})/(av_S)$ shares. Summing over the group of mass $1/N$, the vector of specialist positions is

$$\mathbf{n}_t^{B,\text{stk}} = \frac{1}{Nav_S} \left(\boldsymbol{\mu}_t + \mathbf{g}_t^{\text{stk}} + \mathbf{z}_t \right), \quad (48)$$

and market clearing in all N stocks is

$$\mathbf{n}_t^A + \mathbf{n}_t^{B,\text{stk}} = q. \quad (49)$$

The equicorrelated structure makes it convenient to split any vector $\mathbf{y}_t$ into its market component and its orthogonal complement, $y_{m,t} \equiv q' \mathbf{y}_t$ and $\mathbf{y}_t^\perp \equiv \mathbf{y}_t - y_{m,t} \mathbf{1}$, so that $q' \mathbf{y}_t^\perp = 0$. From equation (46),

$$\Sigma_S \mathbf{1} = N v_{m,S} \mathbf{1}, \quad \Sigma_S \mathbf{y}_t^\perp = (v_S - c_S) \mathbf{y}_t^\perp, \quad (50)$$

so both directions are invariant under Σ_S and Σ_S^{-1} , and $\mathbf{1}' \mathbf{n}_t^A = \mu_{m,t} / (a v_{m,S})$. Premultiplying equation (49) by $\mathbf{1}'$ gives clearing in units of the market portfolio,

$$\frac{\mu_{m,t}}{a v_{m,S}} + \frac{1}{a v_S} \left(\mu_{m,t} + g_{m,t}^{\text{stk}} + z_{m,t} \right) = 1, \quad (51)$$

where $g_{m,t}^{\text{stk}}$ and $z_{m,t}$ are the market states of equation (19). With $\psi_S = v_{m,S} / v_S$ from equation (22),

$$\mu_{m,t} = \bar{\mu}_m - \frac{\psi_S}{1 + \psi_S} \left(g_{m,t}^{\text{stk}} + z_{m,t} \right), \quad \bar{\mu}_m \equiv \frac{a v_{m,S}}{1 + \psi_S}. \quad (52)$$

The behavioral investors' market-level belief distortions reach the required market return with the weight $\psi_S / (1 + \psi_S) < \frac{1}{2}$: their exposure to idiosyncratic risk, $v_S > v_{m,S}$, makes their demand less sensitive to a market-level belief than the arbitrageur's.

Demeaning equation (49) across stocks gives

$$\frac{\boldsymbol{\mu}_t^\perp}{a(v_S - c_S)} + \frac{1}{N a v_S} \left(\boldsymbol{\mu}_t^\perp + \mathbf{z}_t^\perp \right) = \mathbf{0} \implies \boldsymbol{\mu}_t^\perp = -\chi_S(0) \mathbf{z}_t^\perp, \quad \chi_S(0) \equiv \frac{v_S - c_S}{N v_S + (v_S - c_S)}, \quad (53)$$

where $\mathbf{z}_t^\perp$ collects the stock-specific components $z_{i,t}^\perp$ of equation (27). The arbitrageur absorbs stock-specific demand pressure almost completely: $\chi_S(0)$ is of order $1/N$.

B.1.3 The equilibrium price

Conjecture the market price function of equation (20). Using equations (44) and (15), the rational expected one-day market return it implies is

$$\mu_{m,t} = -(R_f - 1)p_S - [R_f - (1 - \alpha_{\text{stk}})] b_{m,S}^F g_{m,t}^{\text{stk}} - (R_f - \lambda) b_{m,S} z_{m,t}, \quad (54)$$

since the dividend terms cancel: $d_{m,t} [1/(R_f - 1) + 1 - R_f/(R_f - 1)] = 0$. Matching coefficients with equation (52) gives

$$p_S = -\frac{\bar{\mu}_m}{R_f - 1}, \quad b_{m,S}^F = \frac{\psi_S}{1 + \psi_S} \frac{1}{R_f - (1 - \alpha_{\text{stk}})}, \quad b_{m,S} = \frac{\psi_S}{1 + \psi_S} \frac{1}{R_f - \lambda}, \quad (55)$$

which are the loadings of equation (21). This establishes Proposition 1.

The two return components stated in the text follow by substitution. The unexpected part of the market return is

$$\Pi_{m,t+1}^e - \mu_{m,t} = \frac{R_f}{R_f - 1} \left[1 - (1 - \alpha_{\text{stk}}) b_{m,S}^F \right] \varepsilon_{m,t+1}^F + b_{m,S} u_{m,t+1}^z, \quad (56)$$

and the expected part is equation (54). Collecting the terms in demand pressure gives $b_{m,S} u_{m,t+1}^z - (R_f - \lambda) b_{m,S} z_{m,t}$, equation (23); collecting the terms in the fundamental belief distortion gives $-\frac{R_f}{R_f-1} (1 - \alpha_{\text{stk}}) b_{m,S}^F \varepsilon_{m,t+1}^F - [R_f - (1 - \alpha_{\text{stk}})] b_{m,S}^F s_{m,t}^{\text{stk}}$, equation (24). The restriction $0 < \lambda < R_f^{-1}$ makes the demand-pressure component generate reversal: the autocovariance of successive market returns that it generates has the sign of $(R_f - \lambda)(R_f \lambda - 1)$ and is negative precisely when $\lambda < R_f^{-1}$. At the daily risk-free rate the bound is essentially nonbinding at the benchmark $\lambda = .90$.

Relative prices clear the remaining $N - 1$ conditions. With $\mathbf{d}_t^\perp \equiv \mathbf{d}_t - d_{m,t} \mathbf{1}$, conjecture

$$\mathbf{P}_t^\perp = \frac{\mathbf{d}_t^\perp}{R_f - 1} + b_{z,S}^\perp \mathbf{z}_t^\perp. \quad (57)$$

The idiosyncratic dividend is observed by the specialists who trade the stock and by the arbitrageur, so it carries its full-information loading and no belief term. The implied expected relative return is $-(R_f - \lambda) b_{z,S}^\perp \mathbf{z}_t^\perp$; matching with equation (53) gives

$$b_{z,S}^\perp = \frac{\chi_S(0)}{R_f - \lambda}. \quad (58)$$

The full price vector $\mathbf{P}_t = P_{m,t} \mathbf{1} + \mathbf{P}_t^\perp$ then satisfies all N conditions in equation (49). Stock-level demand pressure moves individual prices by an amount of order $1/N$.

Equilibrium variances. The variances v_S and $v_{m,S}$ used in the demands must equal those generated by the equilibrium. Equation (56) gives the market condition, and the innovation in equation (57) gives the orthogonal one:

$$v_{m,S} = \left(\frac{R_f}{R_f - 1} \right)^2 \left[1 - (1 - \alpha_{\text{stk}}) b_{m,S}^F \right]^2 \text{Var}(\varepsilon_{m,t+1}^F) + b_{m,S}^2 \text{Var}(u_{m,t+1}^z), \quad (59)$$

$$v_S - c_S = \left(\frac{R_f}{R_f - 1} \right)^2 \sigma_F^2 (1 - \rho_F) + \left(b_{z,S}^\perp \right)^2 \sigma_\eta^2 S^{1-\xi} (1 - \rho_z). \quad (60)$$

Here $\text{Var}(\varepsilon_{m,t+1}^F) = \sigma_F^2 \left(\rho_F + \frac{1-\rho_F}{N} \right)$ and $\text{Var}(u_{m,t+1}^z) = \sigma_\eta^2 S^{1-\xi} \left(\rho_z + \frac{1-\rho_z}{N} \right)$. Because the loadings depend on $\psi_S = v_{m,S}/v_S$ and χ_S , these are two scalar fixed-point conditions in $(v_{m,S}, v_S - c_S)$, which we solve numerically. Although dividend levels are nonstationary, one-day return risk is not: the loading $1/(R_f - 1)$ makes the level of $\mathbf{d}_t$ cancel from returns, so the CARA-normal demands depend only on stationary innovations and belief states.

B.2 Equilibrium with index trading: Proposition 2

A fraction ω of behavioral investors, drawn equally across the stock-specialist groups, now trades the index fund, so each specialist group keeps mass $(1 - \omega)/N$. The fund is a pass-through claim on q , so a position of x units in the fund is the underlying position xq and creates no new supply.

B.2.1 The index investors' forecast gap

The argument of Section B.1.1 applies to the market claim. The index price in equation (34) carries loading $1/(R_f - 1)$ on $d_{m,t}$, so a unit gap in the forecast of $d_{m,t+1}$ shifts the expected index return by $R_f/(R_f - 1)$. The fundamental distortion in equation (28) is therefore

$$g_{m,t}^{\text{idx}} = \frac{R_f}{R_f - 1} \delta_{m,t}^{\text{idx}}, \quad (61)$$

with $\delta_{m,t}^{\text{idx}}$ from equation (32). Subtracting the dividend law from the updating rule in equation (31) gives

$$g_{m,t+1}^{\text{idx}} = (1 - \alpha_{\text{idx}})g_{m,t}^{\text{idx}} - \frac{R_f}{R_f - 1}(1 - \alpha_{\text{idx}})\varepsilon_{m,t+1}^F, \quad E_t^A[g_{m,t+1}^{\text{idx}}] = (1 - \alpha_{\text{idx}})g_{m,t}^{\text{idx}}. \quad (62)$$

Both groups therefore process the same common dividend news; the two rates differ by the claim through which it is acted on (Section B.4 rationalizes the two rates as the outcome of a monitoring choice).

B.2.2 Market clearing with the index group

An index investor has the subjective expected return of equation (28), whose transitory component is $z_{m,t}$ because the stock-specific components of equation (27) sum to zero across the fund's holdings, and faces variance $v_{m,S}$. The group's demand for the underlying stocks is therefore

$$\mathbf{n}_t^{B,\text{idx}} = \frac{\omega}{av_{m,S}} (\mu_{m,t} + g_{m,t}^{\text{idx}} + z_{m,t}) q, \quad (63)$$

while the individual stock investors' positions are $(1 - \omega)$ times equation (48). Market clearing becomes

$$\frac{1}{a} \Sigma_S^{-1} \mu_t + \frac{1 - \omega}{N av_S} (\mu_t + \mathbf{g}_t^{\text{stk}} + \mathbf{z}_t) + \frac{\omega}{av_{m,S}} (\mu_{m,t} + g_{m,t}^{\text{idx}} + z_{m,t}) q = q. \quad (64)$$

Importantly, the same market-level belief produces a larger position when it is expressed through the diversified claim, because the risk weight $1/v_S$ is replaced by $1/v_{m,S}$.

In the market direction, $\mathbf{1}'q = 1$ and equation (50) give

$$\frac{\mu_{m,t}}{av_{m,S}} + \frac{1 - \omega}{av_S} (\mu_{m,t} + g_{m,t}^{\text{stk}} + z_{m,t}) + \frac{\omega}{av_{m,S}} (\mu_{m,t} + g_{m,t}^{\text{idx}} + z_{m,t}) = 1. \quad (65)$$

Multiplying by $av_{m,S}$ and collecting $\mu_{m,t}$ produces the factor $1 + \omega + (1 - \omega)\psi_S = \Delta_S(\omega)$ of equation (33), so

$$\mu_{m,t} = \bar{\mu}_m - \frac{(1 - \omega)\psi_S}{\Delta_S(\omega)} g_{m,t}^{\text{stk}} - \frac{\omega}{\Delta_S(\omega)} g_{m,t}^{\text{idx}} - \frac{\omega + (1 - \omega)\psi_S}{\Delta_S(\omega)} z_{m,t}, \quad \bar{\mu}_m \equiv \frac{av_{m,S}}{\Delta_S(\omega)}. \quad (66)$$

In the orthogonal direction, index demand and supply are both proportional to q and drop out, and only the specialists' mass changes relative to equation (53):

$$\mu_t^\perp = -\chi_S(\omega) \mathbf{z}_t^\perp, \quad \chi_S(\omega) \equiv \frac{(1-\omega)(v_S - c_S)}{Nv_S + (1-\omega)(v_S - c_S)}. \quad (67)$$

Equations (65) and (67) together are equivalent to the N conditions in equation (64).

B.2.3 The equilibrium price

Conjecture the market price function of equation (34). Using equations (44), (62) and (15), the implied rational expected market return is

$$\begin{aligned} \mu_{m,t} = & -(R_f - 1)p_S - [R_f - (1 - \alpha_{\text{stk}})] b_{m,S}^{F,\text{stk}}(\omega) g_{m,t}^{\text{stk}} \\ & - [R_f - (1 - \alpha_{\text{idx}})] b_{m,S}^{F,\text{idx}}(\omega) g_{m,t}^{\text{idx}} - (R_f - \lambda) b_{m,S}(\omega) z_{m,t}. \end{aligned} \quad (68)$$

Matching coefficients with equation (66) gives $p_S = -\bar{\mu}_m / (R_f - 1)$ and the three loadings of equations (35)–(37). This establishes Proposition 2. At $\omega = 0$, $\Delta_S(0) = 1 + \psi_S$ and the loadings reduce to equation (55); at $\omega = 1$, $b_{m,S}(1) = \frac{1}{2} / (R_f - \lambda)$, which is equation (38). The relative price is equation (57) with $b_{z,S}^\perp(\omega) = \chi_S(\omega) / (R_f - \lambda)$. The equilibrium variance conditions are equations (59)–(60), with the fundamental loading replaced by

$$\kappa_F(\omega) \equiv 1 - (1 - \alpha_{\text{stk}}) b_{m,S}^{F,\text{stk}}(\omega) - (1 - \alpha_{\text{idx}}) b_{m,S}^{F,\text{idx}}(\omega), \quad (69)$$

the share of a permanent market innovation reflected in the market price on the day it arrives; $1 - \kappa_F(\omega)$ is the lagged share.

Stock-level objects. Because $\mathbf{P}_t = P_{m,t} \mathbf{1} + \mathbf{P}_t^\perp$ and the relative return is uncorrelated with the market return, every stock has market beta one, and the model's R^2 of Section 4.3 is the market share of a stock's return variance, $\text{Var}(\Pi_{m,t+1}^e) / \text{Var}(\Pi_{i,t+1}^e)$. The arbitrageur's position in stock i decomposes along the same two directions,

$$n_{i,t}^A = \frac{1}{a} \left[\frac{\mu_{m,t}}{Nv_{m,S}} + \frac{\mu_{i,t}^\perp}{v_S - c_S} \right], \quad (70)$$

so the intermediated order flow in equation (39) has a common component, driven by the market states through equation (66), and a stock-specific component of order $1/N$. A large ω is in our calibration associated with a large common component's share, which is the model's counterpart to the rise in market-directed trading.

B.3 An intraday implementation

The main text is solved at the daily horizon. In this subsection, S is explicitly implemented as the number of opportunities to revise positions between two daily boundaries. At each opportunity investors choose the position that would be optimal if held to the next boundary and reoptimize if another opportunity arrives. Sampling this economy at daily boundaries reproduces the daily model exactly.

Timing and state laws. Let $h = 0, \dots, S$ index equally spaced points between daily boundaries t and $t + 1$. Positions can be revised at $h = 0, \dots, S - 1$, while $h = S$ is the next daily boundary, so $S = 1$ is the daily economy. Let

$$R_{h,S} \equiv R_f^{1-h/S} \quad (71)$$

denote the gross risk-free return from opportunity h to the next daily boundary. Fundamental news and fundamental belief updating remain calendar-time processes: no fundamental innovation arrives between t and $t + 1$, so $\mathbf{g}_t^{\text{stk}}$ and $\mathbf{g}_{m,t}^{\text{idx}}$ are fixed during the day and update at the boundary according to equations (44) and (62). Transitory demand pressure instead evolves in trade time. Starting from $\mathbf{z}_{t,0} = \mathbf{z}_t$, with $\lambda_S \equiv \lambda^{1/S}$ and $B_S \equiv \sum_{k=0}^{S-1} \lambda_S^{2k}$,

$$z_{i,t,h+1} = \lambda_S z_{i,t,h} + \eta_{i,t,h+1}^S, \quad \text{Var}(\eta_{i,t,h}^S) = \frac{\sigma_\eta^2 S^{1-\xi}}{B_S}, \quad \text{Corr}(\eta_{i,t,h}^S, \eta_{j,t,h}^S) = \rho_z, \quad i \neq j, \quad (72)$$

for $h = 0, \dots, S - 1$, where the shocks are independent across trading opportunities and calendar days and orthogonal to fundamental news. Iterating to the next boundary,

$$z_{i,t+1} = \lambda z_{i,t} + u_{i,t+1}^z, \quad u_{i,t+1}^z = \sum_{h=1}^S \lambda_S^{S-h} \eta_{i,t,h}^S, \quad \text{Var}(u_{i,t+1}^z | S) = \sigma_\eta^2 S^{1-\xi}, \quad (73)$$

with cross-stock correlation ρ_z : the law of demand pressure sampled at daily boundaries is exactly equations (15)–(16) for every S .

Demands and clearing at opportunity h . Let $\mathbf{Y}_{t+1} \equiv \mathbf{P}_{t+1} + \mathbf{d}_{t+1}$ denote the cum-dividend value of the stocks at the next daily boundary, so that the daily return of the text is $\Pi_{t+1}^e = \mathbf{Y}_{t+1} - R_f \mathbf{P}_t$. An arbitrageur with wealth $W_{t,h}$ who holds $\mathbf{n}$ unchanged to the boundary has terminal wealth

$$\tilde{W}_{t+1|h} = R_{h,S} W_{t,h} + \mathbf{n}'(\mathbf{Y}_{t+1} - R_{h,S} \mathbf{P}_{t,h}). \quad (74)$$

Define, conditional on information at opportunity h ,

$$\boldsymbol{\mu}_{t,h} \equiv E_{t,h}^A[\mathbf{Y}_{t+1}] - R_{h,S} \mathbf{P}_{t,h}, \quad (75)$$

$$\Sigma_{h,S} \equiv \text{Var}_{t,h}^A(\mathbf{Y}_{t+1}) = (v_{h,S} - c_{h,S}) I_N + c_{h,S} \mathbf{1}\mathbf{1}', \quad v_{m,h,S} \equiv q' \Sigma_{h,S} q. \quad (76)$$

With CARA utility and a conditionally Gaussian payoff, wealth drops out and $\mathbf{n}_{t,h}^A = \frac{1}{a} \Sigma_{h,S}^{-1} \boldsymbol{\mu}_{t,h}$. The behavioral demands are the intraday counterparts of equations (48) and (63), with $(\boldsymbol{\mu}_{t,h}, \mathbf{z}_{t,h}, v_{h,S}, v_{m,h,S})$ replacing the daily objects. The fundamental belief distortions carry no h subscript because every intraday decision concerns the same terminal payoff; $z_{i,t,h}$ evolves within the day because demand pressure itself evolves in trade time. Market clearing at every opportunity is

$$\frac{1}{a} \Sigma_{h,S}^{-1} \boldsymbol{\mu}_{t,h} + \frac{1-\omega}{Na v_{h,S}} (\boldsymbol{\mu}_{t,h} + \mathbf{g}_t^{\text{stk}} + \mathbf{z}_{t,h}) + \frac{\omega}{a v_{m,h,S}} (\boldsymbol{\mu}_{m,t,h} + \mathbf{g}_{m,t}^{\text{idx}} + \mathbf{z}_{m,t,h}) q = q. \quad (77)$$

For a given conditional distribution of the terminal payoff, this is a linear system in $\mu_{t,h}$ with a positive-definite coefficient matrix, so the solution is unique and

$$\mathbf{P}_{t,h} = \frac{1}{R_{h,S}} (E_{t,h}^A[\mathbf{Y}_{t+1}] - \mu_{t,h}). \quad (78)$$

The covariance $\Sigma_{h,S}$ depends on the conditional distribution of the terminal payoff, not on the current intraday price, so equation (77) determines the price without an intraday fixed point. As the boundary approaches, the number of demand shocks still to arrive falls while the terminal fundamental innovation is unchanged. The market and relative-price decompositions of Section B.2.2 therefore apply at every opportunity, with $(v_{h,S}, c_{h,S})$ replacing the daily variances.

Exact nesting of the daily equilibrium. At the first opportunity $R_{0,S} = R_f$ and $\mathbf{z}_{t,0} = \mathbf{z}_t$, and by equation (73) together with the calendar-time laws for fundamentals and beliefs, the conditional distribution of the terminal state at $h = 0$ is exactly the one used in the daily model. Since $\mathbf{P}_{t+1}$ is the daily-boundary equilibrium price generated by that state,

$$\Sigma_{0,S} = \Sigma_S, \quad v_{0,S} = v_S, \quad c_{0,S} = c_S, \quad v_{m,0,S} = v_{m,S}, \quad (79)$$

so equation (77) at $h = 0$ is exactly equation (64) and

$$\mathbf{P}_{t,0} = \mathbf{P}_t. \quad (80)$$

At $h = S$ the states update according to the daily laws, so the same argument applies at the next boundary. The intraday economy therefore reproduces the daily prices, returns, variances, and variance ratios exactly at daily boundaries. This gives S a literal trading-time interpretation, while ξ determines how much additional trade-time shock arrival survives at the daily horizon.

B.4 Information acquisition and the updating rates

We rationalize α_{stk} and α_{idx} with a simple monitoring choice. The information friction concerns the common fundamental $d_{m,t}$; stock-specific news is observed immediately.¹⁴ Each investor chooses a standing monitoring intensity $\alpha \in (0, 1]$ and pays a daily flow cost $C(\alpha)$. On each day the current common level is delivered before trading with probability α , independently across investors and days. Monitoring represents costly processing of market-wide information rather than literal access to public news.

¹⁴Allowing a separate stock-specific belief gap would not affect the market direction. In the symmetric N -stock benchmark it enters only relative prices, with loading $\chi_S(\omega)$ of order $1/N$, and is quantitatively negligible at $N = 500$.

Aggregation. A fraction α of each group refreshes each day; non-refreshers retain the previous forecast. The group-average forecast therefore obeys

$$\hat{d}_{m,t} = (1 - \alpha)\hat{d}_{m,t-1} + \alpha d_{m,t}, \quad (81)$$

which is exactly the updating rule used in the model. Subtracting $d_{m,t} = d_{m,t-1} + \varepsilon_{m,t}^F$ gives the corresponding gap laws in equations (44) and (62). The number of days K since an investor's last delivery is geometric, $\Pr(K = k) = \alpha(1 - \alpha)^k$, with $E[K] = (1 - \alpha)/\alpha$.

The value of monitoring. Let m be the current expected one-day return of a claim and m' the return perceived by a stale investor. With claim variance v , the stale investor holds $m'/(av)$ rather than $m/(av)$, generating certainty-equivalent loss $(m - m')^2/(2av)$. Let

$$\sigma_{F,m}^2 \equiv \text{Var}(\varepsilon_{m,t+1}^F) = \sigma_F^2 \left(\rho_F + \frac{1 - \rho_F}{N} \right)$$

denote the variance of common fundamental news. If the common fundamental is k days stale, the expected loss is kB , where

$$B_{\text{stk}} = \frac{(\frac{R_f}{R_f - 1})^2 \sigma_{F,m}^2}{2a v_S}, \quad B_{\text{idx}} = \frac{(\frac{R_f}{R_f - 1})^2 \sigma_{F,m}^2}{2a v_{m,S}}. \quad (82)$$

The stale error concerns the same common fundamental for both groups; what differs is the risk of the claim through which it is acted on. Therefore

$$\frac{B_{\text{idx}}}{B_{\text{stk}}} = \frac{v_S}{v_{m,S}} = \frac{1}{\psi_S}. \quad (83)$$

Thus the same common information is worth $1/\psi_S$ times more when acted on through the diversified claim because the claim carries no idiosyncratic risk.¹⁵

Optimal intensity. Expected staleness costs are $B(1 - \alpha)/\alpha$, so the investor chooses

$$\min_{\alpha \in (0,1]} C(\alpha) + B \frac{1 - \alpha}{\alpha}. \quad (84)$$

For monitoring cost $C(\alpha) = c\alpha^\gamma$ with $0 < \gamma \leq 1$, the first-order condition gives

$$\alpha^*(B) = \min \left\{ 1, \left(\frac{B}{\gamma c} \right)^{1/(1+\gamma)} \right\}. \quad (85)$$

Writing $e \equiv 1/(1 + \gamma)$ for the elasticity of the monitoring rate to the value of information, the two rates satisfy

$$\alpha_{\text{idx}} = \min \left\{ 1, \alpha_{\text{stk}} \left(\frac{1}{\psi_S} \right)^e \right\}, \quad e = \frac{1}{1 + \gamma}. \quad (86)$$

The curvature of monitoring costs determines the size of the difference, but the diversification channel implies $\alpha_{\text{idx}} > \alpha_{\text{stk}}$.

¹⁵Behavioral investors use the objective conditional return variance in their demand; allowing stale investors also to perceive higher variance would not change the ordering.

The benchmark rates. We choose the common cost scale c so that $\alpha_{\text{stk}} = .30$ in the early-sample benchmark, where $\psi_S \simeq 0.211$ and $1/\psi_S \simeq 4.7$. Equation (86) then gives $\alpha_{\text{idx}} = 1$ whenever

$$.30(4.7)^e \geq 1 \iff e \geq \frac{\log(1/0.30)}{\log(1/0.211)} \simeq 0.77 \iff \gamma \lesssim 0.29.$$

Thus a common monitoring technology with sufficiently strong economies of scale delivers the benchmark pair $(\alpha_{\text{stk}}, \alpha_{\text{idx}}) = (.30, 1)$. Under linear monitoring costs ($e = \frac{1}{2}$), re-solving the early-sample equilibrium instead gives $\alpha_{\text{idx}} \simeq .70$; Section B.5 reports this sensitivity. We hold the two updating rates fixed in the main comparative statics so that changes in ω and S are the only sources of variation.

B.5 Variance decomposition and sensitivity to the index updating rate

Fundamental and nonfundamental innovations are independent in the benchmark parameterization, so daily market-return variance decomposes exactly as

$$\text{Var}(r_t) = \text{Var}(r_t^F) + \text{Var}(r_t^z). \quad (87)$$

For the benchmark parameterization in the main text, the annualized variance components are

	Early (1950–70)	Modern (2005–24)
Total variance	0.0132	0.0379
Fundamental component	0.0102	0.0146
Transitory z component	0.0030	0.0233
z share of total	23%	61%

Hence the fraction of the modeled variance increase attributable to the transitory component is

$$\frac{\text{Var}_M(r^z) - \text{Var}_E(r^z)}{\text{Var}_M(r) - \text{Var}_E(r)} = 0.82. \quad (88)$$

Thus the transitory demand component accounts for 82% of the modeled increase in daily market variance.

Sensitivity to the index updating rate. Under linear monitoring costs, re-solving the early benchmark gives $(\alpha_{\text{stk}}, \alpha_{\text{idx}}) = (.30, .70)$. Holding all other primitives fixed, the lower index updating rate leaves more continuation in the model: the variance ratio remains below the data after 1970 and modern volatility falls to 18.1% rather than 19.5%.

Era	Benchmark (.30, 1)		Linear cost (.30, .70)		Data	
	Vol	$VR_{1,21}$	Vol	$VR_{1,21}$	Vol	$VR_{1,21}$
1950–70	11.5%	0.80	10.7%	0.69	10.4%	0.70–0.83
1971–90	13.6%	1.02	12.1%	0.82	14.6%	0.92–1.19
1991–2004	15.3%	1.16	14.0%	0.98	16.3%	1.22
2005–24	19.5%	1.46	18.1%	1.29	19.5%	1.37–1.48

C. Empirical evidence on slow information diffusion

The early-sample variance ratios imply continuation in market returns. The model attributes this continuation to slow diffusion of common information across stocks. We test that mechanism directly, ask whether removing the estimated delay eliminates the continuation, examine whether indexation accelerated information diffusion, and then separate the secular rise in volatility from changes in the timing of permanent news.

C.1 Slow information diffusion

If some stocks incorporate common information before others, returns on these stocks should predict subsequent returns on slower stocks more strongly than the reverse.

We identify leaders and followers within the S&P 500 universe out of sample. At the end of each year $Y - 1$, the candidate universe is the set of current S&P 500 constituents—or, before 1958, the 500 largest stocks by market equity—with valid return and market-equity observations on at least 90% of trading days in the preceding five years. For each stock i we compute

$$\ell_i \equiv \sum_{k=1}^5 \left[\text{Cov}(M_t^{-i}, r_{i,t-k}) - \text{Cov}(r_{i,t}, M_{t-k}^{-i}) \right], \quad (89)$$

where M_t^{-i} is the universe return excluding stock i . The score is positive when stock i predicts the rest of the market more strongly than the reverse; symmetric common comovement drops out of the lead-minus-lag difference. We rank stocks by ℓ_i/β_i , exclude stocks with $|\beta_i| < 0.2$ from the ranking, and form a value-weighted Leader portfolio from the top 50. The remaining constituents form the Follower portfolio. No holding-year return enters any score.

Let L_t and F_t denote the daily Leader and Follower returns. For $j \geq 1$, define

$$d_j = \text{Cov}(F_t, L_{t-j}) - \text{Cov}(F_t, L_{t+j}), \quad (90)$$

the portfolio counterpart of equation (89). A positive d_j means that leader returns predict subsequent follower returns more strongly than the reverse. Symmetric common

Table 20: Out-of-sample leader validation. The stock universe is the S&P 500; before S&P 500 constituent history begins, we use the 500 largest stocks by market equity. At each year-end, the previous five calendar years of daily returns are used to rank stocks by the beta-normalized score of equation (89). The top 50 stocks form the leader portfolio for the following calendar year and the remaining index constituents form the follower portfolio. We report $D = \sum_{j=1}^5 d_j$, where $d_j = \text{Cov}(F_t, L_{t-j}) - \text{Cov}(F_t, L_{t+j})$, scaled as $10^5 D$ for readability. Positive values indicate that leader returns predict subsequent returns of the comparison portfolio more strongly than the reverse. Comparison portfolios are 50 non-leaders matched on size, industry and volatility, 50 matched additionally on beta, and a random 50, all formed from the same S&P 500 universe. Brackets are 95% year-block bootstrap intervals.

Comparison portfolio	1955–70	1971–90	1991–2004	2005–24
$10^5 D$, Leader 50 against each comparison portfolio; 95% intervals in brackets				
Followers	1.23 [0.76, 1.81]	2.62 [1.64, 3.90]	1.77 [0.47, 3.26]	0.61 [-0.00, 1.30]
Size/industry matched	1.07 [0.45, 1.93]	2.36 [1.12, 3.95]	0.90 [0.41, 1.56]	0.29 [-0.11, 0.82]
Beta matched	0.93 [0.24, 1.85]	2.18 [1.12, 3.43]	0.84 [0.34, 1.54]	0.35 [-0.15, 0.93]
Random 50	1.34 [0.71, 2.00]	2.30 [1.33, 3.44]	1.71 [0.56, 2.88]	0.64 [-0.21, 1.58]

comovement drops out, so d_j isolates differential timing between the two portfolios. We report

$$D \equiv \sum_{j=1}^5 d_j, \quad (91)$$

computed within each era, against the Follower portfolio and against three comparison portfolios formed each year from the non-leaders: 50 stocks matched to the leaders on size, industry, and volatility; 50 matched additionally on market beta; and a random 50.

Table 20 shows positive leader-to-follower propagation in every period through 2004, including against matched and random comparison portfolios. The asymmetry disappears in 2005–24. Figure 36 gives the same message in return dynamics: leaders show little continuation in 1955–70 and move into reversal roughly fifteen years before the broad market. By 2005–24 the variance-ratio gap between leaders and followers has largely closed.

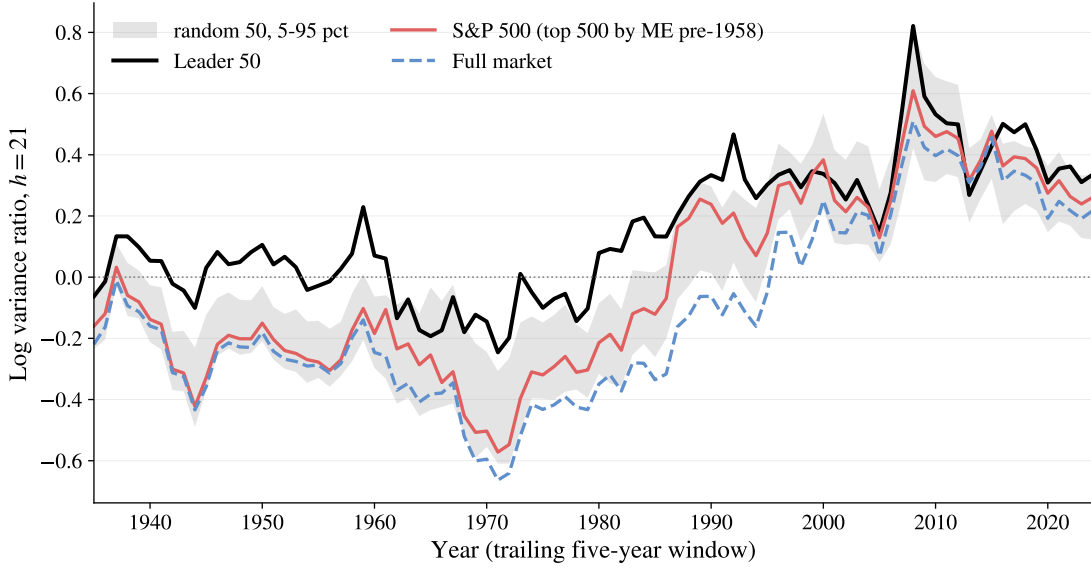


Figure 36: Log variance ratio for the Leader 50, the full market, and the S&P 500 (top 500 stocks by market equity before 1958), using five-year rolling windows. The band is the 5th–95th percentile of 100 random 50-stock portfolios drawn from the same universe each year.

C.2 Slow diffusion explains the early continuation

We use two corrections to assess how much of the early market continuation reflects the measured leader-to-follower propagation. Throughout this section, the market is the value-weighted S&P 500 universe defined above.

Projection synchronization. Regress the daily market return on contemporaneous, lagged, and future Leader 50 returns,

$$r_t = a + \beta_0 L_t + \sum_{j=1}^K (b_j L_{t-j} + c_j L_{t+j}) + e_t, \quad K = 5, \quad (92)$$

and use the coefficient asymmetry, $\lambda_j = b_j - c_j$, to construct

$$r_t^+ = r_t + \sum_{j=1}^K \lambda_j (L_t - L_{t-j}). \quad (93)$$

Applied in the estimation sample, this adjustment removes the asymmetry by construction. We therefore estimate the coefficients separately by year and apply to each year their average from the two preceding and two following years, excluding the year being corrected.

Figure 37 shows a substantial reduction in continuation. Averaged over rolling windows ending in 1955–70 and 1971–90, the variance ratio rises from 0.740 to 0.920

and from 0.830 to 1.084, respectively. The sum of the first five daily autocorrelations falls from +0.197 to +0.058 and from +0.165 to −0.005.

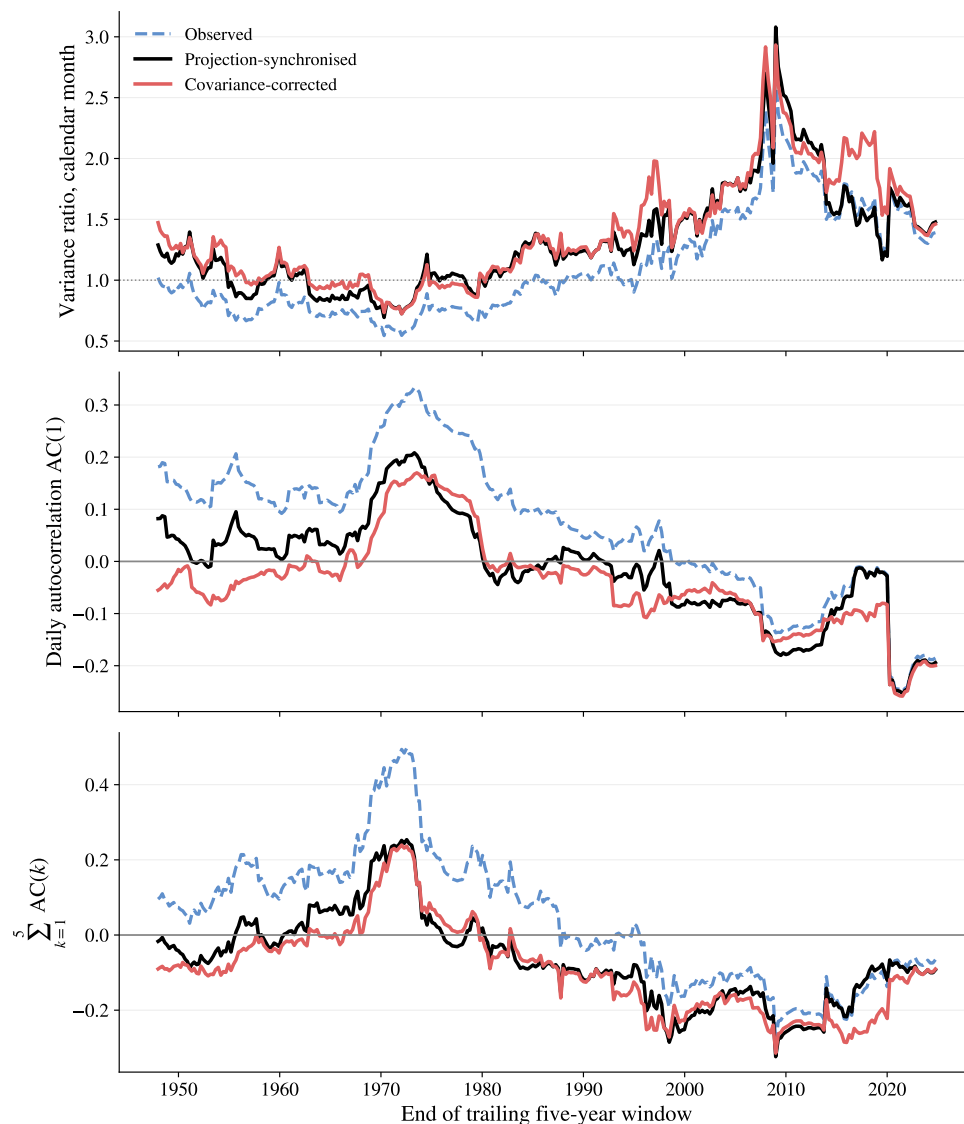


Figure 37: Observed, projection-synchronized, and covariance-corrected return dynamics of the S&P 500 universe in five-year rolling windows. The panels report the calendar-month variance ratio, first-order daily autocorrelation, and sum of the first five daily autocorrelations. The covariance correction uses a delay kernel estimated on trailing 20-year windows.

Covariance correction. The second approach estimates how delayed permanent news contributes to market-return covariances. Normalize leader and follower re-

turns by their contemporaneous market betas, $\tilde{L}_t = L_t/\beta_L$ and $\tilde{F}_t = F_t/\beta_F$. We use these betas as proxies for cumulative permanent-news loadings; the correction is conditional on this approximation.

Suppose a serially uncorrelated permanent innovation ε_t enters $\tilde{L}_t$ on impact with unit loading and reaches $\tilde{F}_t$ with weights $a_0, \dots, a_K$ summing to one. Assume the remaining price components are stationary, orthogonal to ε_t at all leads and lags, and generate symmetric leader-follower cross-covariances. Then the directional covariance in equation (90), computed on normalized returns, identifies

$$d_j = a_j \sigma_\varepsilon^2, \quad j = 1, \dots, K. \quad (94)$$

Stationary price components contribute zero long-run return covariance, so

$$\Omega_{FL} = \sum_{k=-\infty}^{\infty} \text{Cov}(\tilde{F}_t, \tilde{L}_{t-k}) = \sigma_\varepsilon^2. \quad (95)$$

The implied delay weights are therefore

$$a_j = \frac{d_j}{\Omega_{FL}}, \quad a_0 = 1 - \sum_{j=1}^K a_j. \quad (96)$$

Let w be the average market weight of the Leader 50 over the estimation window. Approximating the market by $w\beta_L \tilde{L}_t + (1-w)\beta_F \tilde{F}_t$ gives the permanent-news timing weights

$$m_0 = w\beta_L + (1-w)\beta_F a_0, \quad m_j = (1-w)\beta_F a_j, \quad j = 1, \dots, K. \quad (97)$$

Write $q = \sum_{j=0}^K m_j = w\beta_L + (1-w)\beta_F$, approximately one. Replacing the distributed response with an impact response of size q gives the corrected market autocovariances

$$\Gamma_M^*(0) = \Gamma_M(0) + \Omega_{FL} \left(q^2 - \sum_{j=0}^K m_j^2 \right), \quad (98)$$

$$\Gamma_M^*(k) = \Gamma_M(k) - \Omega_{FL} \sum_{j=0}^{K-k} m_j m_{j+k}, \quad 1 \leq k \leq K, \quad (99)$$

with higher-lag covariances unchanged. Under the maintained normalization, this moves delayed permanent news to impact while preserving its long-run variance.

Estimation and results. The long-run cross-covariance Ω_{FL} is the zero-frequency cross-spectrum under the convention $f_{FL}(\lambda) = \sum_k \text{Cov}(\tilde{F}_t, \tilde{L}_{t-k}) \cos(\lambda k)$. We estimate it by smoothing the real cross-periodogram near zero,

$$\hat{\Omega}_{FL} = \frac{\sum_{s=1}^{T-1} W_T(\lambda_s) I_{FL}(\lambda_s)}{\sum_{s=1}^{T-1} W_T(\lambda_s)}, \quad \lambda_s = \frac{2\pi s}{T}, \quad (100)$$

Table 21: Delayed information and corrected return dynamics of the S&P 500 universe. Starred moments move the estimated delayed permanent-news response to impact under the normalization in the text. The last two columns report the sum of the first five daily autocorrelations.

Era	VR	VR*	AC(1)	AC(1)*	$\sum_{k=1}^5 \text{AC}(k)$	$\sum_{k=1}^5 \text{AC}(k)^*$
1931–45	0.737	0.825	+0.052	-0.018	+0.076	+0.013
1946–54	0.843	1.108	+0.154	+0.004	+0.096	-0.051
1955–70	0.725	0.962	+0.173	+0.048	+0.252	+0.055
1971–90	0.909	1.184	+0.125	+0.007	+0.070	-0.071
1991–2004	1.344	1.619	-0.011	-0.056	-0.099	-0.181
2005–24	1.612	1.711	-0.121	-0.144	-0.148	-0.173

using the Dew-Becker (2017) RQS kernel,

$$W(u) = \frac{1}{2\pi} \left[\frac{9}{8} - \frac{15}{8} \left(\frac{u}{2\pi} \right)^2 \right] \mathbf{1}\{|u| < 2\pi\}.$$

Here W_T is the periodic extension of $B^{-1}W(\lambda/B)$ and $B = 1/H$. The kernel’s zero second moment removes the leading spectral-curvature bias. The denominator normalizes the weights over nonzero Fourier frequencies to sum to one. The kernel permits negative estimates, which we do not truncate.

For the delay weights, we use $H = 50$ trading days, retaining cycles longer than about ten weeks. Estimation uses each era in Table 21 and trailing 20-year windows in Figure 37. At a finite bandwidth, slowly reversing discount-rate movements can enter $\hat{\Omega}_{FL}$ alongside permanent cash-flow news. Section C.4 uses the same estimator with $H = 250$ to measure the market’s long-run variance.

The covariance correction confirms the projection results. In 1955–70, the variance ratio rises from 0.725 to 0.962, first-order autocorrelation falls from +0.173 to +0.048, and the five-lag sum falls from +0.252 to +0.055. In 1971–90, the corresponding changes are 0.909 to 1.184, +0.125 to +0.007, and +0.070 to −0.071. The rolling estimates in Figure 37 likewise show little continuation after correction.

The conclusion is similar across bandwidths and normalizations. Increasing H from 50 to 250 changes the corrected variance ratio from 0.962 to 0.969 in 1955–70 and from 1.184 to 1.177 in 1971–90. Replacing contemporaneous betas with zero-frequency betas, $\hat{\beta}_p^{LR} = \hat{\Omega}_{pM}/\hat{\Omega}_{MM}$ for $p \in \{L, F\}$, estimated by the same RQS method, gives 0.979 and 0.981 at the two bandwidths in 1955–70, and 1.208 and 1.170 in 1971–90. Both corrections therefore attribute most of the early continuation to the measured delay in common information.

C.3 Index membership and information diffusion

We next ask whether indexation helped eliminate the delay. The prediction is that S&P 500 membership should matter for information speed only when index products

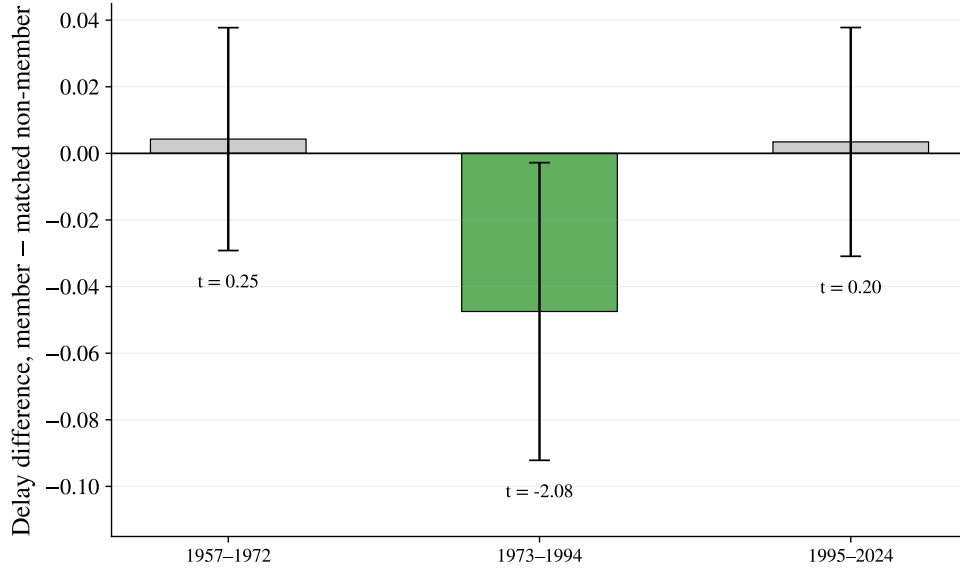


Figure 38: Difference in delayed response to common news, Δ in equation (102), between the smallest 100 S&P 500 members and size-matched non-members, by period. Both stocks in each pair are listed on NYSE or AMEX. Bars show 95% confidence intervals from heteroskedasticity- and autocorrelation-consistent standard errors.

were concentrated in the S&P 500.

We compare the smallest 100 S&P 500 members by month-end market equity with size-matched NYSE/AMEX non-members. Controls cannot be index members within twelve months of the match. Let r_t^{in} and r_t^{out} denote the value-weighted member and non-member returns, where the weights correspond to the member's market equity for each match. For portfolio return r_t^p , delayed response to common news is measured by

$$r_t^p = a + \beta_0 r_t^M + \sum_{j=1}^2 (b_j r_{t-j}^M + c_j r_{t+j}^M) + e_t, \quad \delta^p \equiv \sum_{j=1}^2 (b_j - c_j), \quad (101)$$

where r_t^M is the value-weighted market return. The lead-minus-lag difference removes symmetric common comovement, so δ^p measures delayed response to market-wide news. The test is the member-non-member difference

$$\Delta \equiv \delta^{\text{in}} - \delta^{\text{out}}, \quad (102)$$

which we estimate by applying equation (101) to $r_t^{\text{in}} - r_t^{\text{out}}$ within each period, with heteroskedasticity- and autocorrelation-consistent standard errors.

Figure 38 shows a member-non-member delay difference of +0.004 in 1957-72, -0.047 in 1973-94 ($t = -2.08$), and +0.003 in 1995-2024. Direct comparisons from a regression allowing all coefficients to vary by period show that the middle-period

difference is lower than the early-period difference by 0.052 ($t = -1.82$) and the late-period difference by 0.051 ($t = -1.77$). This pattern is consistent with index membership being associated with faster information incorporation when index products were concentrated in the S&P 500.

C.4 Daily and long-run volatility

Slow information diffusion spreads permanent news across days and therefore depresses measured daily volatility. Its disappearance could consequently account for some of the secular rise documented in the paper. Quantitatively, however, the effect is modest. Applying the covariance correction to the volatility of the S&P 500 universe return, estimated with the specification of Table 1 of the main text, reduces its 1955–2024 increase in annualized daily volatility from 10.75 to 8.43 percentage points (the full-market increase over 1955–2025 in Table 1 of the main text is 11.44). Most of the rise remains.

The long-run variance measures how much return variation survives aggregation over long horizons. Define

$$\Omega^{LR} = \sum_{k=-\infty}^{\infty} \Gamma(k) = \lim_{H \rightarrow \infty} \frac{1}{H} \text{Var} \left(\sum_{h=1}^H r_{t+h} \right). \quad (103)$$

Return components that are first differences of stationary price components have zero long-run variance because their cumulative effects eventually reverse. We estimate Ω^{LR} on trailing 20-year windows using the RQS estimator of Section C.2 with $H = 250$. At this finite bandwidth, slowly reversing discount-rate movements can also contribute to the estimate. We report

$$\sigma_t^{LR} = \sqrt{n_t \Omega_t^{LR}}, \quad (104)$$

where n_t is the number of trading days per year in the corresponding window.

Figure 39 shows the key result. Long-run volatility is nearly flat once Depression observations leave the window, rising from 15.9% in 1970 to 17.3% in 2024. Over the same windows, S&P 500 daily volatility rises from 10.3% to 19.0%, and diffusion-adjusted daily volatility from 12.0% to 19.7%. In variance terms, daily variance rises by 256 percentage-points squared while long-run variance rises by 48; the difference is 208 with a standard error of 99 ($t = 2.11$). The estimated increase in long-run variance is therefore about one fifth of the increase in daily variance.

Most of the rise in daily volatility remains after the diffusion adjustment, while long-run volatility shows a much smaller increase. Together, these findings support the interpretation that the secular rise in market volatility is predominantly transitory.

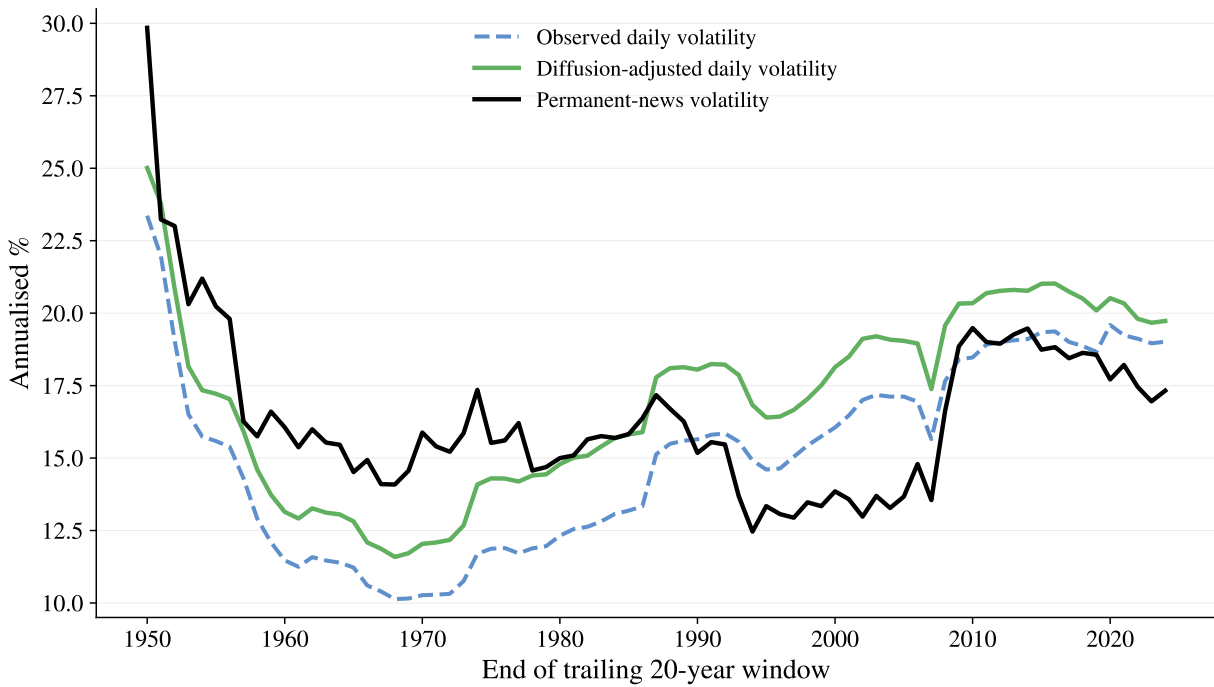


Figure 39: Observed daily volatility, diffusion-adjusted daily volatility, and long-run volatility of the S&P 500 universe return, estimated in trailing 20-year windows and plotted at the end year of each window. The adjusted series applies the covariance correction of Section C.2, with the delay kernel estimated on the same window. Annualization uses the actual number of trading days in each window.